

Supplementary Tables: Origin, Structure, and Composition of a Spider Major Ampullate Silk Fiber Revealed by Genomics, Proteomics, and Single-Cell and Spatial Transcriptomics

Johan

2026-01-28

Contents

Supplementary Table 1	1
Supplementary Table 2	2
Supplementary Table 3	26
Supplementary Table 4	50
Supplementary Table 5	74
Supplementary Table 6	79

Supplementary Table 1

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
resized.

geneName	lascID	signature_accession	nrOfDomains	class
PySpb	LASC011791	G3DSA:1.10.274.60	15	PySp
PySpa	LASC011790	G3DSA:1.10.274.60	9	PySp
TuSpb	LASC008470	G3DSA:1.10.274.60	27	TuSp
TuSpc	LASC022086	G3DSA:1.10.274.60	17	TuSp
TuSpa	LASC002331	G3DSA:1.10.274.60	9	TuSp
TuSp-like	LASC001804	G3DSA:1.10.274.60	2	TuSp
AcSp1c	LASC012315	G3DSA:1.10.274.60	22	AcSp
AcSp1b	LASC012314	G3DSA:1.10.274.60	13	AcSp
AcSpa	LASC004415	G3DSA:1.10.274.60	6	AcSp
MiSpa	LASC007254	G3DSA:1.10.274.60	1	MiSp
MiSpb	LASC007257	G3DSA:1.10.274.60	1	MiSp
MiSpd	LASC022495	G3DSA:1.10.274.60	1	MiSp
AmSp-like2	LASC020272	G3DSA:1.10.274.60	1	AmSp-like
LASC000534	LASC000534	G3DSA:1.10.274.60	5	Other
sp-1339#prd34368.1	LASC020416	G3DSA:1.10.274.60	3	Other
x975_2	LASC020273	G3DSA:1.10.274.60	2	Other
LASC001876	LASC001876	G3DSA:1.10.274.60	1	Other
LASC001879	LASC001879	G3DSA:1.10.274.60	1	Other
LASC004246	LASC004246	G3DSA:1.10.274.60	1	Other

Note:

Table 1: * indicates spacer domains identified via HMM search.

Supplementary Table 2

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
resized.

geneName	Gland	Dope
MiSpb	0.0264021	0.4180026
MiSpa	0.0395967	0.4754890
LASC007034	0.0003730	NA
eef1alpha1_2	0.0156908	0.0011990
lman2l	0.0060777	NA
ef2	0.0046196	0.0000378
MiSpd	0.0022005	0.0264726
sesb	0.0018940	0.0000509
rplp0	0.0034119	NA
rps8	0.0044830	0.0000305
rps15a	0.0023736	NA
hdlbp_2	0.0008963	0.0000018
tg_3	0.0004596	0.0002352
ctsb_2	0.0121978	0.0007408
LASC003944	0.0024064	0.0002440
rpl3	0.0014386	NA
rps6	0.0023943	NA
rpl31	0.0032167	NA
rpl34	0.0037448	NA
rpl27a_2	0.0024247	NA
rpl19	0.0014316	NA
eef1g	0.0017969	0.0000222
tgfb1_4	0.0018449	0.0038838
rpl23_rpl23-a	0.0031444	NA
rpl10	0.0012665	NA
LASC015047	0.0139956	0.0018185
rpp-2_rplp2	0.0056112	0.0000837
rps27l_rps27	0.0016181	NA
rpl21	0.0028443	0.0000504
rpl8	0.0022981	0.0000206
rps11	0.0020844	NA
rpl-36.a	0.0008263	NA
LASC015735	0.0003500	NA
rps14	0.0049201	NA
rps4	0.0029852	NA
SpiCE-LMa3	0.0035932	0.0068701
pabpc1	0.0005376	NA
rplp1	0.0016638	NA
rps25	0.0051929	NA
rps18	0.0026501	NA
rps3	0.0037962	NA
rpl5_rpl5-a	0.0020346	0.0000109
rpl24	0.0019577	0.0000286
rpl27	0.0021103	NA
x975_2	0.0003038	0.0003184

(continued)

geneName	Gland	Dope
rps23	0.0036802	0.0001317
sp-1339#prd34368.1_3	0.0193003	0.0061423
rps9	0.0038885	NA
rps15	0.0021890	NA
gapdh1	0.0044553	0.0005179
rpl11	0.0045191	0.0000811
LASC006378	0.0015474	0.0000617
rpl15	0.0022073	NA
rpl18	0.0032645	0.0001281
rps7	0.0033927	NA
rpl6_rpl-6	0.0039146	0.0000208
rpl18a	0.0017878	NA
inx2_18	0.0028647	NA
rpl17	0.0037895	0.0001069
rpl32	0.0008807	NA
LASC000281	0.0019728	0.0000603
rpl40	0.0024991	0.0007184
LASC014844	0.0041803	NA
rpl36	0.0036393	NA
ssr1	0.0022459	NA
rpl14	0.0041531	NA
rps12	0.0014631	NA
rps27a	0.0024991	0.0007184
rpl23a	0.0049081	0.0001140
eef1alpha1	0.0011442	NA
rps5	0.0021857	NA
rpl7a	0.0042136	NA
rps20	0.0021260	NA
rps2_2	0.0023735	NA
rpl10ab	0.0015423	NA
rpl4	0.0049829	0.0000519
sec61g	0.0003327	NA
LASC000747	0.0006403	NA
rack1	0.0024789	0.0000316
rpl37a_2	0.0008862	NA
rpl39	0.0020653	NA
rpl-7	0.0045146	NA
rpsa	0.0027124	0.0000195
LASC003517	0.0099494	0.0004596
rpl12	0.0042402	0.0000494
fau_2	0.0012991	NA
idh2	0.0010491	NA
LASC019282	0.0139956	0.0018185
rps19	0.0036190	0.0000891
LASC010830	0.0004231	NA
EIF4A1	0.0004995	0.0000446
rps26	0.0023832	NA

(continued)

geneName	Gland	Dope
rpl13	0.0032840	NA
rpl13a	0.0032301	NA
rsf1_3	0.0010782	NA
rps17	0.0020023	NA
cnbp	0.0004499	NA
phgdh	0.0011968	0.0000158
eef1d	0.0012007	NA
ktn1	0.0010873	NA
rps10b	0.0032356	NA
vdac2	0.0015478	NA
gcat	0.0005475	NA
rpl9_2	0.0021665	NA
psat1	0.0018406	0.0000871
ssr2	0.0015517	NA
rpl26_2	0.0017345	NA
rpl35a	0.0012371	NA
rps24	0.0018149	NA
eef1b_2	0.0011408	NA
tdpoz4_15	0.0001209	NA
rpl9	0.0009775	NA
nupr1	0.0019357	NA
rpl28	0.0011829	NA
LASC011384	0.0008800	NA
ece2_2	0.0002971	NA
tsr	0.0021017	0.0003961
slc25a3	0.0003072	NA
14-3-3zeta_ywhab	0.0013150	NA
cotl1	0.0008130	NA
znf346	0.0004452	NA
rpl22	0.0036397	NA
ncl	0.0015872	NA
atp5f1b	0.0021988	0.0002757
tth-1	0.0043405	0.0002407
rpl30	0.0010541	NA
ctsb	0.0064252	0.0002857
surf4	0.0005555	NA
SpiCE-LMa6	0.0007593	NA
tgas006m08.1_2	0.0028019	0.0003879
cp1	0.0027809	NA
LASC005273	0.0007461	NA
LASC015682	0.0073373	0.0002766
b3galt5	0.0001470	NA
dap	0.0028026	NA
rps13	0.0030159	NA
slc25a4	0.0003126	NA
MiSpc	0.0001439	0.0003061
LASC017036	0.0004301	NA
ubc	0.0050368	0.0014545

(continued)

geneName	Gland	Dope
LASC019923	0.0048869	NA
yars	0.0001925	NA
EIF3I_v1g239264_iscu	0.0001088	NA
ahcy-a	0.0003849	NA
blw_atp5f1a	0.0025870	0.0002250
rpl26	0.0004578	NA
eno1	0.0028943	0.0000697
sec61b	0.0024541	NA
EIF6	0.0001520	NA
pro5	0.0011901	0.0001188
cyp1	0.0026261	0.0004229
mdh2	0.0008129	NA
nars	0.0001847	NA
LASC000095	0.0018645	0.0001154
khdrbs3	0.0016498	0.0000764
eef5_tif51a_eif5a1	0.0033106	0.0000935
LASC017035	0.0012371	0.0000656
rpl35_2	0.0008812	NA
cth	0.0003658	NA
LASC007825	0.0028958	0.0002004
atpalpha_4	0.0028241	0.0012697
rpl35	0.0008812	NA
aldh1a2	0.0004402	NA
cher_2	0.0025352	0.0000077
cd63	0.0021642	NA
oat	0.0002059	NA
uqcrfs1	0.0003170	NA
lama2_2	0.0008341	NA
ywrd_5	0.0002695	0.0000056
rtn3-b	0.0004605	NA
cpib	0.0051724	0.0007111
hspa9	0.0002371	NA
actn	0.0002689	NA
jagn	0.0009532	NA
rpl38	0.0028195	0.0000570
maob_3	0.0001897	NA
rps16_2	0.0037131	0.0000432
shmt2	0.0010768	NA
LASC021594	0.0027449	0.0002550
ddx24	0.0006102	NA
slit1_2	0.0006553	NA
LASC019949	0.0001126	NA
LASC020361	0.0005014	NA
uba1	0.0005418	NA
reep5_4	0.0013025	NA
cox6a1	0.0002707	NA
tub2_2	0.0006891	NA

(continued)

geneName	Gland	Dope
nap1l1-b	0.0001819	NA
rpl27a	0.0024247	NA
slc3a1_3	0.0007268	NA
LASC016339	0.0031990	0.0015581
hnrnpdl	0.0025672	0.0000394
sqd	0.0010282	NA
hint1	0.0004586	NA
sar1a_zk180.4_sar1b	0.0004301	NA
arf1_2	0.0005042	0.0001358
aldh1a2_2	0.0001327	NA
cg12163	0.0002422	NA
LASC007830	0.0013379	NA
naca	0.0003359	NA
tubb4b_3	0.0005118	NA
rab1a	0.0010747	NA
pdhb-1_pdhb	0.0002377	NA
tmem214	0.0003836	NA
sec23a	0.0001622	NA
rps21	0.0022463	NA
cct2	0.0002897	NA
arfgap2	0.0001047	NA
tubb4b_4	0.0011092	0.0001449
LASC020761	0.0008883	NA
acadm	0.0001206	NA
acp7	0.0009732	NA
pdi	0.0011476	NA
EIF3-s7	0.0001494	NA
aco2	0.0007771	NA
psap	0.0008218	NA
flkbp12	0.0011142	0.0003664
me1	0.0006123	NA
sdha	0.0004294	NA
lamp1_2	0.0007277	NA
atpsynf	0.0004850	NA
rsf1_2	0.0011776	NA
EIF4e1a EIF4e	0.0001255	NA
ywhae_2	0.0006413	NA
rps28	0.0048841	0.0000957
serbp1	0.0002716	NA
blmh	0.0003226	NA
tktl2_tktl1	0.0003937	NA
ssr4	0.0014960	NA
LASC020665	0.0002257	NA
inx2_2	0.0002459	NA
got2	0.0005852	NA
jafrac1_2	0.0014690	0.0001260
cg1354	0.0004006	NA
sdhb	0.0002737	NA

(continued)

geneName	Gland	Dope
tbl2	0.0006105	NA
psmb7_2	0.0003096	NA
tpi	0.0010151	0.0002008
his1.3	0.0023024	NA
cct7	0.0001059	NA
LASC021646	0.0031073	0.0032556
sars	0.0003442	NA
LASC017315	0.0003602	NA
lec-1	0.0012397	0.0000478
tmed2	0.0003840	NA
actr1b	0.0002282	NA
tpd52	0.0006035	NA
LASC005487	0.0004327	NA
pkm	0.0005033	NA
cox5b	0.0005263	NA
wos2	0.0002982	NA
sdcbp	0.0001358	NA
LASC002459	NA	0.0001030
mlc-c	0.0017804	0.0001228
alyref2_2	0.0003279	NA
agap002499	0.0003457	NA
atpsyngamma	0.0002682	NA
sqstm1	0.0003420	0.0000858
lgals4	0.0001196	NA
LASC021266	0.0024729	NA
ddx4_vasa	0.0001770	NA
LASC015439	0.0053412	NA
moe	0.0009348	NA
rho-1_3	0.0004011	NA
ybx1	0.0004260	NA
aldh1l2	0.0007317	NA
ael006169	0.0015564	0.0001196
alpl_8	0.0005558	NA
dld	0.0002807	NA
atpsyno	0.0005742	NA
cct3	0.0001149	NA
tmed6	0.0002592	NA
ppp2ca	0.0001628	NA
cct6	0.0002892	NA
nas-6	0.0001104	NA
nnmt_v1g200856	0.0003489	NA
cox5a	0.0006546	NA
eca	0.0008039	NA
psmb6	0.0002817	NA
LASC021418	0.0012240	0.0004072
atp5f1d	0.0005531	NA
ppp2r1a	0.0002432	NA

(continued)

geneName	Gland	Dope
aimp1	0.0002893	NA
LASC015247	0.0010865	NA
cndp2	0.0010003	NA
sorbs1_2	0.0001681	NA
pdia3	0.0014050	NA
kars1_2	0.0002244	NA
psme3	0.0002781	NA
psma4	0.0002459	NA
lap-2	0.0005540	NA
cct8	0.0002647	NA
hnrbp_2	0.0002839	NA
clta	0.0003981	NA
ube2l3_2	0.0002049	NA
gdi2_gdi1	0.0006270	0.0000883
elf3j	0.0001606	NA
dlst	0.0002779	NA
eff_2	0.0001129	NA
aldh18a1	0.0001544	NA
hrb27c_2	0.0001394	NA
eff_9	0.0001129	NA
cpa_3	0.0002876	NA
stom	0.0004031	NA
crip1	0.0005375	NA
cat2_2	0.0007562	NA
hrb87f	0.0025039	NA
ndufa4	0.0002572	NA
tars1_tarsl2	0.0002170	NA
LASC022132	0.0002645	NA
sar1b	0.0002542	NA
sp-1339#prd34368.1_4	0.0003782	0.0000630
gyg1_2	0.0001732	NA
atpsynb	0.0004918	NA
cpb_capzb	0.0001703	NA
uqcrb	0.0002258	NA
ndufb9	0.0001430	NA
cct5	0.0001835	NA
LASC003928	0.0001487	NA
ndufa5	0.0002667	NA
lamp1	0.0001363	NA
rho1_7	0.0005550	NA
enpp6_3	0.0001506	NA
llgl1	0.0006022	NA
cycl	0.0001793	NA
uqcrc1	0.0004994	NA
rars	0.0003020	NA
elf3g-b	0.0001110	NA
set	0.0010151	NA
igf2bp1	0.0005127	NA

(continued)

geneName	Gland	Dope
ehd1_2	0.0002107	NA
btf3l4	0.0004985	NA
metap1	0.0001553	NA
aldh7a1	0.0009995	NA
LASC018632	0.0001889	NA
nop58_2	0.0001673	NA
atox1	0.0006088	NA
pitpna	0.0001089	NA
sqh_2	0.0006517	NA
nsf1c	0.0001061	NA
hspa4l_2	0.0001295	NA
acly	0.0002260	NA
fubp3_fubp1	0.0002003	NA
LASC005920	0.0001078	NA
ter94_vcp	0.0008467	0.0000311
LASC016997	0.0001891	0.0003029
lark	0.0002085	NA
LASC010482	0.0002903	NA
spg21	0.0001083	NA
LASC006288	0.0003267	NA
ap2m1	0.0001181	NA
slc7a6_slc7a8	0.0002780	NA
etfa	0.0004427	NA
arcn1	0.0002876	NA
fh	0.0003281	NA
uqcrc2	0.0003354	NA
ago3_ago2	0.0001380	NA
LASC012598	0.0001897	NA
ube2n_ubc-13	0.0002050	NA
cnp_2	0.0002143	NA
snx12	0.0001556	NA
ndufs4	0.0001113	NA
ran-4	0.0001374	NA
aimp2	0.0001507	NA
psma6	0.0003574	NA
nop60b	0.0005188	NA
anxa1l	0.0012920	NA
acat1-b	0.0002264	NA
fib	0.0005352	NA
hadh	0.0001083	NA
prmt8	0.0001884	NA
zip	0.0019633	NA
atpsyncf6	0.0005641	NA
tuba1c	0.0001937	0.0000744
sfpq	0.0002151	NA
erf1	0.0001464	NA
LASC017716	0.0011863	0.0001972

(continued)

geneName	Gland	Dope
phb	0.0004898	NA
sod1	0.0010806	NA
gs2_2	0.0009499	NA
plin2	0.0003049	NA
LASC009674	0.0001255	NA
uqcrq	0.0001444	NA
psma1	0.0002775	NA
l(2)efl_hspb1	0.0004532	NA
cltc	0.0010974	NA
gstd1_2	0.0001841	NA
ugp2_2	0.0001596	NA
lrrc47	0.0001095	NA
psmb5	0.0001953	NA
gbe1	0.0001262	NA
LASC017623	0.0011863	0.0001972
rpn11	0.0002134	NA
cubn_3	0.0007238	NA
snx2	0.0002412	NA
efhd2	0.0001527	NA
qars	0.0004174	NA
ddx1	0.0001571	NA
LASC013621	NA	0.0010738
psmc6	0.0002934	NA
EIF3-s8	0.0001299	NA
cfad	0.0001003	NA
atp5mg	0.0001822	NA
psmc5	0.0003382	NA
septin7	0.0002532	NA
rtca	0.0001723	NA
apip	0.0001124	NA
bai_tmed10	0.0014443	NA
ewsrl	0.0004724	NA
aass	0.0006076	NA
hibadh	0.0001474	NA
ptcd2	0.0001494	NA
zan_srfbp1	0.0005255	NA
ndufs1	0.0001679	NA
ube2v2	0.0003136	0.0000442
LASC007277	0.0005276	NA
hnrnpk	0.0004275	NA
ywhae	0.0032983	0.0000745
psmd8	0.0001637	NA
cd109	0.0003028	NA
vha26	0.0007451	NA
sirt5	0.0002147	NA
dnajc22	0.0008124	NA
atpsynd	0.0003070	NA
ppib_cyn-5	0.0005660	NA

(continued)

geneName	Gland	Dope
ech1_2	0.0007245	NA
srp68	0.0001155	NA
LASC005698	0.0001880	NA
copa	0.0002372	NA
abat	0.0004247	NA
EIF3A_2	0.0001444	NA
ndufs3	0.0001108	NA
ciz1	0.0005435	NA
rpn6	0.0002624	NA
sqh	0.0003265	NA
alpha-spec	0.0009747	NA
srsf12	0.0001445	NA
cbx3	0.0001753	NA
LASC011933	0.0005119	NA
vha55	0.0010823	0.0000491
pebp1	0.0004967	NA
LASC011232	0.0008479	NA
mdh1	0.0011059	NA
gspt1	0.0001225	NA
ptbp1	0.0001150	NA
his2av	0.0001863	0.0000271
tdh	0.0003034	NA
ipgm-1	0.0003108	NA
psma3	0.0004608	NA
rab2a	0.0003055	NA
cap2	0.0001195	NA
ldlr	0.0001577	NA
LASC016532	0.0001227	NA
psmd13	0.0001391	NA
gstM2	0.0001069	NA
psmd2	0.0001838	NA
ppif	0.0003805	0.0000740
hsp83_hsp90aa1	0.0006759	0.0000613
ndufa13	0.0001134	NA
slc9a3r1	0.0002695	NA
aael011789	0.0006483	NA
pkm_3	0.0007369	NA
copg2	0.0003378	NA
echS1	0.0004192	NA
LASC004161	0.0002000	NA
EIF4B	0.0001899	NA
calr_crt	0.0009256	NA
hspb1_3	0.0001061	NA
abce1	0.0002267	NA
LASC011347	0.0003169	NA
phb2	0.0002488	NA
glyp	0.0004418	NA

(continued)

geneName	Gland	Dope
gfpt2	0.0001136	NA
grn_6	0.0006975	NA
snd1	0.0014523	NA
folr3	0.0001280	NA
eprs	0.0003969	NA
prdx6	0.0003881	NA
EIF3F	0.0001026	NA
snrpb	0.0001043	NA
psma5	0.0005029	NA
idh3a	0.0002135	NA
rad23b	0.0002069	NA
SpiCE-LMa4	0.0002376	0.0001751
psmc2	0.0002474	NA
LASC018834	0.0001025	NA
tbb-4_tubb4b	0.0009238	NA
aldh4a1	0.0007698	NA
stip1	0.0002987	NA
hsp60a	0.0005226	NA
lap3	0.0001496	NA
LASC015638	0.0030165	0.0001295
psmd12	0.0001652	NA
abhd12_2	0.0002464	NA
cox6b1	0.0003494	NA
fas2_3	0.0005556	NA
LASC016132	0.0021458	0.0002278
vdac2_2	0.0003748	NA
tth-1_2	0.0006805	NA
sod-2	0.0003935	NA
nurf-38_ipp1	0.0001172	NA
psmb3	0.0001943	NA
hsd17b10_2	0.0002214	NA
ndufv2	0.0001802	NA
lman1	0.0002545	NA
nop10	0.0007052	NA
hce_hcf	0.0000355	0.0009021
aldh2_2	0.0004548	NA
csga	0.0001953	NA
arl8ba_arl8b	0.0001740	NA
LASC006772	0.0003760	NA
ppib_cypb	0.0001872	0.0001284
gbeta1	0.0003373	NA
rab5b	0.0003132	NA
hadha	0.0001904	NA
nhp2	0.0004804	NA
ogdh	0.0001663	NA
rab7_rab7a_2	0.0003713	NA
beta-spec_2	0.0011048	NA
cpt1a_3	0.0001173	NA

(continued)

geneName	Gland	Dope
LASC013620	NA	0.0006039
ran_2	0.0004589	NA
tmco1	0.0001782	NA
LASC011026	0.0001710	NA
copb1	0.0002181	NA
kyat3	0.0002077	NA
tfg	0.0001776	NA
myo6_jar	0.0002158	NA
atp6v0a1	0.0003654	NA
pa2g4	0.0004453	NA
oplah	0.0001688	NA
dpp3_3	0.0001569	NA
nolc1	0.0001244	NA
anp32a_2	0.0004344	NA
sacm1l_sacm1lb	0.0002004	NA
sme	0.0001306	NA
cct4	0.0002297	NA
gpx4	0.0001247	NA
rala	0.0001346	NA
sec22b	0.0003655	NA
vapa	0.0001652	NA
tufm	0.0001118	NA
vat1l_2	0.0001336	NA
LASC013486	0.0002424	0.0000108
rpt2_2	0.0003709	NA
gcdh	0.0001858	NA
chmp4b	0.0002059	NA
ndufa8	0.0001996	NA
LASC018874	0.0031073	0.0032556
nop58	0.0001673	NA
serca_2	0.0003437	NA
psmd4	0.0001920	NA
smo-1	0.0007249	NA
edf1	0.0001282	NA
ndufs7	0.0001157	NA
sfpq_2	0.0003353	NA
serbp1_2	0.0001605	NA
sgtb	0.0001634	NA
ndufa7	0.0001218	NA
pgk	0.0008747	NA
snx6	0.0001284	NA
pdcd4	0.0001531	NA
nd-42	0.0001235	NA
chmp2a	0.0001021	NA
nucb2	0.0002633	NA
f42h10.3_lasp1	0.0001249	NA
dars	0.0002699	NA

(continued)

geneName	Gland	Dope
gdh	0.0001790	NA
uso1	0.0001348	NA
ssb	0.0001490	NA
txnl1	0.0001224	NA
cct1	0.0001744	NA
fk506-bp2_fkbp1b	0.0001667	NA
psma2	0.0003630	NA
psma8_2	0.0002934	NA
dsp1	0.0003823	NA
rtcb	0.0001122	NA
impa1	0.0001466	NA
cd109_4	0.0003784	NA
zmpste24	0.0001753	NA
plekhf2	0.0001424	NA
hadhb	0.0002516	NA
etfb	0.0005443	NA
blvrb	0.0006293	NA
stx7	0.0001029	NA
prdx4_prdx1	0.0001604	NA
hspa5_2	0.0010573	0.0000602
tecr_art-1	0.0001165	NA
eif4ebp2_eif4ebp1	0.0002412	NA
scsalpha1_os07g0577700	0.0003146	NA
ehd1_3	0.0001162	NA
psmb2	0.0002740	NA
atpalpha	0.0011540	NA
LASC016362	0.0008355	NA
hnrnpf_hnrnp2	0.0002731	NA
slc8a2	0.0001044	NA
ndufs6	0.0001263	NA
rbm4_rbm14	0.0005442	NA
LASC013967	0.0000819	0.0003821
cs	0.0004367	NA
hibch	0.0003241	NA
LASC010684	0.0003306	NA
LASC009347	0.0000839	0.0003569
cg3556_15	0.0005410	NA
uchl3_3	0.0001876	NA
park7	0.0004220	0.0000761
elavl2	0.0002254	NA
cnm3	0.0009323	NA
LASC001129	0.0002484	NA
snu13	0.0011172	NA
gstm3	0.0034767	0.0001505
nnt	0.0003239	NA
hspa5	0.0002025	NA
cope	0.0002141	NA
hnrnp_3	0.0004703	NA

(continued)

geneName	Gland	Dope
jafrac1	0.0006307	NA
lrp2_6	0.0002769	NA
snr-4	0.0001818	NA
esd_2	0.0001025	NA
prep	0.0001033	NA
acat1_2	0.0004178	NA
psmd1_rpn2	0.0001033	NA
ndufb4	0.0001714	NA
dcxr_2	0.0001082	NA
ppp1cc_ppp1cc-b	0.0001721	NA
kpnb1	0.0002035	NA
glyp_2	0.0005530	0.0000371
ppp1cb	0.0001263	NA
cpt1a_2	0.0001379	NA
rpn1	0.0002899	NA
LASC002299	0.0004620	NA
hnrnp1	0.0002289	NA
mapre3	0.0002868	NA
cdc42_2	0.0001475	NA
ap2b1	0.0001292	NA
lrpap1	0.0001192	NA
nop56	0.0005478	NA
cand1	0.0001150	NA
tm_fabg	0.0002676	NA
mtap	0.0001666	NA
c05d11.1	0.0001292	NA
idh1	0.0004889	NA
acads	0.0001759	NA
suc1g2	0.0002589	NA
itga9	0.0001653	NA
LASC005335	0.0003053	NA
pxn	0.0001235	NA
glrx3	0.0001418	NA
scfd1	0.0001094	NA
ap1m1	0.0001723	NA
rap1b	0.0004236	NA
pebp1_2	0.0001293	NA
caps1_2	0.0008509	0.0003987
ablim1_ablim2	0.0001455	NA
aprt	0.0001624	NA
acadvl	0.0001714	NA
LASC001932	0.0003924	NA
igf2r	0.0001981	NA
spac4h3.07c	0.0001301	NA
anp32a	0.0004313	NA
hcd	0.0000322	0.0004829
ehd1	0.0003540	NA

(continued)

geneName	Gland	Dope
txn2	0.0001300	NA
hcg	0.0000680	0.0012106
hca	0.0000296	0.0005083
twf2	0.0001243	NA
vha14	0.0002576	NA
arf4	0.0004605	NA
alkbh3	0.0001213	NA
der11	0.0001120	NA
pgd	0.0001264	NA
rab18	0.0001037	NA
hspe1	0.0007189	NA
rab7_rab7a	0.0004975	NA
zk1073.1_2	0.0005217	0.0000243
slc25a11	0.0001069	NA
plxdc1	0.0002378	NA
srsf4_3	0.0002210	NA
ivd	0.0002103	NA
acdh-6_psmf1	0.0001176	NA
lgmn_2	0.0003441	NA
pdcd6	0.0001084	NA
hsp90b1	0.0003961	NA
prdx3	0.0001962	NA
spa	0.0002778	NA
EIF1A	0.0001030	NA
IMMT	0.0001402	NA
COXB2	0.0001154	NA
DAG1	0.0001871	NA
LASC009784	0.0006971	NA
EIF3K	0.0001164	NA
GAR1	0.0006094	NA
TIN2	0.0002767	NA
LRRC59	0.0002032	NA
VHA36-1	0.0001973	NA
ACAA2	0.0002755	NA
CYT-B5	0.0005714	NA
C1QBP	0.0002607	NA
PIWIL1	0.0001722	NA
NDUF7	0.0001830	NA
EIF2A	0.0001733	NA
B52	0.0002210	NA
HM13	0.0003511	NA
MTHFS_2	0.0003104	NA
DHC64C	0.0001464	NA
FK506-bp1_fpr4	0.0002596	NA
AHCYL1	0.0001634	NA
CBR1	0.0002329	NA
IGF2BP1_2	0.0001052	NA
DPM1	0.0001690	NA

(continued)

geneName	Gland	Dope
snrpg	0.0001261	NA
arhgdib_arhgdia	0.0007390	NA
arp3	0.0001064	NA
pcd_pcbd2	0.0004936	NA
vhaac39-1	0.0003499	NA
paip1	0.0001626	NA
rtraf	0.0001157	NA
dlat	0.0001164	NA
npc2a_2	0.0003292	NA
acad8	0.0001198	NA
hcb	NA	0.0002836
tmem68	0.0001069	NA
scpep1	0.0001242	NA
ist1	0.0001641	NA
rpt2	0.0001029	NA
nxnl2	0.0001118	NA
myg1	0.0002204	NA
hsp70b2_7	0.0006330	NA
syncrip_2	0.0004060	NA
xirp2	0.0003218	NA
fkbp7_fkbp3_fpr2	0.0011596	0.0000752
psmc3	0.0002061	NA
g3bp2	0.0001521	NA
EIF2S1	0.0001893	NA
gstz1_2	0.0001516	NA
psmb4	0.0003502	NA
hts	0.0003969	NA
dhtkd1	0.0001224	NA
rbm4	0.0001985	NA
tmx1	0.0005711	NA
ghra_2	0.0002098	NA
lypla2	0.0001586	NA
mtch2	0.0001141	NA
mys_2	0.0001667	NA
l(2)efl_5	0.0001260	NA
mecr	0.0001134	NA
canx	0.0001933	NA
ndufs2	0.0001017	NA
exc-4	0.0003530	NA
ctsa	0.0001338	NA
plxna4_2	0.0001077	NA
ddost	0.0001897	NA
hnrbp	0.0006074	NA
gpd1	0.0001739	NA
LASC019081	0.0002562	NA
arl1	0.0001176	NA
vhaa	0.0011718	NA

(continued)

geneName	Gland	Dope
LASC013820	0.0002234	NA
hcc	0.0000296	0.0002480
eif4a2	0.0002989	NA
crat	0.0002461	NA
lin7b	0.0004249	NA
hsp90ab1	0.0006759	0.0000613
taldo1_taldo	0.0004936	NA
alpha-cat	0.0001046	NA
actg1	0.0056560	0.0013910
snrpd3_2	0.0001913	NA
rpn2	0.0002383	NA
mroh1	0.0001491	NA
larpl	0.0001091	NA
gly3	0.0001010	NA
epb41l3_2	0.0003002	NA
cpm_2	0.0003204	NA
lman2	0.0001704	NA
myef2_2	0.0001199	NA
grn_7	0.0021017	NA
pdia6	0.0016633	NA
dctn2	0.0001238	NA
LASC019392	0.0001163	NA
LASC006421	0.0001761	NA
col4a1_2	0.0006306	NA
adk2	0.0001140	NA
gstml_3	0.0002424	0.0000108
t05h10.6	0.0002426	NA
tmed7	0.0001647	NA
lcp1	0.0006573	NA
FlSpC	0.0004015	0.0003107
dcd2b	0.0001140	NA
ubxn1_ubxn1-a	0.0001065	NA
pura	0.0003055	NA
ddx39b	0.0003215	NA
cisd1	0.0003302	NA
vps35	0.0001537	NA
hts_2	0.0004243	NA
gmpr_gmpr2	0.0002013	NA
reep6_reep5	0.0001019	NA
rpia	0.0001096	NA
hspl1_2	0.0001176	NA
erlin2-b	0.0001788	NA
h2afv	0.0001863	0.0000271
metap2	0.0001177	NA
asna	0.0005575	NA
ddx3x	0.0001220	NA
LASC010290	0.0001009	NA
parn	0.0001429	NA

(continued)

geneName	Gland	Dope
FlSpb	0.0064473	0.0026180
mif	0.0002649	NA
eci1	0.0002413	NA
atl	0.0001006	NA
dpys	0.0005136	NA
nrg	0.0005822	NA
vhasfd	0.0001994	NA
apmap_3	0.0003746	NA
letm1	0.0001011	NA
ap2a2_2	0.0002022	NA
txn	0.0008091	NA
nipsnap_2	0.0001056	NA
awd	0.0004075	NA
etfdh	0.0001026	NA
usp5	0.0001710	NA
pgls	0.0001111	NA
LASC010973	0.0001461	NA
lam_2	0.0006485	NA
LASC010012	0.0096775	0.0016009
mrpl12	0.0001693	NA
snb-1_vamp3_syb	0.0001076	NA
LASC005883	0.0001536	NA
manf_2	0.0001597	NA
EIF4H	0.0001317	NA
sep2	0.0001625	NA
LASC005807	0.0001227	NA
mpp6	0.0004255	NA
bcap31	0.0001791	NA
nrg_2	0.0001119	NA
vinc	0.0005565	NA
unc-94	0.0002277	NA
smf	0.0002530	NA
bcat2_v1g246094	0.0001445	NA
sec11c	0.0003285	NA
tp02	0.0001697	NA
bola2	0.0001166	NA
myef2	0.0002663	NA
prtp	0.0001120	NA
ylpm1	0.0001111	NA
oxr1_2	0.0002024	NA
LASC022131	0.0001086	NA
srp14	0.0001740	NA
sfxn1	0.0002426	NA
LASC001730	0.0030770	NA
camki	0.0003299	NA
hexa	0.0001154	NA
ranbp1	0.0001308	NA

(continued)

geneName	Gland	Dope
syncrip	0.0002016	NA
hyou1	0.0001026	NA
LASC016483	0.0002189	NA
mtpn	0.0004409	NA
rplp1_2	0.0001455	NA
usp14	0.0001085	NA
xpnpep1	0.0001052	NA
LASC016310	0.0047743	0.0005751
f26e4.3_2	0.0003806	NA
ppt1	0.0001507	NA
cse1l	0.0004624	NA
qdpr	0.0002682	NA
LASC004844	0.0002671	NA
sub1	0.0003551	NA
slc27a1_2	0.0001139	NA
LASC016530	0.0001765	NA
lana	0.0001422	NA
AgSp2b	0.0001604	0.0001988
FlSpa	0.0010670	0.0010524
idh3g	0.0001703	NA
lanb1	0.0001380	NA
adk_adk2	0.0001593	NA
nudt16	0.0001409	NA
pdcd6_2	0.0002039	NA
hnrnpul1	0.0002406	NA
lamtor3	0.0001326	NA
atp6ap1	0.0004026	NA
slc25a3_2	0.0003072	NA
LASC003279	0.0001942	NA
hspg2	0.0020658	NA
spcs2	0.0001957	NA
pgm1_2	0.0002165	NA
pnliprp1	0.0003379	NA
emc1	0.0001029	NA
lam	0.0003101	NA
jana	0.0001376	NA
rac1_2	0.0001235	NA
nrx-iv	0.0002784	NA
khsrp	0.0001403	NA
tgas006m08.1_3	0.0028019	0.0003879
tom1	0.0001049	NA
m6pr	0.0001139	NA
pccb	0.0002098	NA
LASC011699	0.0031073	0.0032556
LASC017285	0.0003068	NA
golim4	0.0001250	NA
flo2_2	0.0001304	NA
sptbn5	0.0007664	NA

(continued)

geneName	Gland	Dope
ergic1	0.0002598	NA
SpiCE-LMa2	0.0004824	0.0000544
dek	0.0001747	NA
LASC020142	0.0003306	NA
cg3556_2	0.0009319	NA
LASC017000	0.0001527	NA
cana-14f	0.0001475	NA
LASC002717	0.0001736	NA
LASC015168	0.0002088	NA
flr	0.0001362	NA
ube2l3	0.0001593	NA
tpd52l2	0.0001042	NA
pde12_4	0.0001739	NA
gpr158	0.0004304	NA
vat1l	0.0001282	NA
rab14	0.0001456	NA
cont	0.0005380	NA
meltf	0.0002989	NA
LASC003125	0.0001584	NA
cav1_3	0.0004213	NA
futsch	0.0001354	NA
ank3_8	0.0003164	NA
col4a1_let-2	0.0002629	NA
ccdc58	0.0001202	NA
LASC019275	0.0001566	NA
mvp	0.0002528	NA
atpsynd_2	0.0003070	NA
gyg1	0.0001176	NA
pabp2	0.0001093	NA
rab6a	0.0008995	NA
tgfb1_3	0.0001938	NA
LASC002397	0.0005333	NA
brafldraft	0.0001550	NA
LASC007511	0.0001400	NA
pka-c1	0.0001241	NA
sep1	0.0001833	NA
LASC020300	0.0011300	0.0003008
LASC017532	0.0010455	NA
csrp2	0.0001277	NA
tom1l2	0.0001455	NA
SpiCE-LMa1	0.0010371	NA
LASC021415	0.0031073	0.0032556
hpgd_14	0.0001975	0.0001220
lsm10	0.0001113	NA
stt3a	0.0001035	NA
lmf2	0.0001440	NA
sh3beta	0.0001190	NA

(continued)

geneName	Gland	Dope
LASC005973	0.0001211	NA
dpp3_4	0.0001070	NA
pcca	0.0002093	NA
LASC002879	0.0003508	NA
ppm1e	0.0001286	NA
AcSpa	0.0006370	0.0049870
AmSp-like1	0.0003299	NA
macrodl_macrodl2	0.0001642	NA
csga_5	0.0001554	NA
rheb	0.0001103	NA
txndc17	0.0001690	NA
LASC011075	0.0003292	NA
arl6ip5	0.0002400	NA
pld3	0.0003319	0.0001263
capn5	0.0001328	NA
isoc2	0.0001011	NA
arf1	0.0004783	NA
lbr	0.0001425	NA
gstml	0.0002424	0.0000108
LASC005034	0.0001037	0.0004924
lac_7	0.0002535	NA
itpr1	0.0007754	NA
jhamt_7	0.0010998	NA
lanb2_2	0.0001361	NA
mlec	0.0001285	NA
grhpr_7	0.0010964	0.0002257
h2afy	0.0005067	NA
pka-cl_2	0.0001241	NA
bpnt1	0.0001451	NA
ncam1	0.0001108	NA
AmSp-like2	0.0001933	NA
galphaq	0.0002784	NA
trip11	0.0013937	NA
MaSp1a	0.0048159	0.0041661
CaAn_20	0.0007019	0.0003647
LASC004484	0.0195565	0.0003792
gstml_10	0.0002424	0.0000108
adpgk	0.0001027	NA
pdia4	0.0001106	NA
MaSp1b	0.0009722	NA
AgSp1	0.0000971	0.0002607
pgm2	0.0001398	NA
hspa8_hsp70	0.0015474	0.0000617
LASC018595	0.0020125	NA
ndrg3_ndrg4-a	0.0007556	0.0002260
LASC001726	0.0167368	0.0000799
ftcd	0.0001403	NA
naga_3	0.0001268	NA

(continued)

geneName	Gland	Dope
LASC004993	0.0018371	0.0000305
aldh5a1	0.0001322	NA
nt1	0.0001155	NA
LASC015175	0.0000934	0.0018520
colla2	0.0003845	NA
SpiCE-LMa5	0.0002618	0.0005442
lev-9_cr1_csmd3	0.0001643	NA
LASC009239	0.0001066	NA
mttp_2	0.0001883	NA
LASC016433	0.0003599	NA
LASC006408	0.0099523	0.0000460
LASC000038	0.0035913	NA
MaSp2c	0.0006364	0.0012708
MaSp1c	0.0015840	NA
ykvo_tm_fabg	0.0002676	NA
ywrd_2	0.0002695	0.0000056
LASC004995	0.0005096	NA
erp44_3	0.0001235	NA
cmpk	0.0001353	NA
ptrhd1	0.0001423	NA
hdlbp_3	0.0008963	0.0000018
LASC007099	0.0001924	NA
dst_2	0.0001678	NA
hsd17b6_hsd17b2	0.0002384	NA
lox12	0.0004864	NA
uchl3	0.0001876	NA
LASC012887	0.0002223	NA
pka-r2_2	0.0003434	NA
h2bc1	0.0055492	0.0013549
nop58_3	0.0001673	NA
mas_ppaf2	0.0003685	NA
MaSp2f	0.0012685	NA
LASC010337	0.0001076	NA
mys	0.0001667	NA
LASC005488	0.0001316	NA
pdcd6_bche	0.0001084	NA
LASC009994	0.0027124	0.0000195
fasn_8	0.0001009	NA
LASC019814	0.0055492	0.0013549
zbtb48_3	0.0001593	NA
LASC003331	0.0001218	0.0035152
LASC001732	0.0005262	NA
ywrd_3	0.0002695	0.0000056
erp44_2	0.0001235	NA
LASC018776	0.0019284	NA
rpl10a	0.0015423	NA
MaSp3b	0.0010178	NA

(continued)

geneName	Gland	Dope
hsp70b2_4	0.0006330	NA
LASC001731	0.0008799	NA
pnliprp3_pnlip	0.0002937	NA
nutf2	0.0001374	NA
LASC020883	0.0001751	NA
LASC003330	0.0001489	0.0021808
beta-spec	0.0002296	NA
gstd1_gst1	0.0001841	NA
alphanub85e_tuba1c	0.0005015	0.0004572
cd109_5	0.0003028	NA
gstm1_5	0.0002424	0.0000108
LASC015292	0.0015556	NA
epb41l3	0.0003002	NA
cmpk_2	0.0001353	NA
LASC015173	0.0001428	0.0021020
ywrd_6	0.0002695	0.0000056
vit-6_vg	0.0001722	0.0031508
LASC000397	0.0001244	NA
fib_2	0.0005352	NA
cut-1_6	0.0001612	NA
LASC003332	0.0001579	0.0013617
LASC005598	0.0000316	0.0018911
LASC006407	0.0018371	0.0000305
LASC007604	NA	0.0022619
LASC022503	0.0001428	0.0021020
LASC022504	0.0000934	0.0018520
TuSpb	0.0002945	0.0081969
hpgd_12	0.0001975	0.0001220
ywrd_7	0.0002695	0.0000056
LASC008951	0.0001458	NA
LASC019165	0.0011146	NA
atpalph_3	0.0001398	NA
psmb1	0.0002450	NA
psmb1_2	0.0002450	NA
AgSp2a	0.0001604	0.0001988
LASC019470	0.0055492	0.0013549
LASC021431	0.0001863	0.0000271
LASC022267	0.0055492	0.0013549
LASC022852	0.0028019	0.0003879
LASC022854	0.0047743	0.0005751
LASC022855	0.0028019	0.0003879
LASC022856	0.0047743	0.0005751
LASC022857	0.0031073	0.0032556
LASC022858	0.0055492	0.0013549
glyp_3	0.0005530	0.0000371
h4-viii	0.0028019	0.0003879
h4-viii_2	0.0028019	0.0003879
hspa8_2	0.0002076	0.0001034

(continued)

geneName	Gland	Dope
pxt_3	0.0002376	0.0001751
tgas006m08.1	0.0028019	0.0003879
tubb4b	0.0003745	0.0001720
ankrd50_2	0.0012834	NA
ap1b1	0.0001292	NA
ctsa_2	0.0001338	NA
eef1b	0.0011408	NA
esd	0.0001025	NA
flo2	0.0001304	NA
his1	0.0001400	NA
his1.3_2	0.0001400	NA
hrb27c_4	0.0001394	NA
hsp70b2_3	0.0006330	NA
hsp70b2_5	0.0006330	NA
hsp70b2_hsp70a2	0.0006330	NA
inx2_15	0.0002459	NA

Note:

Table 2: Proteomics data. * Fold change calculated relative to control.

Supplementary Table 3

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
resized.

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
MiSpb	4.845141e+03	8.213742e+00	1.324123e+00	11.7566016	0.000
MiSpa	2.718288e+03	7.061585e+00	9.818891e-01	11.3432524	0.000
LASC007034	8.198473e+03	3.620634e+03	1.569918e+01	9.0524622	0.000
eef1alpha1_2	5.538035e+03	3.932024e+03	1.486394e+03	1.8582699	0.000
lman2l	3.473966e+03	1.758073e+03	1.060943e+01	8.3912317	0.000
ef2	5.373922e+03	4.175083e+03	1.751056e+03	1.6231090	0.000
MiSpd	1.626753e+03	2.742999e+00	5.060979e-01	11.5538636	0.000
sesb	5.317414e+03	4.598409e+03	3.179717e+03	0.7285321	0.013
rplp0	3.385902e+03	2.910151e+03	1.415828e+03	1.2424615	0.000
rps8	4.468031e+03	4.814924e+03	1.294031e+03	1.8431505	0.000
rps15a	4.477650e+03	3.763552e+03	1.733554e+03	1.3884885	0.000
hdlbp_2	3.987372e+03	2.838649e+03	5.497123e+02	2.8213265	0.000
tg_3	5.257847e+03	2.057134e+03	9.895628e+01	5.7428245	0.000
ctsb_2	9.858882e+03	3.910438e+02	7.005113e+02	3.7914994	0.000
LASC003944	4.201858e+03	2.874860e+03	1.304413e+03	1.6757522	0.000
rpl3	2.268860e+03	1.995350e+03	1.251216e+03	0.8656821	0.000
rps6	3.175161e+03	2.457148e+03	1.199750e+03	1.4028283	0.000
rpl31	3.069921e+03	2.102721e+03	1.319148e+03	1.2064734	0.000
rpl34	3.379109e+03	2.977152e+03	1.062691e+03	1.7131502	0.000
rpl27a_2	2.057216e+03	1.557351e+03	6.924432e+02	1.5855950	0.000
rpl19	2.649821e+03	2.675350e+03	7.738873e+02	1.8354915	0.000
eef1g	5.557923e+03	3.740279e+03	1.718239e+03	1.7047801	0.000
tgfb1_4	4.828915e+03	9.474382e+02	1.499477e+01	8.3221612	0.000
rpl23_rpl23-a	1.724100e+03	1.610615e+03	6.868441e+02	1.3447047	0.000
rpl10	6.911923e+03	1.170455e+04	1.852721e+02	5.2631140	0.000
LASC015047	4.521849e+03	2.199505e+03	1.712551e+03	1.3756111	0.000
rpp-2_rplp2	3.142861e+03	2.887062e+03	1.125293e+03	1.4887637	0.000
rps27l_rps27	3.940081e+03	3.272371e+03	1.367136e+03	1.5337689	0.000
rpl21	2.069554e+03	1.605102e+03	6.128357e+02	1.7508449	0.000
rpl8	2.083839e+03	1.582800e+03	8.109553e+02	1.3632251	0.000
rps11	3.535123e+03	2.876320e+03	8.614092e+02	2.0952830	0.000
rpl-36.a	3.588456e+03	3.841015e+03	7.308940e+02	2.3332136	0.000
LASC015735	2.313181e+03	1.005755e+03	5.130482e+00	8.8101998	0.000
rps14	1.912751e+03	1.633959e+03	1.150257e+03	0.7414056	0.003
rps4	1.849764e+03	1.667704e+03	9.940887e+02	0.8919110	0.000
SpiCE-LMa3	2.552414e+03	2.341899e+03	6.244300e+00	8.7180282	0.000
pabpc1	2.171889e+03	2.088746e+03	2.137997e+03	-0.0283270	0.913
rplp1	3.427015e+03	3.571274e+03	9.551799e+02	1.8729471	0.000
rps25	3.277321e+03	3.092021e+03	7.151832e+02	2.2511632	0.000
rps18	2.086027e+03	1.698368e+03	8.932671e+02	1.2257174	0.000
rps3	1.832058e+03	1.593959e+03	9.259982e+02	0.9857883	0.000
rpl5_rpl5-a	2.756767e+03	2.338162e+03	1.410657e+03	0.9759239	0.000
rpl24	2.295967e+03	1.932796e+03	1.138748e+03	1.0178567	0.000
rpl27	2.185356e+03	1.640546e+03	7.576855e+02	1.5514509	0.000
x975_2	9.490727e+02	6.094618e+02	3.265522e+00	8.2045581	0.000

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
rps23	2.186366e+03	1.587821e+03	8.455808e+02	1.3783952	0.000
sp-1339#prd34368.1_3	1.585911e+04	2.649457e+00	2.265246e+00	12.7484196	0.000
rps9	3.535104e+03	3.542951e+03	9.491052e+02	1.9465368	0.000
rps15	2.331320e+03	1.787047e+03	9.938644e+02	1.2406397	0.000
gapdh1	2.372280e+03	2.040085e+03	5.295218e+03	-1.2318671	0.000
rpl11	2.619834e+03	1.774070e+03	1.158979e+03	1.1641233	0.000
LASC006378	2.422419e+03	1.649363e+03	1.998802e+03	0.2554992	0.552
rpl15	1.563222e+03	1.377280e+03	7.055737e+02	1.1380498	0.000
rpl18	1.868203e+03	1.318008e+03	7.766161e+02	1.2526204	0.000
rps7	3.761768e+03	3.583645e+03	7.341254e+02	2.3970569	0.000
rpl6_rpl-6	1.415349e+03	1.331627e+03	7.755270e+02	0.8563684	0.000
rpl18a	1.774770e+03	1.414743e+03	6.347810e+02	1.4769313	0.000
inx2_18	1.538004e+03	1.606526e+03	6.294796e+02	1.2624930	0.000
rpl17	1.611498e+03	1.271276e+03	6.311013e+02	1.3552249	0.000
rpl32	1.953499e+03	1.441430e+03	7.560641e+02	1.3945419	0.000
LASC000281	4.027082e+03	3.363973e+03	2.977876e+03	0.4306201	0.022
rpl40	2.826949e+03	2.403890e+03	9.504274e+02	1.5991111	0.000
LASC014844	1.839336e+03	1.598224e+03	9.268766e+02	0.9851485	0.000
rpl36	1.376040e+03	9.762769e+02	6.154562e+02	1.1521211	0.000
ssr1	1.073956e+03	9.196191e+02	3.931951e+02	1.4366745	0.000
rpl14	1.845357e+03	1.414563e+03	7.419650e+02	1.3119011	0.000
rps12	2.011712e+03	2.127992e+03	5.888670e+02	1.8028641	0.000
rps27a	1.880657e+03	1.720418e+03	7.285687e+02	1.3820341	0.000
rpl23a	1.209733e+03	1.086629e+03	5.758418e+02	1.0616651	0.000
eef1alpha1	3.366773e+03	2.901197e+03	1.681357e+03	1.0586058	0.000
rps5	1.649106e+03	1.305242e+03	6.773337e+02	1.2783282	0.000
rpl7a	1.071175e+03	9.227775e+02	5.301460e+02	1.0272401	0.000
rps20	2.309217e+03	1.803394e+03	8.970415e+02	1.3584330	0.000
rps2_2	8.909429e+02	1.188189e+03	4.482701e+02	1.0119820	0.000
rpl10ab	1.323838e+03	1.149694e+03	6.288890e+02	1.1063716	0.000
rpl4	2.073560e+03	1.758550e+03	6.483007e+02	1.7053074	0.000
sec61g	4.000239e+03	4.150326e+03	5.400732e+02	2.9525022	0.000
LASC000747	1.478696e+03	1.230649e+03	1.774501e+03	-0.2845745	0.263
rack1	1.822726e+03	1.522975e+03	5.369605e+02	1.7994153	0.000
rpl37a_2	1.369850e+03	1.035181e+03	4.749640e+02	1.5937379	0.000
rpl39	4.059809e+03	4.520847e+03	2.597354e+03	0.6843811	0.043
rpl-7	1.877427e+03	1.424540e+03	8.337256e+02	1.1774840	0.000
rpsa	1.381907e+03	1.943243e+03	6.889648e+02	1.0042669	0.000
LASC003517	1.181809e+03	1.586928e+03	2.622792e+02	2.1048616	0.000
rpl12	1.309017e+03	1.291800e+03	5.620502e+02	1.2487906	0.000
fau_2	1.346183e+03	9.525388e+02	3.567317e+02	1.9465279	0.000
idh2	9.384861e+02	5.926080e+02	1.471361e+02	2.6388152	0.000
LASC019282	1.959453e+03	1.537727e+03	7.655801e+03	-2.0408665	0.000
rps19	1.039857e+03	9.292497e+02	4.808491e+02	1.1023840	0.000
LASC010830	2.284443e+03	9.117656e+02	2.385191e+00	9.9446344	0.000
eif4a1	1.917606e+03	1.880406e+03	1.322936e+03	0.5371138	0.022
rps26	7.501948e+02	6.285899e+02	3.923537e+02	0.9568475	0.000

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
rpl13	9.357150e+02	7.800002e+02	4.310402e+02	1.1366341	0.000
rpl13a	1.023964e+03	9.473600e+02	4.931464e+02	1.0528494	0.000
rsf1_3	1.275569e+03	9.335472e+02	6.881268e+02	0.8724813	0.013
rps17	1.125422e+03	9.191473e+02	4.750454e+02	1.2554979	0.000
cnbp	8.248930e+02	9.129715e+02	2.822334e+02	1.4826310	0.000
phgdh	1.457184e+03	1.092288e+03	3.816712e+02	1.8523939	0.000
eef1d	1.852137e+03	1.408644e+03	7.667388e+02	1.2706693	0.000
ktn1	1.456452e+03	1.192248e+03	1.233917e+02	3.5434500	0.000
rps10b	7.694814e+02	7.032242e+02	3.941415e+02	0.9571482	0.000
vdac2	1.980042e+03	1.927358e+03	1.644077e+03	0.2534775	0.470
gcat	7.208688e+02	5.853165e+02	4.425046e+00	7.3687923	0.000
rpl9_2	1.253902e+03	1.197340e+03	1.749619e+02	2.8946903	0.000
psat1	1.616762e+03	1.450799e+03	3.847670e+02	2.0809897	0.000
ssr2	7.574038e+02	6.100056e+02	2.572909e+02	1.5577226	0.000
rpl26_2	6.927117e+02	6.810180e+02	4.268996e+02	0.6813092	0.000
rpl35a	1.177493e+03	1.238126e+03	6.163189e+02	0.9366360	0.000
rps24	1.708682e+03	1.667291e+03	5.962418e+02	1.5548220	0.000
eef1b_2	1.621045e+03	1.095448e+03	6.886180e+02	1.2345958	0.000
tdpoz4_15	1.488561e+02	1.958337e+02	1.100522e+01	3.5594959	0.000
rpl9	1.136416e+03	9.814648e+02	4.680550e+02	1.2974100	0.000
nupr1	2.631370e+03	3.563649e+03	5.016982e+02	2.4556589	0.000
rpl28	1.830803e+03	1.806508e+03	6.221928e+02	1.6119772	0.000
LASC011384	8.531490e+02	4.113876e+02	8.089950e+01	3.3333796	0.000
ece2_2	5.951310e+02	4.133513e+02	1.835209e+02	1.6460553	0.000
tsr	1.008080e+03	8.026918e+02	1.076663e+03	-0.1258963	0.613
slc25a3	7.782855e+02	6.453799e+02	8.659170e+02	-0.1776943	0.510
14-3-3zeta_ywhab	1.116975e+03	9.606703e+02	1.176439e+03	-0.0895185	0.640
cotl1	4.289320e+02	3.625019e+02	2.072291e+01	4.3197914	0.000
znf346	2.359932e+02	3.744055e+02	2.057434e+02	0.0567468	0.960
rpl22	2.479989e+03	2.617213e+03	5.618928e+02	2.2103272	0.000
ncl	8.425460e+02	5.281848e+02	3.892759e+02	1.0641374	0.000
atp5f1b	5.988504e+02	5.304471e+02	1.206936e+03	-1.0566586	0.000
tth-1	1.922370e+03	9.516487e+02	3.651224e+02	2.3732115	0.000
rpl30	7.927538e+02	7.174600e+02	2.016982e+02	1.9908313	0.000
ctsb	1.585991e+03	2.341341e+02	2.032071e+03	-0.4690532	0.620
surf4	5.955797e+02	4.694092e+02	1.341349e+02	2.1328385	0.000
SpiCE-LMa6	8.343751e+02	2.525369e+03	8.240879e+00	6.7342328	0.000
tgas006m08.1_2	1.062948e+03	9.705720e+02	1.412089e+03	-0.4049326	0.040
cp1	1.883604e+03	6.578230e+02	4.342501e+02	2.1393513	0.000
LASC005273	6.601793e+03	2.956187e+03	4.400017e+00	10.6117031	0.000
LASC015682	6.479841e+02	3.178179e+02	1.288432e+03	-0.9310568	0.000
b3galt5	8.094424e+02	6.699509e+02	7.639597e+01	3.3115321	0.000
dap	5.709038e+02	1.004381e+03	1.619819e+02	1.8812872	0.000
rps13	5.828212e+02	4.806155e+02	2.336255e+02	1.3184709	0.000
slc25a4	2.113516e+03	2.594488e+03	8.127928e+03	-1.9615595	0.000
MiSpc	2.411582e+02	2.137296e+00	2.895130e-02	12.8955533	0.000
LASC017036	2.789876e+02	2.516507e+02	4.780204e+01	2.5301663	0.000
ubc	8.367950e+02	7.213887e+02	1.307146e+03	-0.6245540	0.000

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
LASC019923	1.049064e+03	1.029290e+03	3.629112e+00	8.1990662	0.000
yars	1.039168e+03	9.406920e+02	3.495173e+02	1.5770684	0.000
eif3i_v1g239264_iscu	6.149423e+02	4.490995e+02	3.780319e+02	0.6890073	0.000
ahcy-a	9.874973e+02	7.726804e+02	9.242718e+02	0.0400347	0.920
blw_atp5f1a	4.608591e+02	3.821711e+02	9.703815e+02	-1.1286901	0.000
rpl26	6.370881e+02	4.964815e+02	2.913450e+02	1.1472603	0.000
eno1	6.690352e+02	8.108733e+02	2.288854e+03	-1.7824929	0.000
sec61b	4.208688e+02	6.076192e+02	1.531365e+02	1.4786742	0.000
eif6	5.044163e+02	3.300842e+02	2.455118e+02	1.0246719	0.000
pro5	5.791174e+02	4.257937e+02	3.617100e+02	0.6522380	0.000
cyp1	8.203773e+02	5.740563e+02	4.289855e+02	0.9363760	0.000
mdh2	6.157121e+02	4.696993e+02	8.780697e+02	-0.5808150	0.073
nars	9.219392e+02	6.549068e+02	2.689200e+02	1.8359415	0.000
LASC000095	6.534555e+02	4.907166e+02	1.993306e+02	1.6703475	0.000
khdrbs3	4.265158e+02	3.169678e+02	4.452538e+02	-0.0743065	0.820
eef5_tif51a_eif5a1	4.381252e+02	4.356809e+02	2.813309e+02	0.6610034	0.000
LASC017035	2.682204e+02	2.290088e+02	4.760862e+01	2.4670993	0.000
rpl35_2	1.066834e+03	6.508141e+02	4.054692e+02	1.4524629	0.120
cth	7.076506e+02	2.220085e+02	9.354542e+02	-0.4606436	0.370
LASC007825	7.549248e+02	6.423771e+02	4.420849e+03	-2.6142170	0.000
atpalph4_4	3.922287e+02	4.390550e+02	4.345054e+02	-0.1844460	0.600
rpl35	1.846493e+02	3.215112e+02	5.043632e+01	1.8989832	0.330
aldh1a2	5.101105e+02	5.683580e+02	6.225070e+02	-0.3963490	0.350
cher_2	4.034230e+02	2.134844e+02	4.242813e+02	-0.1802448	0.760
cd63	1.074217e+03	7.199840e+02	1.139650e+03	-0.1318652	0.650
oat	1.515581e+02	1.538814e+02	9.482827e+01	0.6443899	0.100
uqcrfs1	3.052198e+02	3.011043e+02	8.121571e+02	-1.4563501	0.000
lama2_2	4.680615e+02	3.250812e+02	5.792953e+02	-0.3566527	0.290
ywrd_5	4.653702e+02	3.973144e+02	3.306522e+00	7.0733984	0.000
rtn3-b	4.581523e+02	4.239940e+02	5.951141e+02	-0.4102619	0.100
cpib	1.140317e+03	7.426665e+02	2.719461e+02	2.0880666	0.000
hspa9	5.598872e+02	4.160684e+02	3.364060e+02	0.7388327	0.000
actn	3.849513e+02	2.408838e+02	8.359044e+02	-1.2060197	0.000
jagn	4.257940e+02	4.698279e+02	1.036896e+02	2.0417271	0.000
rpl38	1.378095e+03	1.444884e+03	5.957558e+02	1.2334629	0.000
maob_3	4.500289e+02	5.577553e+02	1.813510e+02	1.2441660	0.000
rps16_2	4.290137e+02	3.703477e+02	1.676510e+02	1.3529810	0.000
shmt2	3.102833e+02	2.917081e+02	3.800030e+01	2.9621221	0.000
LASC021594	3.542621e+02	2.970354e+02	9.084492e+01	2.0252945	0.000
ddx24	1.015008e+02	1.601239e+02	3.567451e+01	1.4120463	0.070
slit1_2	6.750611e+02	7.234602e+02	8.397381e+02	-0.3942236	0.600
LASC019949	2.421980e+02	1.712090e+02	2.213339e+01	3.4167447	0.000
LASC020361	2.148113e+02	2.032490e+02	7.140834e+02	-1.7872827	0.000
uba1	3.185488e+02	2.941551e+02	2.418406e+02	0.3976423	0.170
reep5_4	5.033999e+02	4.161773e+02	6.740296e+02	-0.5454861	0.170
cox6a1	5.425794e+02	5.049442e+02	8.974081e+02	-0.6515394	0.140
tub2_2	4.581284e+02	2.984197e+02	3.742629e+02	0.2648591	0.310

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
nap1l1-b	4.043800e+02	3.677539e+02	3.235304e+02	0.3436304	0.2511100
rpl27a	4.216960e+02	3.234763e+02	1.340553e+02	1.6417120	0.0001100
slc3a1_3	3.862218e+02	3.006062e+02	1.864890e+02	1.0252100	0.0001100
LASC016339	4.237424e+02	4.181931e+02	9.477058e+02	-1.2530322	0.0211100
hnrnpl	3.758730e+02	3.057510e+02	3.277799e+02	0.2132634	0.3111100
sqd	4.535794e+02	3.645574e+02	4.496269e+02	0.0133281	0.9611100
hint1	2.799777e+02	2.117684e+02	2.158880e+02	0.3490398	0.1911100
sar1a_zk180.4_sar1b	3.308505e+02	2.615801e+02	1.561590e+02	1.1252895	0.0001100
arf1_2	6.741387e+02	5.435300e+02	4.024083e+02	0.7356527	0.0001100
aldh1a2_2	1.327578e+02	1.151088e+02	2.910060e+02	-1.1669317	0.0211100
cg12163	4.726444e+02	4.457529e+02	7.405268e+02	-0.6190548	0.0311100
LASC007830	3.860260e+02	3.840797e+02	3.335376e+02	0.2512710	0.5411100
naca	3.512225e+02	2.951389e+02	4.954030e+02	-0.5234413	0.0111100
tubb4b_3	5.504234e+02	2.991692e+02	4.434639e+02	0.2780100	0.3711100
rab1a	4.978247e+02	4.346486e+02	3.780218e+02	0.3836337	0.0311100
pdhb-1_pdhb	2.907275e+02	2.541993e+02	3.155297e+02	-0.1298369	0.6311100
tmem214	3.004964e+02	2.708558e+02	1.013033e+02	1.5203325	0.0001100
sec23a	2.103210e+02	1.789250e+02	8.761022e+01	1.2674718	0.0001100
rps21	3.072697e+02	2.687598e+02	7.178527e+01	2.1739110	0.0001100
cct2	1.444160e+02	1.107791e+02	9.908921e+01	0.5454154	0.1011100
arfgap2	2.493401e+02	1.771478e+02	6.328429e+01	1.9883121	0.0001100
tubb4b_4	7.140014e+02	5.339086e+02	5.570876e+02	0.3539821	0.3511100
LASC020761	3.317297e+02	4.296164e+02	2.104633e+02	0.7148171	0.0311100
acadm	1.893627e+02	1.313824e+02	3.424861e+02	-0.8990933	0.0001100
acp7	9.191713e+02	8.068501e+02	2.441434e+02	1.9176221	0.0001100
pdi	5.213940e+02	4.878357e+02	1.970907e+02	1.4543563	0.0001100
EIF3-s7	2.436823e+02	2.199304e+02	1.489381e+02	0.7399677	0.0001100
aco2	2.227006e+02	2.498778e+02	4.704012e+02	-1.1145488	0.0001100
psap	2.764312e+02	3.038207e+02	9.667779e+02	-1.8516396	0.0001100
fkbp12	4.025958e+02	2.600739e+02	2.189807e+02	0.8812679	0.0001100
me1	1.543681e+02	2.034435e+02	7.501659e+02	-2.3423859	0.0001100
sdha	1.761613e+02	1.491784e+02	3.878807e+02	-1.1797439	0.0001100
lamp1_2	5.769963e+02	3.905327e+02	4.580385e+02	0.3054479	0.4311100
atpsynf	5.949640e+02	5.892126e+02	8.758762e+02	-0.5650367	0.0311100
rsf1_2	3.691169e+02	3.187481e+02	3.438562e+02	0.0747377	0.8211100
EIF4e1a EIF4e	2.538437e+02	2.823792e+02	1.650256e+02	0.6095487	0.0001100
ywhae_2	2.521023e+02	2.013020e+02	3.157880e+02	-0.3222487	0.2111100
rps28	1.590929e+02	1.345589e+02	3.946782e+01	2.0421497	0.0001100
serbp1	1.521987e+02	1.568007e+02	2.534943e+02	-0.7327105	0.0001100
blmh	2.889576e+02	2.606571e+02	2.343786e+02	0.2489110	0.2111100
tktl2_tktl1	3.871556e+02	2.108140e+02	4.958611e+02	-0.4364862	0.2911100
ssr4	5.034615e+02	7.052792e+02	2.254356e+02	1.2093421	0.0001100
LASC020665	2.279682e+02	1.108221e+02	1.054976e+01	4.3512492	0.0001100
inx2_2	1.471095e+02	1.875289e+02	1.893046e+01	2.9226018	0.0001100
got2	3.048418e+02	2.809200e+02	8.958638e+02	-1.6209382	0.0001100
jafrac1_2	2.814386e+02	3.031506e+02	2.838221e+02	0.0069940	0.9711100
cg1354	2.019719e+02	1.664076e+02	6.266229e+01	1.7057746	0.0001100
sdhb	2.263276e+02	2.434264e+02	4.140045e+02	-0.9037642	0.0001100

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
tbl2	3.481450e+02	2.171017e+02	4.332743e+01	3.0064269	0.000
psmb7_2	2.103842e+02	1.552481e+02	2.107167e+02	-0.0541543	0.850
tpi	3.172360e+02	2.960337e+02	1.063249e+03	-1.7968227	0.000
his1.3	1.433967e+02	1.250600e+02	1.913244e+02	-0.4444894	0.290
cct7	2.326977e+02	1.709606e+02	1.418456e+02	0.7314305	0.020
LASC021646	3.611738e+02	3.316839e+02	6.084931e+02	-0.7637120	0.000
sars	1.372372e+02	1.380787e+02	3.088742e+01	2.2129622	0.000
LASC017315	3.278644e+02	1.463363e+02	2.840062e+02	0.1907470	0.760
lec-1	1.875407e+02	1.578594e+02	2.326854e+02	-0.3205226	0.500
tmed2	3.184545e+02	2.641769e+02	1.197374e+02	1.4141634	0.000
actr1b	1.374513e+02	1.156300e+02	1.066698e+02	0.3603730	0.090
tpd52	1.949234e+02	1.862260e+02	3.656641e+01	2.3801004	0.000
LASC005487	3.389471e+02	2.832523e+02	1.824380e+02	0.8159212	0.020
pkm	2.389680e+02	2.042997e+02	2.463083e+02	-0.0146881	0.950
cox5b	3.013265e+02	3.751991e+02	8.852817e+02	-1.5458107	0.000
wos2	4.323131e+02	2.897852e+02	2.660732e+02	0.6297033	0.100
sdcbp	1.800281e+02	1.261355e+02	4.018867e+02	-1.2309350	0.000
LASC002459	5.010175e+02	3.441518e+02	7.364701e+03	-3.7532439	0.000
mlc-c	3.393851e+02	3.513344e+02	1.770607e+02	0.9501704	0.000
alyref2_2	1.154175e+02	7.928814e+01	5.036845e+01	1.1795134	0.000
agap002499	1.836948e+02	1.394841e+02	2.111113e+02	-0.2879772	0.350
atpsyngamma	2.115253e+02	2.014289e+02	5.845776e+02	-1.5256268	0.000
sqstm1	1.985265e+02	1.794284e+02	1.681347e+02	0.2556168	0.200
lgals4	1.939080e+02	8.837600e+01	2.839801e+02	-0.6590117	0.250
LASC021266	3.040982e+02	2.907221e+02	2.095697e+02	0.5164686	0.000
ddx4_vasa	1.220536e+02	8.667067e+01	9.931206e+01	0.1955828	0.620
LASC015439	2.856669e+02	1.943132e+02	5.397788e+02	-0.9300884	0.000
moe	2.183086e+02	2.051681e+02	1.740759e+02	0.3094752	0.060
rho-1_3	1.198643e+02	5.350405e+01	1.168871e+02	-0.0023437	0.990
ybx1	1.342967e+02	1.028329e+02	1.426648e+02	-0.0956943	0.690
aldh1l2	7.070894e+01	1.083709e+02	1.042656e+01	2.8608547	0.000
ael006169	3.927155e+02	2.616437e+02	5.606070e+02	-0.5375415	0.190
apl_8	2.252196e+02	2.773720e+02	4.502581e+01	2.4977209	0.000
dld	1.848475e+02	1.632528e+02	1.964397e+02	-0.1286737	0.540
atpsyno	2.950091e+02	2.988693e+02	5.310864e+02	-0.8615688	0.000
cct3	1.577779e+02	1.206434e+02	1.268396e+02	0.3132889	0.350
tmed6	5.321565e+01	5.872739e+01	5.748631e+00	3.1843771	0.000
ppp2ca	2.851179e+02	3.079083e+02	2.376914e+02	0.2686440	0.120
cct6	1.206019e+02	8.921005e+01	1.319396e+02	-0.1464109	0.660
nas-6	8.417423e+01	9.868829e+01	1.186976e+02	-0.5175387	0.500
nnmt_vlg200856	1.025654e+02	1.032383e+02	5.889949e+01	0.7835187	0.000
cox5a	2.600973e+02	2.676295e+02	8.303228e+02	-1.7049593	0.000
eca	2.064162e+02	1.937354e+02	1.355947e+02	0.6100570	0.000
psmb6	1.139609e+02	9.473155e+01	1.458405e+02	-0.3850494	0.080
LASC021418	2.094775e+02	2.722764e+02	4.620522e+02	-1.0780313	0.000
atp5f1d	2.008757e+02	2.344928e+02	2.750206e+02	-0.4266080	0.280
ppp2r1a	2.263391e+02	1.776341e+02	1.859797e+02	0.2737188	0.200

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
aimp1	1.300540e+02	1.185817e+02	9.757190e+01	0.4002100	0.12
LASC015247	1.393645e+03	1.167790e+03	6.647100e+03	-2.2976167	0.00
cndp2	2.450879e+02	1.984188e+02	1.155705e+02	1.0970135	0.00
sorbs1_2	1.929459e+02	1.601080e+02	6.419095e+01	1.5526706	0.00
pdia3	3.958322e+02	3.999098e+02	4.568466e+02	-0.1874190	0.21
kars1_2	1.533534e+02	1.329273e+02	1.052891e+02	0.5785698	0.03
psme3	2.281913e+02	1.672929e+02	2.288354e+02	-0.0338460	0.90
psma4	2.264994e+02	1.875448e+02	1.893570e+02	0.2435079	0.19
lap-2	2.753471e+02	1.246896e+02	9.279729e+01	1.3566651	0.03
cct8	1.384995e+02	1.094949e+02	1.072491e+02	0.4026218	0.12
hnrrnp_2	2.297273e+02	1.356597e+02	1.814274e+02	0.3235770	0.34
clta	1.718950e+02	1.771212e+02	2.631412e+02	-0.5889549	0.00
ube2l3_2	9.806395e+01	1.134793e+02	2.660163e+02	-1.4087113	0.00
gdi2_gdi1	1.616289e+02	1.721910e+02	2.653809e+02	-0.6933980	0.00
elf3j	2.320728e+02	2.000249e+02	9.448656e+01	1.3419149	0.00
dlst	1.449495e+02	1.527986e+02	1.728693e+02	-0.3005995	0.27
eff_2	1.995443e+02	2.214054e+02	2.550887e+02	-0.3520770	0.07
aldh18a1	5.000012e+01	1.581657e+02	6.490064e+00	3.0105735	0.00
hrb27c_2	1.797047e+02	1.509286e+02	1.815555e+02	-0.0131670	0.96
eff_9	1.332928e+02	1.115142e+02	1.809620e+02	-0.4346759	0.02
cpa_3	2.480613e+02	2.049337e+02	1.709597e+02	0.5821666	0.03
stom	1.726001e+02	1.589244e+02	1.754165e+02	-0.0530718	0.88
crip1	3.194019e+02	4.007096e+02	1.828879e+02	0.8132903	0.00
cat2_2	1.687872e+02	1.417976e+02	2.877940e+02	-0.8038423	0.00
hrb87f	1.445677e+02	9.988153e+01	6.476745e+01	1.1062607	0.00
ndufa4	3.668167e+02	3.701267e+02	9.093123e+02	-1.3072948	0.00
tars1_tarsl2	1.879631e+02	1.426724e+02	7.535566e+01	1.3280719	0.00
LASC022132	1.136969e+02	1.046949e+02	1.289435e+02	-0.2160690	0.46
sar1b	1.681403e+02	1.799942e+02	1.521205e+02	0.1272604	0.62
sp-1339#prd34368.1_4	1.162247e+03	2.211685e+03	2.069893e+00	9.1681398	0.00
gygl_2	5.192364e+01	4.754784e+01	1.094432e+02	-1.1531573	0.01
atpsynb	1.959421e+02	2.363798e+02	4.922627e+02	-1.3751301	0.00
cpb_capzb	1.231790e+02	9.619168e+01	1.322476e+02	-0.1562940	0.58
uqcrb	3.215749e+02	3.411952e+02	5.746645e+02	-0.7441374	0.12
ndufb9	2.404650e+02	2.298537e+02	3.557149e+02	-0.5762075	0.04
cct5	1.365362e+02	1.136164e+02	1.370433e+02	0.0064576	0.97
LASC003928	2.487479e+02	2.430424e+02	4.283201e+02	-0.7968034	0.00
ndufa5	1.783493e+02	1.486670e+02	3.974439e+02	-1.1723397	0.00
lamp1	2.429854e+02	2.087932e+02	3.256137e+02	-0.4287930	0.02
rho1_7	1.644823e+02	1.865572e+02	3.088850e+02	-0.8849570	0.00
enpp6_3	1.836229e+02	1.304564e+02	6.304980e+00	4.8494398	0.00
llgl1	1.055296e+02	9.601234e+01	1.155838e+02	-0.1996619	0.56
cyc1	8.572857e+01	7.536472e+01	2.008821e+02	-1.2646916	0.00
uqcrc1	1.432059e+02	1.292796e+02	2.182316e+02	-0.5887485	0.00
rars	2.455009e+02	2.061025e+02	9.321278e+01	1.3736969	0.00
elf3g-b	1.217166e+02	1.199296e+02	8.733307e+01	0.4837455	0.00
set	2.208350e+02	1.442681e+02	1.579737e+02	0.4603501	0.02
igf2bp1	1.225047e+02	1.138723e+02	1.369050e+02	-0.2084849	0.35

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
ehd1_2	7.072178e+01	6.821875e+01	2.824729e+01	1.2750759	0.000
btbf3l4	1.868100e+02	1.968605e+02	2.029766e+02	-0.0985966	0.691
metap1	1.581091e+02	1.130111e+02	9.042586e+01	0.8206991	0.000
aldh7a1	1.276825e+02	1.296647e+02	5.317416e+01	1.2998778	0.000
LASC018632	9.408195e+01	9.675366e+01	5.534607e+01	0.7780834	0.000
nop58_2	1.103070e+02	7.218635e+01	3.856673e+01	1.5150800	0.000
atox1	5.925455e+02	6.860255e+02	3.950929e+02	0.6537677	0.260
pitpna	2.435445e+02	1.949523e+02	1.898361e+02	0.4053690	0.200
sqh_2	3.005006e+02	2.524345e+02	1.198950e+02	1.3298531	0.000
nsf1lc	8.118223e+01	6.040429e+01	6.866899e+01	0.2485067	0.330
hspa4l_2	1.682963e+02	1.683748e+02	1.392118e+02	0.2973092	0.270
acly	6.979079e+01	6.191099e+01	6.127858e+01	0.1924913	0.510
fubp3_fubp1	8.908099e+01	8.616762e+01	1.040551e+02	-0.2136314	0.420
LASC005920	8.097213e+01	4.386329e+01	6.867890e+02	-3.0476490	0.000
ter94_vcp	1.031189e+02	1.162519e+02	1.908360e+02	-0.8750622	0.000
LASC016997	1.656392e+02	1.953874e+02	1.765058e+02	-0.0544985	0.910
lark	1.068577e+02	1.105198e+02	1.001608e+02	0.1160095	0.620
LASC010482	2.164432e+02	2.486344e+02	3.319116e+02	-0.5361343	0.260
spg21	1.374595e+02	1.040717e+02	7.533733e+01	0.7918185	0.090
LASC006288	1.667226e+02	1.051425e+02	6.785801e+01	1.2408227	0.000
ap2m1	7.496549e+01	6.421676e+01	1.649852e+02	-1.1356680	0.000
slc7a6_slc7a8	7.240075e+01	4.626077e+01	1.745116e+01	2.0341337	0.000
etfa	1.305044e+02	1.215098e+02	1.806594e+02	-0.5016200	0.000
arcn1	1.025812e+02	9.272977e+01	9.548805e+01	0.1001339	0.640
fh	1.359531e+02	1.057885e+02	1.100483e+02	0.2554220	0.350
uqcrc2	1.135625e+02	1.037211e+02	2.166934e+02	-0.9367472	0.000
ago3_ago2	3.823724e+01	3.531497e+01	5.176938e+01	-0.5202044	0.260
LASC012598	1.252652e+02	9.807734e+01	1.492830e+02	-0.3000352	0.330
ube2n_ubc-13	7.331258e+01	6.957096e+01	1.759293e+02	-1.2436570	0.000
cnp_2	2.052771e+02	1.358759e+02	7.988537e+01	1.2007621	0.060
snx12	1.598825e+02	1.164819e+02	9.148376e+01	0.8330801	0.010
ndufs4	2.033740e+02	1.775450e+02	2.923029e+02	-0.5393180	0.100
ran-4	2.389689e+02	1.636302e+02	1.485744e+02	0.6214729	0.010
aimp2	1.705089e+02	1.272163e+02	1.105266e+02	0.5928716	0.040
psma6	1.634808e+02	1.582045e+02	1.775307e+02	-0.1392773	0.610
nop60b	1.110363e+02	8.081680e+01	4.781947e+01	1.2212028	0.000
anxa11	1.535543e+02	1.773763e+02	1.539891e+02	-0.0248810	0.930
acat1-b	7.906918e+01	6.179136e+01	4.655107e+01	0.7284267	0.000
fib	6.975953e+01	6.929199e+01	3.841620e+01	0.8771362	0.010
hadh	1.736491e+02	1.346870e+02	3.677258e+02	-1.1202884	0.000
prmt8	2.321545e+02	2.151944e+02	1.669823e+02	0.4902904	0.080
zip	1.144765e+02	1.058922e+02	4.789128e+01	1.2537515	0.000
atpsyncf6	1.563554e+02	2.139809e+02	3.642552e+02	-1.2070303	0.000
tuba1c	2.157798e+02	1.717586e+02	3.268434e+02	-0.5834049	0.550
sfpq	1.103104e+02	8.264818e+01	8.010801e+01	0.4470524	0.120
erf1	2.146227e+02	1.664243e+02	8.804129e+01	1.3019483	0.000
LASC017716	9.628773e+01	9.159340e+01	1.525116e+02	-0.5897166	0.030

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
phb	1.267196e+02	8.993065e+01	1.353522e+02	-0.1266062	0.640
sod1	1.437897e+02	1.013864e+02	1.792940e+02	-0.3387962	0.133
gs2_2	1.019172e+02	1.064043e+02	8.684868e+02	-3.1363293	0.000
plin2	3.006327e+02	3.227800e+02	1.045391e+04	-5.1033153	0.000
LASC009674	8.659639e+01	7.621690e+01	8.897471e+01	-0.0308987	0.909
uqcrq	2.776041e+02	2.798075e+02	4.952443e+02	-0.8120344	0.023
psma1	1.058522e+02	9.609819e+01	1.660334e+02	-0.6668571	0.000
l(2)efl_hspb1	8.041761e+01	5.167455e+01	4.511907e+03	-5.7640669	0.000
cltc	7.949654e+01	7.098822e+01	1.166140e+02	-0.5501893	0.014
gstd1_2	1.049389e+02	1.181174e+02	3.160179e+02	-1.6345598	0.000
ugp2_2	2.747118e+01	2.425551e+01	2.355626e+01	0.1653192	0.613
lrrc47	9.836981e+01	8.029225e+01	6.621141e+01	0.5415465	0.040
psmb5	1.101209e+02	9.786930e+01	1.373823e+02	-0.3623656	0.102
gbe1	9.667030e+01	7.009350e+01	1.323525e+02	-0.4969695	0.034
LASC017623	1.972618e+02	1.247556e+02	2.797416e+02	-0.4580564	0.109
rpn11	1.053709e+02	9.030308e+01	9.597613e+01	0.1444246	0.544
cubn_3	1.946810e+02	1.212153e+02	1.723088e+02	0.0514158	0.920
snx2	9.461097e+01	7.819430e+01	1.435029e+02	-0.6356243	0.008
efhd2	3.708522e+01	2.874260e+01	5.419502e+01	-0.5723419	0.122
qars	1.275880e+02	1.117477e+02	4.225582e+01	1.5716589	0.000
ddx1	6.946001e+01	5.404927e+01	3.522000e+01	0.9487401	0.000
LASC013621	1.004317e+02	1.055269e+02	9.109180e+02	-3.1384008	0.000
psmc6	6.103865e+01	6.894654e+01	9.905822e+01	-0.6929928	0.000
elf3-s8	8.120345e+01	8.139128e+01	7.602190e+01	0.1259233	0.613
cfad	5.950996e+01	1.255657e+02	2.728632e+02	-2.1202125	0.000
atp5mg	9.710800e+01	1.022951e+02	1.804433e+02	-0.9017854	0.000
psmc5	9.538420e+01	8.795797e+01	1.259187e+02	-0.4083437	0.000
septin7	7.434384e+01	6.368957e+01	3.283546e+01	1.1966335	0.000
rtca	4.863447e+01	3.618824e+01	1.310009e+01	1.8736027	0.000
apip	1.398239e+02	1.291117e+02	1.611576e+02	-0.2455940	0.243
bai_tmed10	1.222796e+02	1.908243e+02	9.199118e+01	0.4664623	0.250
ewsr1	1.243376e+02	1.035695e+02	1.126602e+02	0.1120351	0.690
aass	4.466445e+01	4.650874e+01	4.038946e+01	0.1636376	0.473
hibadh	1.096754e+02	8.685859e+01	1.576018e+02	-0.5866698	0.014
ptcd2	1.159181e+02	8.429806e+01	9.104061e+01	0.3722912	0.077
zan_srfbp1	9.936722e+01	8.536674e+01	2.897949e+02	-1.5220138	0.000
ndufs1	6.757061e+01	6.449645e+01	1.627932e+02	-1.2932987	0.000
ube2v2	9.127403e+01	8.972847e+01	1.162089e+02	-0.3560153	0.014
LASC007277	3.045838e+02	3.125959e+02	8.515566e+01	1.9473370	0.000
hnrnpk	8.869576e+01	6.120541e+01	1.026791e+02	-0.1919807	0.360
ywhae	9.918298e+01	1.141402e+02	1.822951e+02	-0.8370013	0.000
psmd8	7.070527e+01	8.302711e+01	1.255423e+02	-0.8205320	0.000
cd109	2.414817e+02	1.809756e+02	1.286948e+02	0.8261181	0.057
vha26	2.775950e+02	2.247949e+02	1.791490e+02	0.7306145	0.053
sirt5	6.466258e+01	7.483536e+01	1.702499e+02	-1.4247250	0.000
dnajc22	4.868533e+01	6.510726e+01	5.770825e+00	3.0643954	0.000
atpsynd	8.571216e+01	1.090783e+02	2.026735e+02	-1.1992499	0.190
ppib_cyn-5	1.589252e+02	1.474794e+02	1.570456e+02	0.0416285	0.869

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
ech1_2	1.603404e+02	2.232330e+02	1.375203e+02	0.2178555	0.330
srp68	9.194302e+01	9.183879e+01	6.437226e+01	0.4866136	0.050
LASC005698	1.601248e+02	1.177793e+02	6.794517e+01	1.1903959	0.000
copa	6.211310e+01	7.971999e+01	6.321604e+01	0.0187955	0.950
abat	3.524282e+01	5.875561e+01	1.377211e+02	-1.8763176	0.000
eif3a_2	6.547028e+01	6.408863e+01	6.359408e+01	0.0677897	0.790
ndufs3	2.066234e+02	2.453778e+02	3.564363e+02	-0.7542059	0.070
ciz1	1.046182e+02	7.090177e+01	9.368088e+01	0.1422588	0.480
rpn6	7.335823e+01	5.508828e+01	6.010449e+01	0.3164160	0.120
sqh	1.010063e+02	8.154502e+01	7.296773e+01	0.4727028	0.120
alpha-spec	8.816122e+01	7.324181e+01	1.546249e+02	-0.8077707	0.000
srsf12	5.288231e+01	4.202907e+01	4.696951e+01	0.1831005	0.630
cbx3	1.018457e+02	8.603604e+01	8.440181e+01	0.2646155	0.090
LASC011933	1.045008e+02	1.358482e+02	4.057208e+01	1.3601287	0.000
vha55	1.469326e+02	1.404623e+02	1.301786e+02	0.2235955	0.480
pebp1	9.938006e+01	8.724109e+01	1.633779e+02	-0.7176605	0.000
LASC011232	8.435236e+01	4.047906e+01	1.778840e+01	2.1523825	0.000
mdh1	1.091688e+02	1.097303e+02	3.169375e+02	-1.5945752	0.000
gspt1	1.023494e+02	9.706456e+01	8.203752e+01	0.3254273	0.070
ptbp1	1.507829e+02	1.088763e+02	8.676250e+01	0.7703962	0.000
his2av	7.246626e+01	4.575125e+01	5.093812e+01	0.4777886	0.060
tdh	3.460148e+01	1.000516e+02	4.029985e-01	6.5196065	0.000
ipgm-1	8.604497e+01	8.811598e+01	5.120367e+02	-2.5574256	0.000
psma3	9.772751e+01	1.021301e+02	1.620026e+02	-0.7125679	0.010
rab2a	6.618299e+01	6.586972e+01	5.683841e+01	0.2413004	0.190
cap2	1.340680e+02	1.053884e+02	1.903294e+02	-0.5202184	0.060
ldlr	9.402989e+01	8.137104e+01	1.031064e+01	3.3079798	0.000
LASC016532	9.889305e+01	5.381161e+01	1.533146e+02	-0.7226770	0.190
psmd13	8.968556e+01	8.965451e+01	8.542179e+01	0.0804491	0.690
gstm2	1.622837e+01	3.482178e+01	2.907486e+01	-0.8178057	0.170
psmd2	8.003635e+01	7.732979e+01	1.269072e+02	-0.6874323	0.000
ppif	5.730909e+01	9.470129e+01	2.100140e+02	-1.7684283	0.000
hsp83_hsp90aa1	1.858612e+02	1.337709e+02	3.326485e+02	-0.7552707	0.280
ndufa13	1.748365e+02	1.633909e+02	1.952716e+02	-0.1124802	0.770
slc9a3r1	6.298153e+01	7.529271e+01	7.314960e+01	-0.1985291	0.450
ael011789	7.424619e+01	8.172370e+01	2.180306e+02	-1.5916229	0.000
pkm_3	1.117972e+02	7.965309e+01	2.945544e+02	-1.4250655	0.000
copg2	4.443914e+01	5.397081e+01	4.854408e+01	-0.1131791	0.640
echs1	8.261028e+01	6.763692e+01	1.414862e+02	-0.7951732	0.000
LASC004161	9.488066e+01	9.347364e+01	9.383794e+01	0.0087569	0.970
eif4b	4.380862e+01	4.208350e+01	5.031447e+01	-0.2144964	0.430
calr_crt	2.696433e+02	3.438498e+02	4.822496e+02	-0.7356985	0.030
hspl1_3	5.163586e+01	6.699193e+01	2.480825e+02	-2.3697601	0.000
abce1	6.896255e+01	6.226991e+01	4.204399e+01	0.7696546	0.000
LASC011347	2.460975e+02	3.899547e+02	1.929603e+01	3.7021247	0.000
phb2	5.685783e+01	5.562602e+01	8.023981e+01	-0.5003488	0.000
glyp	1.390031e+02	1.056623e+02	3.619597e+02	-1.4516695	0.010

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
gfpt2	4.266273e+01	3.157795e+01	2.972854e+01	0.5077216	0.010
grn_6	9.477521e+01	1.100404e+02	1.060959e+03	-3.5024912	0.000
snd1	5.640595e+01	6.901505e+01	4.190407e+01	0.4649005	0.218
folr3	9.106948e+01	6.722605e+01	8.734503e+00	3.3372682	0.000
eprs	5.001651e+01	4.631301e+01	2.168378e+01	1.2202703	0.000
prdx6	9.321771e+01	1.057289e+02	1.291427e+02	-0.4547583	0.118
EIF3F	1.179145e+02	1.340186e+02	9.896587e+01	0.2700762	0.287
snrpb	4.801818e+01	4.086831e+01	3.737043e+01	0.3363060	0.300
psma5	7.555738e+01	6.711353e+01	9.134107e+01	-0.2607853	0.217
idh3a	7.349684e+01	6.956106e+01	1.945783e+02	-1.4446020	0.000
rad23b	6.352644e+01	5.877307e+01	7.541311e+01	-0.2607684	0.104
SpICE-LMa4	2.241378e+02	1.534526e+02	2.354090e+01	3.2222415	0.000
psmc2	6.632684e+01	8.106111e+01	1.197348e+02	-0.8205873	0.012
LASC018834	4.544933e+01	4.478431e+01	1.131402e+02	-1.3831713	0.000
tbb-4_tubb4b	1.017322e+02	7.338793e+01	1.929090e+02	-0.8304482	0.063
aldh4a1	7.149700e+01	5.474909e+01	4.621419e+01	0.6275076	0.029
stip1	1.190700e+02	9.239494e+01	1.989778e+02	-0.7766681	0.000
hsp60a	6.821750e+01	4.541950e+01	7.880943e+01	-0.1888873	0.649
lap3	5.102095e+01	3.696380e+01	5.474917e+01	-0.0960176	0.740
LASC015638	1.149233e+02	8.935963e+01	1.557535e+01	2.8389080	0.000
psmd12	7.940693e+01	6.368197e+01	9.271252e+01	-0.2215969	0.139
abhd12_2	9.041871e+01	7.714659e+01	7.290604e+01	0.3577885	0.443
cox6b1	6.638179e+01	7.053508e+01	1.736675e+02	-1.3995432	0.000
fas2_3	9.437145e+01	9.448309e+01	6.320351e+01	0.6321180	0.080
LASC016132	5.013030e+01	3.732594e+01	1.418971e+02	-1.4572648	0.000
vdac2_2	5.840701e+01	6.054810e+01	1.373053e+02	-1.2178483	0.000
tth-1_2	1.423076e+02	1.598183e+02	3.169071e+02	-1.1134391	0.000
sod-2	7.800379e+01	7.909772e+01	1.598134e+02	-1.0566060	0.000
nurf-38_ipp1	8.837796e+01	6.556747e+01	1.346418e+02	-0.6067429	0.080
psmb3	1.104343e+02	9.200256e+01	1.181718e+02	-0.0953800	0.700
hsd17b10_2	5.781887e+01	4.553240e+01	9.681672e+01	-0.7681527	0.012
ndufv2	4.223967e+01	4.070728e+01	1.444023e+02	-1.8014987	0.000
lman1	5.379461e+01	5.669818e+01	4.322174e+01	0.3449185	0.279
nop10	1.205484e+02	9.373239e+01	8.378327e+01	0.4901227	0.063
hce_hcf	6.350025e+02	1.158723e+03	2.835532e+03	-2.2639074	0.067
aldh2_2	7.557139e+01	5.995149e+01	6.985727e+01	0.1193608	0.627
csga	8.250136e+01	6.113857e+01	7.964160e+01	0.0058944	0.979
arl8ba_arl8b	1.281475e+02	1.213957e+02	1.146420e+02	0.2019158	0.550
LASC006772	5.578572e+01	3.217300e+01	9.065229e+01	-0.7790271	0.032
ppib_cypb	1.992964e+02	2.939275e+02	9.744731e+02	-2.2421212	0.000
gbeta1	8.049565e+01	7.323930e+01	1.234013e+02	-0.6120755	0.000
rab5b	8.674762e+01	9.677509e+01	1.196061e+02	-0.4606149	0.063
hadha	6.454163e+01	5.291526e+01	7.185507e+01	-0.1914872	0.360
nhp2	6.595159e+01	6.077978e+01	4.132278e+01	0.7187577	0.120
ogdh	6.322604e+01	6.866265e+01	2.094878e+02	-1.7411648	0.000
rab7_rab7a_2	1.240003e+02	1.165358e+02	1.445773e+02	-0.1936933	0.583
beta-spec_2	7.693418e+01	6.813377e+01	1.292694e+02	-0.7403904	0.000
cpt1a_3	4.943959e+01	6.059075e+01	6.225659e+01	-0.3794160	0.417

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
LASC013620	2.632188e+01	3.863396e+01	3.486872e+02	NA	
ran_2	9.731486e+01	7.919893e+01	5.916621e+01	0.7289680	0.010
tmco1	1.419510e+02	1.576253e+02	8.000595e+01	0.8844932	0.010
LASC011026	6.105247e+01	5.856153e+01	1.307012e+02	-1.0993335	0.000
copb1	5.862596e+01	6.854612e+01	5.737665e+01	0.0633485	0.793
kyat3	6.447483e+01	7.776390e+01	7.150516e+01	-0.1718584	0.600
tfg	4.900850e+01	5.303890e+01	6.015023e+01	-0.2708131	0.133
myo6_jar	3.763780e+01	4.195309e+01	1.719817e+01	1.0866349	0.000
atp6v0a1	1.186822e+02	8.411370e+01	4.958014e+01	1.2535489	0.000
pa2g4	6.146169e+01	6.336592e+01	5.336624e+01	0.2311904	0.480
oplah	2.673118e+01	2.829768e+01	1.952038e+01	0.3999129	0.173
dpp3_3	8.085968e+01	7.863334e+01	6.180711e+01	0.3601973	0.163
nolc1	5.388322e+01	4.526875e+01	3.706240e+01	0.5291803	0.043
anp32a_2	8.963625e+01	8.176733e+01	1.005939e+02	-0.1662808	0.423
sacm1l_sacm1lb	4.539576e+01	4.011925e+01	4.891553e+01	-0.0907127	0.723
sme	1.568196e+02	1.216659e+02	9.522354e+01	0.7693004	0.023
cct4	7.295964e+01	7.449541e+01	9.769063e+01	-0.4083904	0.063
gpx4	1.310840e+02	1.772876e+02	2.605448e+02	-0.9761707	0.000
rala	7.724427e+01	6.238252e+01	6.747722e+01	0.1484203	0.623
sec22b	3.798939e+01	4.252517e+01	3.525902e+01	0.0844491	0.743
vapa	6.707279e+01	6.088795e+01	1.154766e+02	-0.8105167	0.000
tufm	5.360119e+01	4.393991e+01	7.791105e+01	-0.5584782	0.007
vat1l_2	4.226681e+01	2.496211e+01	7.895105e+01	-0.8532909	0.143
LASC013486	1.047906e+02	7.218006e+01	1.690261e+03	-4.0019117	0.000
rpt2_2	6.668939e+01	8.243106e+01	1.127379e+02	-0.7195158	0.003
gcdh	7.223111e+01	7.574349e+01	5.920976e+01	0.2796303	0.113
chmp4b	7.851065e+01	6.902343e+01	1.179766e+02	-0.5874807	0.003
ndufa8	5.478504e+01	4.148466e+01	7.878604e+01	-0.5000071	0.140
LASC018874	3.313856e+01	2.655538e+01	4.520884e+01	-0.4773623	0.193
nop58	6.165753e+01	4.296730e+01	2.587751e+01	1.3136799	0.000
serca_2	4.525158e+01	5.529032e+01	8.554465e+01	-0.8781642	0.000
psmd4	5.613815e+01	5.489896e+01	7.949825e+01	-0.4817426	0.014
smo-1	1.049006e+02	1.219793e+02	8.936956e+01	0.2669520	0.423
edf1	5.921584e+01	6.859427e+01	8.729440e+01	-0.5757791	0.014
ndufs7	4.782409e+01	4.098091e+01	2.147432e+02	-2.2244682	0.000
sfpq_2	5.030655e+01	4.198626e+01	7.539688e+01	-0.5512939	0.013
serbp1_2	7.020163e+01	5.949641e+01	1.153916e+02	-0.7435550	0.000
sgtb	8.194204e+01	1.204192e+02	1.071806e+02	-0.3526430	0.063
ndufa7	7.100318e+01	6.538140e+01	1.111167e+02	-0.6301629	0.043
pgk	7.121235e+01	8.775188e+01	2.244400e+02	-1.6263625	0.000
snx6	5.475816e+01	3.934609e+01	7.007702e+01	-0.3840765	0.133
pdcd4	6.773762e+01	7.210923e+01	9.275006e+01	-0.4947996	0.093
nd-42	5.521671e+01	6.563580e+01	2.009125e+02	-1.8867728	0.000
chmp2a	6.677058e+01	6.072374e+01	7.916305e+01	-0.2495759	0.323
nucb2	7.999736e+01	9.138232e+01	9.553810e+01	-0.2342866	0.193
f42h10.3_lasp1	3.868315e+01	3.974831e+01	5.771313e+01	-0.5622096	0.003
dars	5.770235e+01	5.818698e+01	6.309308e+01	-0.1344283	0.560

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
gdh	6.037495e+01	5.456147e+01	2.841692e+02	-2.2309597	0.000
uso1	3.672133e+01	3.861622e+01	2.823186e+01	0.3547936	0.147
ssb	5.348919e+01	4.391502e+01	3.787730e+01	0.5225661	0.067
txnl1	4.850834e+01	4.435902e+01	4.710524e+01	0.0082941	0.977
cct1	6.226402e+01	6.169679e+01	8.264904e+01	-0.3924765	0.093
fk506-bp2_fkbp1b	8.761854e+01	1.088634e+02	1.795025e+02	-1.0265254	0.000
psma2	8.205709e+01	5.981333e+01	8.029368e+01	0.0240250	0.934
psma8_2	1.002903e+02	1.017556e+02	1.886881e+02	-0.9223355	0.000
dsp1	1.022513e+02	9.845173e+01	1.923179e+02	-0.9116837	0.000
rtcb	3.454411e+01	2.478794e+01	2.157072e+01	0.6788352	0.011
impa1	6.148615e+01	4.710853e+01	3.580883e+01	0.7800362	0.000
cd109_4	3.988153e+01	4.025576e+01	6.140526e+00	2.7122575	0.000
zmpste24	7.401917e+01	5.099483e+01	8.721684e+01	-0.2648913	0.223
plekhf2	3.289566e+01	3.030811e+01	1.020449e+02	-1.6530140	0.000
hadhb	6.698973e+01	7.036286e+01	2.143165e+02	-1.7115470	0.000
etfb	6.388657e+01	6.586898e+01	1.097643e+02	-0.7461507	0.004
blvrb	1.034006e+02	1.109674e+02	9.481040e+01	0.1232846	0.667
stx7	6.530043e+01	5.885150e+01	1.146186e+02	-0.8365808	0.003
prdx4_prdx1	7.075263e+01	5.229133e+01	1.082274e+02	-0.5472724	0.033
hsa5_2	1.599082e+02	1.935423e+02	1.699814e+02	0.0469691	0.933
tecr_art-1	9.703168e+01	1.474901e+02	4.221198e+02	-2.2215403	0.000
eif4ebp2_eif4ebp1	8.645779e+01	1.062728e+02	9.829280e+01	-0.2260467	0.521
scsalpha1_os07g0577700	5.307243e+01	5.089632e+01	1.202807e+02	-1.1673634	0.000
ehd1_3	4.667951e+01	4.802277e+01	7.008396e+01	-0.6136679	0.009
psmb2	1.271639e+02	1.192275e+02	1.455787e+02	-0.1447136	0.744
atpalpha	7.004355e+01	7.335699e+01	1.987153e+02	-1.3798072	0.000
LASC016362	2.777033e+03	4.594832e+03	1.218977e+01	7.9115277	0.000
hnrnpf_hnrnp2	3.773549e+01	3.336316e+01	6.322378e+01	-0.7116390	0.000
slc8a2	2.706200e+01	2.246967e+01	1.808911e+01	0.5657131	0.283
ndufs6	3.224226e+01	3.525953e+01	9.022451e+01	-1.4552236	0.000
rbm4_rbm14	4.010640e+01	3.410890e+01	3.047778e+01	0.3845559	0.019
LASC013967	2.220062e+01	8.556751e+01	9.527140e+01	-2.1183091	0.003
cs	5.693953e+01	5.001248e+01	2.912254e+01	0.9571121	0.000
hibch	4.017061e+01	3.939865e+01	9.140505e+01	-1.2270669	0.000
LASC010684	3.820319e+01	3.710629e+01	3.447044e+01	0.3615320	0.657
LASC009347	2.052091e+02	1.694234e+02	1.021322e+03	-2.2762944	0.003
cg3556_15	3.031859e+01	2.684396e+01	3.284031e+00	3.1061478	0.000
uchl3_3	5.773025e+01	5.925640e+01	8.068011e+01	-0.4638674	0.009
park7	6.569504e+01	5.960657e+01	8.543365e+01	-0.3811777	0.037
elavl2	7.471225e+01	4.949851e+01	5.850049e+01	0.3501247	0.123
cnm3	5.790208e+01	6.791028e+01	2.337215e+01	1.2895325	0.000
LASC001129	3.053236e+01	3.462095e+01	2.493638e+01	0.1952527	0.823
snu13	9.212896e+01	1.198578e+02	5.157296e+01	0.9236907	0.104
gstm3	1.355522e+02	1.497704e+02	1.302132e+02	0.0520705	0.810
nnt	1.561913e+01	1.956022e+01	1.397249e+01	0.1731668	0.699
hsa5	3.361011e+01	4.379115e+01	3.972668e+01	-0.1298436	0.764
cope	1.076126e+02	1.327566e+02	8.014071e+01	0.4073080	0.049
hnrnp_3	6.414731e+01	5.748939e+01	1.031613e+02	-0.6885851	0.003

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
jafrac1	2.372670e+02	3.912732e+02	2.492282e+02	-0.0240480	0.964
lrp2_6	5.134925e+01	4.718248e+01	7.382665e+00	2.7996102	0.000
snr-4	8.249026e+01	6.501626e+01	3.341786e+01	1.3662123	0.000
esd_2	7.596108e+01	4.430193e+01	6.031950e+01	0.3080931	0.437
prep	4.563731e+01	4.205272e+01	4.207062e+01	0.1035640	0.673
acat1_2	5.451410e+01	5.272431e+01	9.181962e+01	-0.7690444	0.000
psmd1_rpn2	3.663220e+01	4.900389e+01	7.910882e+01	-1.0794633	0.000
ndufb4	8.576259e+01	8.584753e+01	2.047909e+02	-1.2255222	0.000
dcxr_2	8.854624e+01	6.483138e+01	1.350001e+02	-0.6785827	0.029
ppp1cc_ppp1cc-b	5.751954e+01	4.583999e+01	6.127377e+01	-0.1148335	0.641
kpnb1	4.171497e+01	3.431208e+01	2.665754e+01	0.6561824	0.000
glyp_2	1.165857e+02	1.347117e+02	8.689639e+02	-2.8972645	0.000
ppp1cb	5.710818e+01	5.740990e+01	1.164550e+02	-1.0337161	0.000
cpt1a_2	6.184584e+01	5.164932e+01	6.054938e+01	0.0721796	0.911
rpn1	8.139156e+01	1.096154e+02	1.090120e+02	-0.3593388	0.229
LASC002299	9.953017e+01	1.240887e+02	5.066719e+01	1.0074880	0.000
hnrnpl	4.874565e+01	4.584043e+01	8.536775e+01	-0.7842692	0.000
mapre3	6.967134e+01	5.908071e+01	7.239880e+01	-0.0512524	0.821
cdc42_2	5.472421e+01	5.380676e+01	1.049635e+02	-0.9712824	0.000
ap2b1	2.061967e+01	2.313570e+01	3.998732e+01	-0.9385933	0.000
lrpap1	1.276384e+02	1.077711e+02	4.515227e+01	1.4770695	0.000
nop56	5.550581e+01	5.004945e+01	3.739829e+01	0.5726495	0.000
cand1	2.597134e+01	2.367250e+01	4.324260e+01	-0.7323127	0.000
tm_fabg	6.155973e+01	7.136621e+01	3.485034e+01	0.8511954	0.000
mtap	6.765650e+01	4.526769e+01	5.283611e+01	0.2549035	0.547
c05d11.1	3.578578e+01	3.804364e+01	4.041265e+01	-0.1546494	0.380
idh1	1.125101e+02	9.172458e+01	4.145616e+02	-1.9363604	0.000
acads	5.381073e+01	7.639785e+01	1.343195e+02	-1.2845582	0.000
suc1g2	2.259684e+01	2.334005e+01	2.788424e+01	-0.2822650	0.371
itga9	3.765494e+01	4.277890e+01	1.351364e+01	1.4960884	0.000
LASC005335	1.029047e+02	1.064080e+02	3.671358e+02	-1.8213526	0.000
pxn	3.066973e+01	4.309884e+01	2.507811e+02	-3.0727917	0.000
glrx3	9.348512e+01	1.030069e+02	8.238858e+01	0.1700201	0.544
scfd1	4.923728e+01	4.514592e+01	3.578455e+01	0.4557934	0.000
ap1m1	5.678303e+01	4.352799e+01	6.768067e+01	-0.2491891	0.173
rap1b	6.228313e+01	6.989421e+01	1.292516e+02	-1.0717828	0.000
pebp1_2	1.028986e+02	1.274966e+02	2.550347e+02	-1.2957627	0.000
capsl_2	4.996612e+01	9.607330e+01	1.317379e+02	-1.2647705	0.000
ablim1_ablim2	2.491849e+01	2.589537e+01	5.448517e+01	-1.1207488	0.000
aprt	5.894453e+01	5.765096e+01	4.679755e+01	0.3190444	0.134
acadvl	2.723039e+01	2.494503e+01	7.053788e+01	-1.3822788	0.000
LASC001932	9.682311e+01	9.066382e+01	5.355547e+01	0.8706637	0.000
igf2r	5.621397e+01	5.622004e+01	4.163653e+01	0.3546366	0.417
spac4h3.07c	5.211366e+01	5.775255e+01	7.497639e+01	-0.5178013	0.067
anp32a	1.675557e+02	1.052921e+02	1.410595e+02	0.2548445	0.547
hcd	3.160833e+02	5.396753e+02	1.411335e+03	-2.2663344	0.063
ehd1	3.683740e+01	2.500359e+01	2.661603e+01	0.4484602	0.133

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
txn2	9.930734e+01	1.025056e+02	5.880353e+01	0.8631959	0.063
hcg	3.609564e+02	6.434649e+02	1.592476e+03	-2.2514688	0.063
hca	3.573540e+02	6.462796e+02	1.603199e+03	-2.2825471	0.063
twf2	3.484541e+01	3.489316e+01	2.942195e+01	0.2571527	0.233
vha14	1.324475e+02	1.172744e+02	1.332898e+02	0.0504770	0.883
arf4	5.934235e+01	9.393828e+01	6.777605e+01	-0.1946946	0.566
alkbh3	4.326789e+01	3.559463e+01	2.981205e+01	0.4562524	0.164
derl1	6.983696e+01	7.928285e+01	6.457210e+01	0.0710441	0.843
pgd	7.396772e+01	6.876408e+01	2.975684e+02	-2.0291389	0.000
rab18	5.567062e+01	6.282277e+01	7.543694e+01	-0.4386226	0.053
hspe1	5.754353e+01	5.087719e+01	7.426752e+01	-0.3594853	0.311
rab7_rab7a	8.990608e+01	7.124863e+01	8.828686e+01	0.0837165	0.843
zk1073.1_2	1.001452e+02	1.130319e+02	4.738324e+02	-2.3037460	0.000
slc25a11	4.530867e+01	5.461025e+01	1.145686e+02	-1.3600457	0.000
plxdc1	5.523804e+01	5.694365e+01	1.408684e+01	1.9630544	0.000
srsf4_3	3.234696e+01	3.804017e+01	6.195030e+01	-0.9251086	0.000
ivd	2.857315e+01	2.754389e+01	8.550407e+01	-1.6064559	0.000
acd6-6_psmf1	4.879572e+01	4.763333e+01	9.765508e+01	-1.0423950	0.000
lgmn_2	1.952701e+02	1.960704e+02	2.094241e+02	-0.0857214	0.863
pdc6	7.105003e+01	5.070831e+01	7.711689e+01	-0.1159959	0.733
hsp90b1	1.226847e+02	1.159298e+02	1.298835e+02	0.0644938	0.911
prdx3	4.992139e+01	4.434486e+01	1.296543e+02	-1.3726943	0.000
spa	4.941449e+03	1.928969e+04	2.120320e+01	7.8889880	0.000
elf1a	5.226381e+01	5.265904e+01	1.359465e+02	-1.3831146	0.000
immt	2.938553e+01	3.058041e+01	5.647347e+01	-0.9681624	0.000
copb2	4.945819e+01	5.390595e+01	5.259926e+01	-0.0505903	0.833
dag1	1.849899e+01	1.696897e+01	3.253268e+01	-0.8530441	0.000
LASC009784	7.176135e+02	4.116265e+02	5.670844e-01	10.2661328	0.000
elf3k	8.893997e+01	1.163813e+02	4.274051e+01	1.1317299	0.000
gar1	2.152951e+01	1.968591e+01	2.021764e+01	0.0645509	0.893
tl2	2.385598e+01	2.459072e+01	4.232874e+01	-0.8364581	0.000
lrrc59	5.233954e+01	4.882332e+01	4.350131e+01	0.2628027	0.200
vha36-1	8.109811e+01	7.346527e+01	7.997829e+01	0.0378176	0.903
aca2	4.453960e+01	3.898214e+01	9.493921e+01	-1.1019950	0.000
cyt-b5	6.415372e+01	9.637328e+01	1.157126e+02	-0.7949151	0.040
clqbp	9.855040e+01	6.965126e+01	4.870009e+01	1.0585233	0.000
piwil1	1.864572e+01	1.818781e+01	1.607939e+01	0.1904426	0.350
ndufaf7	3.447655e+01	3.278448e+01	1.196524e+02	-1.8110979	0.000
elf2a	3.083390e+01	2.890473e+01	1.444894e+01	1.0979813	0.000
b52	4.746328e+01	4.510335e+01	5.575939e+01	-0.2539965	0.473
hm13	3.538075e+01	3.677871e+01	1.396463e+01	1.3434989	0.000
mthfs_2	3.193436e+01	1.639986e+01	2.738042e+01	0.2033319	0.643
dhc64c	2.328730e+01	2.590543e+01	5.213417e+01	-1.1336529	0.000
fk506-bp1_fpr4	5.153000e+01	3.673992e+01	3.545735e+01	0.5512241	0.013
ahcy11	2.825129e+01	2.705016e+01	3.253900e+01	-0.1927615	0.633
cbr1	4.179713e+01	4.079687e+01	7.693650e+01	-0.9229211	0.010
igf2bp1_2	1.411302e+01	1.439841e+01	3.936995e+01	-1.5009644	0.000
dpm1	4.556053e+01	4.827276e+01	5.960983e+01	-0.3560064	0.253

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
snrpg	6.419908e+01	7.241347e+01	4.468444e+01	0.5708480	0.273
arhgdib_arhgdia	6.970014e+01	7.478009e+01	8.762439e+01	-0.3342954	0.173
arp3	4.376664e+01	3.595175e+01	6.248777e+01	-0.4982182	0.023
pcd_pcbd2	1.859720e+01	3.117402e+01	1.660048e+01	0.2028218	0.543
vhaac39-1	5.090989e+01	5.641331e+01	5.716100e+01	-0.1011777	0.774
paip1	2.714532e+01	3.090557e+01	5.323552e+01	-0.9557900	0.000
rtraf	5.127069e+01	5.569998e+01	4.433387e+01	0.2415262	0.452
dlat	1.992813e+01	2.176537e+01	6.191712e+01	-1.6185000	0.000
npc2a_2	1.748466e+02	4.153115e+02	3.331865e+03	-4.3307312	0.000
acad8	2.486676e+01	3.310140e+01	2.987163e+01	-0.3350432	0.253
hcb	2.046059e+02	3.509935e+02	8.877064e+02	-2.2199401	0.064
tmem68	6.540156e+01	1.103251e+02	2.048966e+01	1.7024594	0.000
scepe1	9.296365e+01	8.356300e+01	2.077594e+02	-1.2491646	0.000
ist1	3.550031e+01	4.102389e+01	8.570501e+01	-1.2548683	0.000
rpt2	1.199335e+01	1.320618e+01	1.923113e+01	-0.6563210	0.130
nxn12	8.933471e+01	1.158493e+02	1.595677e+02	-0.8471438	0.000
myg1	3.444767e+01	3.492362e+01	3.196542e+01	0.0816878	0.797
hsp70b2_7	1.784202e+01	9.981905e+00	1.027776e+00	4.0543602	0.000
syncrip_2	3.717495e+01	4.795143e+01	6.502790e+01	-0.7833464	0.037
xirp2	2.544026e+01	3.101977e+01	8.751963e+01	-1.8086444	0.000
fkbp7_fkbp3_fpr2	5.777730e+01	7.866162e+01	6.262185e+01	-0.0195800	0.964
psmc3	2.617670e+01	2.405149e+01	3.761591e+01	-0.5204113	0.003
g3bp2	1.862130e+01	2.226234e+01	3.756890e+01	-1.0278063	0.000
elf2s1	6.784322e+01	6.292728e+01	5.006738e+01	0.4634879	0.000
gstz1_2	6.662322e+01	2.327757e+01	8.864124e+01	-0.5310425	0.240
psmb4	5.356240e+01	4.274729e+01	5.130338e+01	0.0500186	0.889
hts	1.968327e+01	1.873742e+01	4.202504e+01	-1.1105152	0.000
dhtkd1	2.564508e+01	2.597229e+01	1.321837e+01	0.9332468	0.000
rbm4	2.445804e+01	2.117214e+01	5.064656e+01	-0.9994225	0.000
tmx1	4.101218e+01	3.614292e+01	4.438947e+01	-0.1223856	0.640
ghra_2	3.593957e+01	2.820515e+01	3.281877e+01	0.0958910	0.733
lypla2	3.030918e+01	4.028190e+01	5.141191e+01	-0.7268445	0.010
mtch2	1.565051e+01	1.926963e+01	4.242072e+01	-1.4277594	0.000
mys_2	2.604476e+01	2.855929e+01	4.911729e+01	-0.9526224	0.000
l(2)efl_5	5.554504e+01	1.971931e+01	1.583451e+02	-1.6194076	0.007
mecr	3.420929e+01	2.319413e+01	2.834386e+01	0.2570597	0.311
canx	3.984015e+01	4.291850e+01	8.236565e+01	-1.0434142	0.000
ndufs2	2.212845e+01	2.447581e+01	7.387668e+01	-1.7147480	0.000
exc-4	2.905543e+01	4.066923e+01	3.340344e+01	-0.1883293	0.633
ctsa	4.808601e+01	3.585111e+01	1.577587e+02	-1.6480346	0.003
plxna4_2	1.448602e+01	1.341651e+01	1.322584e+01	0.1103543	0.703
ddost	7.871466e+01	9.266259e+01	9.138660e+01	-0.1717579	0.621
hnrrp	2.251855e+01	1.687553e+01	3.358824e+01	-0.5841313	0.000
gpd1	4.051848e+01	3.068905e+01	1.385362e+02	-1.8690891	0.000
LASC019081	3.276305e+01	4.721024e+01	4.631741e+01	-0.4773369	0.253
arl1	2.498625e+01	2.694687e+01	2.241462e+01	0.1824415	0.360
vhaa	8.622587e+01	6.681817e+01	8.865383e+01	0.0695646	0.900

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
LASC013820	4.050840e+01	3.938455e+01	5.093620e+01	-0.3788244	0.307
hcc	1.772774e+02	3.191712e+02	7.353926e+02	-2.1564969	0.080
eif4a2	2.842770e+01	3.455054e+01	4.477929e+01	-0.6627222	0.023
crat	2.088465e+01	2.357287e+01	5.173810e+01	-1.3617208	0.000
lin7b	2.566794e+01	2.956185e+01	3.475547e+01	-0.3986786	0.080
hsp90ab1	3.533903e+01	2.571937e+01	5.027286e+01	-0.4577995	0.560
taldo1_taldo	5.024067e+01	4.362904e+01	7.212970e+01	-0.5317286	0.023
alpha-cat	3.505210e+01	3.087432e+01	3.289731e+01	0.1207720	0.607
actg1	1.274580e+01	1.286702e+01	1.754567e+04	-10.6055510	0.000
snrpd3_2	2.713490e+01	3.193479e+01	1.944339e+01	0.5522232	0.283
rpn2	7.807106e+01	9.494260e+01	9.654587e+01	-0.2608308	0.380
mrh1	1.568007e+01	1.541179e+01	1.908554e+01	-0.2976908	0.353
larpl	2.438799e+01	2.850382e+01	2.504923e+01	-0.0790183	0.813
gly3	4.606448e+01	4.957184e+01	5.812364e+01	-0.3344429	0.273
epb41l3_2	6.679519e+01	6.146313e+01	6.763219e+01	0.0053702	0.977
cpm_2	1.409614e+01	1.734994e+01	3.061733e+01	-1.0576028	0.063
lman2	2.292990e+01	1.891645e+01	3.727104e+01	-0.7164673	0.002
myef2_2	2.193579e+01	2.177335e+01	3.353849e+01	-0.5798564	0.013
grn_7	3.164521e+01	3.335005e+01	4.358887e+01	-0.4387921	0.140
pdia6	3.800013e+01	4.070263e+01	4.739056e+01	-0.2006318	0.673
dctn2	2.993291e+01	3.865775e+01	3.523082e+01	-0.1987238	0.420
LASC019392	3.451235e+01	3.131494e+01	1.057724e+01	1.6971363	0.000
LASC006421	2.844869e+01	3.402178e+01	9.591883e+00	1.5260310	0.000
col4a1_2	4.340741e+01	6.982916e+01	2.006603e+02	-2.2092537	0.000
adk2	1.108723e+01	9.282419e+00	1.132953e+01	-0.0409232	0.909
gstml_3	9.334171e+00	1.185622e+01	1.228731e+01	-0.2938771	0.740
t05h10.6	1.282975e+01	1.586141e+01	6.013341e+01	-2.1963418	0.000
tmed7	4.634556e+01	6.448727e+01	3.879747e+01	0.2541532	0.553
lcp1	3.964371e+01	4.436730e+01	2.822866e+01	0.4573286	0.093
FlSpc	3.143613e+02	2.732279e+02	4.255470e-01	NA	
dcd2b	3.415227e+02	3.920232e+02	1.360541e+02	1.2663939	0.000
ubxn1_ubxn1-a	2.553473e+01	2.738327e+01	4.345873e+01	-0.7488871	0.000
pura	1.721869e+01	2.086929e+01	3.152016e+01	-0.8948780	0.000
ddx39b	3.630628e+01	3.418468e+01	4.400352e+01	-0.2692515	0.290
cisd1	1.297957e+02	1.809898e+02	2.104520e+02	-0.6203564	0.183
vps35	2.767363e+01	2.853117e+01	5.713767e+01	-1.0079424	0.000
hts_2	4.394668e+01	4.771697e+01	1.017885e+02	-1.2137676	0.000
gmpr_gmpr2	3.063170e+01	3.042303e+01	3.610627e+01	-0.1630273	0.597
reep6_reep5	5.638712e+01	5.293150e+01	1.692599e+02	-1.6135857	0.000
rpia	4.254048e+01	4.391724e+01	3.217432e+01	0.4302146	0.023
hspl1_2	4.229173e+01	5.020068e+01	4.342389e+02	-3.3578364	0.000
erlin2-b	4.692644e+01	4.599828e+01	6.880959e+01	-0.5525121	0.000
h2afv	1.295788e+01	1.294680e+01	1.488948e+01	-0.2298103	0.397
metap2	1.502423e+01	1.640750e+01	3.748258e+01	-1.3341101	0.000
asna	2.281230e+01	3.403729e+01	5.060046e+00	2.3056010	0.000
ddx3x	1.162325e+01	1.276163e+01	1.784442e+01	-0.6214467	0.033
LASC010290	1.578806e+01	1.514577e+01	6.568976e+01	-1.9931465	0.000
parn	2.792007e+01	2.922296e+01	3.587373e+01	-0.3218549	0.090

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
FlSpb	5.676590e+02	2.378777e+02	6.179345e-01	NA	
mif	4.763706e+01	5.744907e+01	7.644934e+01	-0.6630346	0.07
eci1	3.196527e+01	3.391855e+01	6.704241e+01	-1.0753381	0.00
atl	1.316909e+01	1.405483e+01	2.417206e+01	-0.9389701	0.00
dpys	2.178154e+01	1.747512e+01	4.028284e+01	-0.8256094	0.06
nrg	2.537628e+01	2.493875e+01	2.218858e+01	0.1762213	0.47
vhasfd	4.645684e+01	4.889952e+01	3.843801e+01	0.3549603	0.29
apmap_3	2.546824e+01	2.218193e+01	5.579581e+01	-1.1694151	0.00
letm1	1.647732e+01	1.263844e+01	2.243348e+01	-0.5096391	0.15
ap2a2_2	1.770703e+01	1.615225e+01	4.250436e+01	-1.2768733	0.00
txn	4.346656e+01	5.772768e+01	7.137971e+01	-0.7110866	0.09
nipsnap_2	2.231314e+01	2.224547e+01	7.143685e+01	-1.6972007	0.00
awd	4.514225e+01	4.819288e+01	1.053474e+02	-1.1374910	0.00
etfdh	1.804569e+01	1.615995e+01	3.027968e+01	-0.7761994	0.00
usp5	1.461563e+01	1.250917e+01	1.539512e+01	-0.0655060	0.81
pgls	8.531059e+01	7.383618e+01	1.351582e+02	-0.6720955	0.06
LASC010973	1.160965e+01	1.237191e+01	3.021755e+00	1.8634779	0.00
lam_2	2.408189e+01	1.627120e+01	1.934257e+01	0.3136094	0.22
LASC010012	5.930271e+03	2.365996e+00	3.744055e-01	14.0176820	0.00
mrpl12	3.034404e+01	3.582161e+01	3.596814e+01	-0.2313099	0.52
snb-1_vamp3_syb	2.018938e+01	2.184642e+01	3.276877e+01	-0.7194326	0.03
LASC005883	7.190423e+01	7.883851e+01	8.554709e+00	3.0517812	0.00
manf_2	8.006179e+01	9.535272e+01	6.159360e+01	0.5264163	0.43
eif4h	1.528784e+01	1.779996e+01	2.995923e+01	-0.9754483	0.00
sep2	3.155636e+01	2.768204e+01	4.705279e+01	-0.5691223	0.02
LASC005807	2.455312e+01	3.000076e+01	1.048999e+02	-2.0537184	0.00
mpp6	2.067984e+01	2.449702e+01	7.249076e+00	1.5095109	0.00
bcap31	1.987109e+01	2.142803e+01	4.643981e+01	-1.2578786	0.00
nrg_2	1.441285e+01	1.515011e+01	3.954042e+01	-1.4055504	0.00
vinc	2.810815e+01	2.286908e+01	4.518759e+01	-0.7606268	0.02
unc-94	2.636746e+01	2.602138e+01	1.081861e+02	-2.0474873	0.00
smf	7.838198e+01	1.064143e+02	3.771799e+01	1.1476464	0.05
bcat2_vlg246094	2.479508e+01	2.055941e+01	4.329173e+01	-0.7656602	0.07
sec11c	3.071923e+01	4.672517e+01	3.012878e+01	0.0799582	0.84
tp02	2.442439e+01	3.011105e+01	1.837402e+01	0.4094818	0.31
bola2	5.134132e+01	4.294466e+01	2.776706e+01	0.9036337	0.00
myef2	9.053109e+00	8.307663e+00	1.896353e+01	-1.0175129	0.00
prtp	1.349047e+01	1.487786e+01	2.563407e+01	-0.9160685	0.00
ylpm1	1.302875e+01	1.981638e+01	2.874540e+01	-1.1098566	0.00
oxr1_2	4.869974e+01	6.055898e+01	4.223083e+01	0.2650619	0.61
LASC022131	1.250862e+01	8.545522e+00	1.089114e+01	0.2733542	0.53
srp14	4.150924e+01	4.609266e+01	2.404649e+01	0.8506180	0.02
sfxn1	1.584528e+01	1.814010e+01	1.700069e+01	-0.1509271	0.65
LASC001730	4.496938e+03	2.670287e+03	4.655548e+00	10.0421742	0.00
camki	1.948058e+01	1.945165e+01	1.740257e+01	0.1694688	0.74
hexa	3.808290e+01	4.182804e+01	2.619323e+02	-2.7884591	0.00
ranbp1	1.633268e+01	1.696474e+01	1.469965e+01	0.1898106	0.67

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
syncrip	1.549235e+01	1.373124e+01	2.287020e+01	-0.5054289	0.14
hyou1	1.380082e+01	1.942497e+01	2.151511e+01	-0.5130172	0.29
LASC016483	3.734869e+01	1.165685e+02	2.131164e+02	-2.5596923	0.00
mtpn	3.177652e+01	5.165026e+01	4.090044e+01	-0.2629492	0.60
rplp1_2	3.761340e+01	3.836400e+01	2.020607e+02	-2.4055802	0.00
usp14	2.259851e+01	2.326898e+01	3.525456e+01	-0.6359719	0.00
xpnpep1	1.578056e+01	1.006171e+01	1.297998e+01	0.2967720	0.23
LASC016310	1.111473e+01	9.007473e+00	1.429526e+00	3.2510215	0.08
f26e4.3_2	2.383633e+01	2.658884e+01	8.974549e+01	-1.8899614	0.00
ppt1	2.353350e+01	2.544222e+01	7.118817e+01	-1.6127289	0.00
cse1l	1.040105e+01	8.394020e+00	1.156846e+01	-0.1472772	0.54
qdpr	1.312208e+01	1.555079e+01	2.683401e+01	-0.9263071	0.11
LASC004844	3.818428e+01	4.894942e+01	1.872753e+01	1.0210186	0.01
sub1	2.347726e+01	2.566921e+01	5.729685e+01	-1.2577833	0.00
slc27a1_2	8.963940e+00	1.605183e+01	4.413596e+01	-2.3107399	0.00
LASC016530	1.567198e+01	1.577981e+01	4.251277e+01	-1.3583777	0.00
lana	2.348401e+01	2.889005e+01	7.932613e+01	-1.6430305	0.00
AgSp2b	1.573297e+02	2.797796e+02	4.359068e+00	5.4277741	0.00
FlSpa	9.330085e+01	6.160052e+02	8.927715e-01	NA	
idh3g	7.185709e+00	6.850601e+00	2.413090e+01	-1.7320638	0.00
lanb1	1.322783e+01	1.451906e+01	3.320048e+01	-1.3193070	0.00
adk_adk2	1.135038e+01	6.299600e+00	3.504438e+01	-1.6985305	0.00
nudt16	4.478845e+01	5.126741e+01	4.075821e+01	0.2054685	0.55
pdc6_2	3.807854e+01	4.510899e+01	3.761421e+01	0.0947624	0.86
hnrnpul1	8.747611e+00	8.358211e+00	1.228405e+01	-0.4862738	0.21
lamtor3	4.127976e+01	3.282130e+01	3.725240e+01	0.2404503	0.60
atp6ap1	2.905788e+01	3.106717e+01	3.186574e+01	-0.0710936	0.85
slc25a3_2	2.049335e+01	2.519622e+01	2.443280e+01	-0.2565875	0.44
LASC003279	2.127112e+01	2.143478e+01	7.301939e+00	1.5309180	0.00
hspg2	1.724646e+01	1.671326e+01	7.504234e+01	-2.1105577	0.00
spcs2	6.194728e+01	8.332426e+01	7.054107e+01	-0.0608833	0.92
pgm1_2	1.059078e+01	1.252618e+01	6.610868e+01	-2.6867786	0.00
pnliprp1	8.154914e+02	1.861811e+03	2.906238e+00	8.1573000	0.00
emc1	7.325477e+00	1.336149e+01	2.094185e+01	-1.4172696	0.00
lam	5.454531e+00	4.138175e+00	1.300540e+01	-1.3307622	0.00
jana	1.379263e+01	1.619482e+01	2.986802e+01	-1.0683425	0.00
rac1_2	8.659481e+00	1.146234e+01	2.325238e+01	-1.4424432	0.00
nrx-iv	2.104014e+01	1.808570e+01	6.908682e+00	1.6018948	0.00
khsrp	1.398402e+01	1.355236e+01	1.829501e+01	-0.3526803	0.39
tgas006m08.1_3	8.490192e+00	9.124107e+00	2.900273e+00	1.8162809	0.31
tom1	1.189201e+01	1.258671e+01	2.125323e+01	-0.8418116	0.00
m6pr	1.455442e+01	1.707434e+01	1.031271e+01	0.5012878	0.29
pccb	1.489160e+01	2.237852e+01	3.128597e+01	-1.0186429	0.00
LASC011699	4.398155e+00	6.354800e+00	7.983185e-01	2.7852565	0.13
LASC017285	3.882935e+01	3.555139e+01	6.745269e+00	2.5161029	0.00
golim4	2.388701e+01	3.369051e+01	2.191165e+01	0.1758161	0.65
flo2_2	9.935116e+00	1.149402e+01	2.112967e+01	-1.1247060	0.00
sptbn5	1.266442e+01	1.070874e+01	1.568508e+01	-0.3085956	0.23

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
ergic1	1.911235e+01	2.340353e+01	7.745196e+00	1.3376102	0.000
SpiCE-LMa2	5.426611e+02	9.382672e+03	9.947424e+00	5.8943686	0.000
dek	1.727373e+01	1.449128e+01	2.819058e+01	-0.6754179	0.000
LASC020142	3.142545e+01	3.097012e+01	3.469933e+01	-0.0757780	0.850
cg3556_2	1.705947e+01	3.052450e+01	2.114600e+02	-3.4343144	0.000
LASC017000	2.730260e+01	3.773455e+01	2.695296e+01	0.0174072	0.970
cana-14f	8.254679e+00	9.730724e+00	2.470538e+01	-1.5731045	0.000
LASC002717	2.136402e+01	2.268711e+01	6.627165e+00	1.6599892	0.000
LASC015168	4.463507e+01	4.544087e+01	1.690756e+01	1.3607572	0.000
flr	1.186392e+01	1.253389e+01	3.249106e+01	-1.4754339	0.000
ube2l3	1.703698e+01	2.423863e+01	3.189827e+01	-0.9192114	0.010
tpd52l2	1.176255e+01	1.136697e+01	5.116443e+01	-2.0173809	0.000
pde12_4	1.121721e+01	1.174734e+01	2.038270e+01	-0.8882323	0.000
gpr158	1.526378e+01	1.650481e+01	4.628926e+00	1.7019136	0.000
vat1l	1.786689e+01	1.886252e+01	4.604782e+01	-1.2662303	0.000
rab14	1.148036e+01	1.319548e+01	2.384557e+01	-1.0550045	0.000
cont	1.654827e+01	1.439698e+01	6.426183e+00	1.3177486	0.000
meltf	1.383123e+01	1.421532e+01	4.222756e+00	1.6910535	0.000
LASC003125	7.836332e+00	8.783126e+00	3.202332e+00	1.2769862	0.000
cav1_3	3.408894e+01	4.529710e+01	5.579413e+01	-0.6862474	0.190
futsch	5.990857e+00	8.299807e+00	2.373860e+01	-1.8364202	0.000
ank3_8	6.773656e+00	7.180242e+00	7.007915e+01	-3.3576356	0.000
col4a1_let-2	1.697286e+01	2.850258e+01	1.334624e+02	-2.8758297	0.000
ccdc58	2.274805e+01	2.379923e+01	2.129886e+01	0.1738969	0.760
LASC019275	6.356752e+02	2.524343e+02	9.205447e-01	9.4731300	0.000
mvp	1.011700e+01	7.970192e+00	2.870885e+01	-1.5442037	0.000
atpsynd_2	1.443985e+02	7.819164e+01	2.292405e+02	-0.7014013	0.570
gyg1	1.120982e+01	9.537518e+00	1.584655e+01	-0.4604073	0.130
pabp2	9.148115e+00	5.585527e+00	9.851321e+00	-0.1261815	0.740
rab6a	3.754436e+00	3.534497e+00	1.753958e+01	-2.0940767	0.000
tgfb1_3	2.421889e+01	3.741255e+01	1.708950e+02	-2.6670234	0.000
LASC002397	1.224917e+01	1.316612e+01	2.207633e+01	-0.8530883	0.000
braflldraft	1.792518e+01	1.192389e+01	4.256328e+02	-4.4406874	0.000
LASC007511	1.040571e+01	1.247382e+01	1.807816e+01	-0.7739607	0.020
pka-c1	2.979878e+01	4.418732e+01	8.172699e+01	-1.4325548	0.000
sep1	9.054349e+00	1.249938e+01	1.407663e+01	-0.6598618	0.030
LASC020300	2.423290e+02	8.567694e+01	2.758804e+03	-3.5561531	0.000
LASC017532	1.154722e+03	4.941216e+03	5.114599e+00	7.8656062	0.000
csrp2	7.619544e+00	9.659726e+00	2.786223e+01	-1.8474469	0.000
tom1l2	1.227994e+01	1.645281e+01	1.218215e+01	0.0733091	0.830
SpiCE-LMa1	1.016607e+03	3.462429e+04	3.802945e+01	4.9953780	0.000
LASC021415	1.051318e+00	1.990022e+00	6.330892e-01	1.0240494	0.640
hpgd_14	8.495649e+02	8.781302e+02	1.032622e+00	9.7266768	0.000
lsm10	6.282362e+00	5.333605e+00	6.774550e+00	-0.1037435	0.860
stt3a	2.337458e+01	3.549993e+01	7.299334e+01	-1.5324905	0.000
lmf2	8.259135e+00	8.801206e+00	2.204320e+01	-1.3614604	0.000
sh3beta	1.118833e+01	1.221030e+01	1.438238e+01	-0.3098620	0.490

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
LASC005973	1.470474e+03	5.468885e+02	1.457895e+00	10.0635269	0.000
dpp3_4	2.704177e+01	3.457409e+01	5.633409e+01	-0.9948139	0.000
pcca	7.496433e+00	1.089581e+01	1.221915e+01	-0.6566935	0.151
LASC002879	8.783385e+02	1.522411e+03	4.989704e+02	0.9121007	0.231
ppm1e	6.823027e+00	8.697339e+00	1.736207e+01	-1.3629816	0.000
AcSpa	1.554122e+00	1.802743e-01	2.432008e-01	2.7835482	0.023
AmSp-like1	7.591754e+01	1.361637e+02	2.669154e-01	8.0824558	0.000
macrodl_macrodl2	8.457206e+00	9.371283e+00	1.605713e+01	-0.9156862	0.000
csga_5	9.693293e+00	1.118331e+01	3.401356e+01	-1.7979526	0.000
rheb	1.486154e+01	1.839833e+01	2.478581e+01	-0.7520618	0.031
txndc17	5.881849e+00	6.084890e+00	1.512900e+01	-1.3002689	0.000
LASC011075	7.178222e+01	5.552737e+01	8.692246e+00	3.0451417	0.000
arl6ip5	1.699916e+01	1.527414e+01	2.234617e+01	-0.3988634	0.351
pld3	1.535934e+02	1.905546e+02	4.315936e+01	1.8424406	0.003
capn5	9.743956e+00	8.104575e+00	1.287411e+01	-0.3462928	0.223
isoc2	6.521414e+00	9.167432e+00	1.792059e+01	-1.4458761	0.000
arf1	5.715658e+00	6.749443e+00	3.279420e+00	0.8305947	0.020
lbr	5.930416e+00	4.842589e+00	7.474461e+00	-0.3827598	0.311
gstml	6.386693e+00	1.430008e+01	1.523144e+01	-1.0379909	0.243
LASC005034	5.769818e+02	3.866492e+02	2.816103e+01	4.4323100	0.000
lac_7	1.044270e+01	1.128592e+01	3.523696e+00	1.5360304	0.000
itpr1	5.981549e+00	7.305903e+00	1.222519e+01	-1.0325135	0.000
jhamt_7	3.923983e+02	6.463788e+02	7.217331e-01	9.1014218	0.000
lanb2_2	7.095169e+00	1.145484e+01	2.612975e+01	-1.8570165	0.000
mlec	3.724907e+00	6.053174e+00	3.477496e+00	0.1190632	0.820
grhpr_7	4.933003e+01	7.801734e+01	4.409499e+02	-3.1386779	0.000
h2afy	2.614656e+00	3.192958e+00	4.431284e+00	-0.7427683	0.064
pka-cl_2	3.176490e+00	1.911424e+00	1.839808e+01	-2.6374335	0.000
bpnt1	5.186587e+00	6.716035e+00	1.699156e+01	-1.6503269	0.000
ncam1	4.106608e+00	7.913716e+00	4.640813e+01	-3.3846941	0.000
AmSp-like2	7.751322e+01	1.038318e+02	1.197435e-01	9.3622442	0.000
galphaq	3.243080e+00	3.553760e+00	7.806779e+00	-1.2114380	0.000
trip11	2.995135e+00	4.942985e+00	3.405133e+00	-0.1569779	0.683
MaSp1a	2.570912e+01	1.615975e+03	1.546984e+01	1.0209114	0.463
CaAn_20	2.980737e+01	6.146322e+01	2.715355e+01	0.1958248	0.711
LASC004484	4.690673e+02	3.486658e+02	6.391603e+00	6.2237610	0.000
gstml_10	1.443096e+00	1.487654e+00	1.400872e+02	-6.6685041	0.000
adpgk	1.540428e+00	2.490696e+00	1.938824e+00	-0.2608370	0.607
pdia4	1.402226e+01	1.694622e+01	1.819537e+01	-0.3209611	0.543
MaSp1b	2.330467e+01	9.971849e+02	3.259825e+00	3.1210535	0.000
AgSp1	3.183153e+01	4.611383e+01	6.806286e-01	5.8317163	0.000
pgm2	3.240210e+00	4.463755e+00	5.684152e+00	-0.7778869	0.067
hspa8_hsp70	1.891355e-01	9.178996e-01	3.003709e+00	-4.0873753	0.000
LASC018595	1.045134e+03	1.837244e+03	2.816314e+00	8.6117739	0.000
ndrg3_ndrg4-a	9.919037e+01	2.277038e+02	2.042460e+01	2.3902039	0.000
LASC001726	3.836195e+01	2.891982e+02	7.590944e-01	5.8093281	0.000
ftcd	1.687091e+00	1.487201e+00	2.463451e+00	-0.5192961	0.297
naga_3	2.260192e+01	8.357696e+00	1.190003e+02	-2.4019791	0.000

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
LASC004993	6.071202e+02	4.601339e+02	8.642817e-01	9.5542904	0.000
aldh5a1	6.408527e+00	7.338559e+00	1.141196e+01	-0.8264808	0.000
nt1	4.000380e+02	4.805262e+02	2.643028e+00	7.2134543	0.000
LASC015175	3.098712e+01	2.735304e+01	3.844845e-01	NA	0.000
coll1a2	5.768716e+00	8.399082e+00	5.257909e+01	-3.2639377	0.000
SpiCE-LMa5	1.932291e+01	9.290272e+02	4.473058e+00	2.3955590	0.033
lev-9_cr1_csmd3	3.157956e+00	3.575149e+00	3.893795e+00	-0.2934991	0.533
LASC009239	6.892011e+00	1.035634e+01	2.526587e+01	-1.9203442	0.000
mttp_2	2.032376e+02	2.148994e+02	4.216806e+01	2.4394914	0.000
LASC016433	2.239233e+01	2.318425e+01	9.462855e+01	-2.1376336	0.023
LASC006408	3.163859e+02	2.524352e+02	4.760937e-01	9.4623143	0.000
LASC000038	1.378507e+02	4.770173e+01	9.645610e-02	10.5165092	0.000
MaSp2c	1.455441e+01	1.614962e+03	1.179791e+01	0.5802609	0.683
MaSp1c	1.018638e+01	2.839469e+02	6.476295e-01	4.2346852	0.000
ykho_tm_fabg	3.262400e+00	2.365868e+00	1.274457e+01	-2.0202163	0.000
ywrd_2	7.729377e-01	5.023809e-01	8.142583e+00	-3.3445316	0.000
LASC004995	4.407007e+02	4.229643e+02	1.538042e+00	8.2302505	0.000
erp44_3	5.889943e+00	1.096657e+01	2.069133e+00	1.4050430	0.123
cmpk	7.122692e+00	1.074512e+01	2.278163e+01	-1.6062087	0.000
ptrhd1	1.570493e+01	3.749447e+01	7.365240e+00	1.1616816	0.033
hdlbp_3	2.898999e+00	3.330363e+00	4.766288e+00	-0.7543581	0.093
LASC007099	3.155744e+02	3.198522e+02	1.300368e-01	11.3734496	0.000
dst_2	1.622585e+01	1.441688e+01	3.157769e+01	-0.9604523	0.000
hsd17b6_hsd17b2	7.955345e+01	1.116023e+02	2.852038e-01	8.1298589	0.000
lox12	1.002959e+01	1.234350e+01	2.573846e+01	-1.2396942	0.100
uchl3	2.607332e+00	2.267512e+00	5.193361e+00	-0.9456099	0.123
LASC012887	6.864422e+01	6.952554e+01	1.247360e-01	9.1856140	0.000
pka-r2_2	4.564850e+00	5.200441e+00	1.314677e+01	-1.4243769	0.010
h2bc1	1.095280e+00	3.249548e-01	5.269590e-02	3.6990310	0.233
nop58_3	3.950555e+00	1.385972e+00	2.254535e+00	0.6384344	0.733
mas_ppaf2	1.001863e+02	1.956555e+01	8.551354e-01	6.8741525	0.000
MaSp2f	2.294970e+01	3.149694e+03	2.640244e+01	0.0699124	0.963
LASC010337	1.155727e+00	2.061431e+00	2.815375e+01	-4.4807206	0.000
mys	3.692773e+00	4.763305e+00	4.876805e+01	-3.7280150	0.000
LASC005488	4.198981e+00	8.151802e+00	5.206698e+01	-3.5255417	0.000
pdcd6_bche	2.624777e+00	3.063423e+00	2.480999e+00	0.0649444	0.903
LASC009994	8.115965e-01	7.274717e-01	8.056159e+00	-3.2301875	0.000
fasn_8	3.997370e+01	7.876412e+01	1.092970e-01	8.5729510	0.000
LASC019814	5.123614e-01	5.698313e-01	5.574332e+01	-6.8164164	0.000
zbtb48_3	1.390443e+00	1.711136e+00	1.924214e+00	-0.4368972	0.353
LASC003331	2.014665e+03	7.678816e+02	2.643473e+03	NA	0.000
LASC001732	1.448747e+01	8.829893e+01	1.445149e-01	6.6223178	0.000
ywrd_3	1.143197e+00	3.588018e+00	3.300667e+00	-1.5032932	0.403
erp44_2	4.336367e+00	3.661539e+00	5.467814e+00	-0.2648466	0.790
LASC018776	7.123463e+01	1.705997e+02	5.090644e-01	7.0999413	0.000
rpl10a	5.601966e-01	6.565837e-01	1.417533e+01	-4.4801821	0.000
MaSp3b	6.112027e+01	8.430313e+02	1.194422e+00	5.8290188	0.000

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
hsp70b2_4	4.150343e-01	4.048409e-01	8.202999e-01	-0.9232554	0.560
LASC001731	1.108306e+02	2.546989e+02	5.185133e-01	7.8007122	0.000
pnlipr3_pnlip	6.309668e+01	1.044072e+02	1.253660e+00	5.6118465	0.000
nutf2	1.826409e+00	3.074228e+00	5.933051e+00	-1.6829391	0.000
LASC020883	6.372482e-01	1.109905e+00	2.025923e+01	-4.9549337	0.000
LASC003330	1.619285e+03	6.099860e+02	1.557158e+03	NA	
beta-spec	1.127226e+00	1.414751e+00	3.521326e+01	-4.9440863	0.000
gstd1_gst1	2.304319e-01	7.093000e-02	8.614477e+00	-5.1213901	0.000
alphatub85e_tuba1c	2.572857e+00	2.292110e+00	1.444193e+00	0.9047075	0.097
cd109_5	4.544483e+00	5.111389e+00	1.348984e+01	-1.5166811	0.000
gstm1_5	5.011836e-01	1.378238e-01	7.600396e+00	-4.0277376	0.000
LASC015292	2.519454e+01	5.517816e+00	5.071130e-02	8.9466895	0.000
epb41l3	1.673563e-01	1.716727e-01	3.254870e+00	-4.3100700	0.000
cmpk_2	2.970496e-01	4.883552e-01	1.891842e+00	-2.4364389	0.004
LASC015173	2.855527e+01	3.301852e+01	3.190677e-01	NA	
ywrp_6	6.419100e-02	5.034460e-02	1.585461e-01	-1.2260528	0.677
vit-6_vg	3.225867e+02	1.356888e+02	1.137706e+03	NA	
LASC000397	6.936137e-01	4.424142e-01	0.000000e+00	4.1324719	0.082
fib_2	9.665970e-02	1.739306e-01	8.439000e-02	0.3223236	0.918
cut-1_6	6.513771e+00	5.923854e+00	1.343926e+00	2.3214561	0.000
LASC003332	5.339123e+02	2.020132e+02	1.941595e+02	NA	
LASC005598	1.664644e+02	6.822854e+01	9.442205e+02	NA	
LASC006407	1.237485e+00	3.227192e+00	0.000000e+00	6.6819810	0.000
LASC007604	1.934714e+03	6.553947e+02	1.328933e+03	NA	
LASC022503	3.545228e+00	3.362306e+00	1.402837e-01	NA	
LASC022504	2.401657e-01	3.025368e-01	9.866780e-02	1.4137596	0.782
TuSpb	1.013235e+00	6.485465e+00	4.144140e-02	NA	
hpgd_12	1.367095e-01	5.456094e+00	0.000000e+00	3.4029341	0.010
ywrp_7	0.000000e+00	1.550700e-02	1.507903e-01	-2.7642090	0.580
LASC008951	2.296985e-01	5.823191e-01	1.336457e-01	1.0174471	0.710
LASC019165	5.764807e-01	3.813236e-01	1.099015e+01	-4.2225016	0.003
atpalpha_3	3.312870e-02	2.561530e-02	2.256095e+02	-12.7878963	0.000
psmb1	2.424688e+00	3.061921e+00	1.824062e+00	0.4187224	0.482
psmb1_2	4.500787e+00	3.291739e+00	3.139654e+00	0.5033316	0.242
AgSp2a	8.116000e-03	2.076170e-02	2.876280e-02	NA	
LASC019470	1.255707e-01	0.000000e+00	0.000000e+00	NA	
LASC021431	0.000000e+00	0.000000e+00	0.000000e+00	NA	
LASC022267	0.000000e+00	3.047090e-02	1.465743e-01	NA	
LASC022852	0.000000e+00	0.000000e+00	0.000000e+00	NA	
LASC022854	0.000000e+00	0.000000e+00	4.731090e-02	NA	
LASC022855	0.000000e+00	0.000000e+00	0.000000e+00	NA	
LASC022856	0.000000e+00	0.000000e+00	0.000000e+00	NA	
LASC022857	0.000000e+00	2.350490e-02	5.185280e-02	NA	
LASC022858	4.196840e-02	0.000000e+00	0.000000e+00	NA	
glyp_3	0.000000e+00	0.000000e+00	0.000000e+00	NA	
h4-viii	0.000000e+00	0.000000e+00	0.000000e+00	NA	
h4-viii_2	0.000000e+00	0.000000e+00	0.000000e+00	NA	
hspa8_2	0.000000e+00	0.000000e+00	0.000000e+00	NA	

(continued)

geneName	Minor	Major	Head	log2FC_Minor_vs_Head	padj_Minor_vs_Head
pxt_3	0.000000e+00	0.000000e+00	0.000000e+00		NA
tgas006m08.1	0.000000e+00	0.000000e+00	1.127234e-01		NA
tubb4b	1.377614e-01	1.984452e-01	3.602724e-01		NA
ankrd50_2	5.399900e-02	6.588590e-02	1.912100e+00		NA
ap1b1	0.000000e+00	4.000890e-02	3.759060e-02		NA
ctsa_2	5.395400e-02	3.141911e-01	2.504296e-01		NA
eef1b	1.052872e-01	8.946600e-02	4.739980e-02		NA
esd	3.530525e-01	3.173947e-01	6.643759e-01		NA
flo2	2.587226e-01	3.585464e-01	1.197250e+00		NA
his1	0.000000e+00	0.000000e+00	0.000000e+00		NA
his1.3_2	1.660111e-01	1.352144e-01	2.613550e-02		NA
hrb27c_4	4.103825e-01	3.485541e-01	6.276534e-01		NA
hsp70b2_3	2.325070e-02	3.742980e-02	1.925939e-01		NA
hsp70b2_5	0.000000e+00	0.000000e+00	3.192599e-01		NA
hsp70b2_hsp70a2	8.170130e-02	2.540728e-01	7.733402e-01		NA
inx2_15	0.000000e+00	0.000000e+00	2.959630e-02		NA

Note:

Table 3: Bulk RNA expression. Normalized counts used for analysis.

Supplementary Table 4

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
resized.

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
MiSpb	4.808264e+04	3.319369e+04	1.742311e+04	2.905666e+02	
MiSpa	4.110487e+04	2.586276e+04	4.780462e+03	2.781541e+02	
LASC007034	1.573747e+04	1.564032e+04	5.560319e+03	1.231027e+03	
eef1alpha1_2	1.458878e+04	1.516634e+04	1.016343e+04	4.726675e+03	
lman2l	1.313190e+04	1.247209e+04	5.624520e+03	7.336428e+02	
ef2	1.342139e+04	1.230828e+04	7.011710e+03	2.857194e+03	
MiSpd	1.605683e+04	1.204483e+04	6.043778e+03	1.213376e+02	
sesb	1.172676e+04	1.180122e+04	7.595147e+03	4.108111e+03	
rplp0	9.816942e+03	8.544801e+03	4.817232e+03	1.849587e+03	
rps8	1.142173e+04	7.887351e+03	3.611797e+03	1.125186e+03	
rps15a	9.926149e+03	7.781469e+03	4.125964e+03	1.892989e+03	
hdlbp_2	9.998900e+03	7.779982e+03	3.950665e+03	1.321445e+03	
tg_3	4.241202e+03	7.206892e+03	9.172734e+03	1.520283e+03	
ctsb_2	4.674636e+03	7.031663e+03	1.054222e+04	6.131874e+02	
LASC003944	7.417521e+03	6.726852e+03	3.648622e+03	1.886077e+03	
rpl3	6.788597e+03	6.595376e+03	4.174808e+03	1.599583e+03	
rps6	7.936077e+03	6.557585e+03	3.393001e+03	1.209998e+03	
rpl31	6.595794e+03	6.495538e+03	3.841877e+03	1.447336e+03	
rpl34	7.329060e+03	6.058096e+03	3.318291e+03	1.161398e+03	
rpl27a_2	6.539458e+03	5.626771e+03	3.279723e+03	1.085218e+03	
rpl19	7.328183e+03	5.501943e+03	2.570452e+03	6.325332e+02	
eef1g	5.912581e+03	5.344803e+03	3.124716e+03	1.632425e+03	
tgfb1_4	3.789866e+03	5.295148e+03	3.487366e+03	6.434329e+02	
rpl23_rpl23-a	6.500832e+03	5.225287e+03	2.610540e+03	1.015591e+03	
rpl10	4.736576e+03	5.219743e+03	7.050769e+03	4.195796e+03	
LASC015047	2.909587e+03	5.176863e+03	6.503569e+03	1.972506e+03	
rpp-2_rplp2	6.852774e+03	5.173422e+03	2.531228e+03	1.087270e+03	
rps27l_rps27	5.800913e+03	5.124123e+03	3.242790e+03	1.216694e+03	
rpl21	5.511020e+03	5.120697e+03	2.789568e+03	9.451000e+02	
rpl8	5.730093e+03	5.025270e+03	2.700128e+03	8.188925e+02	
rps11	5.996777e+03	4.996677e+03	2.733820e+03	9.856911e+02	
rpl-36.a	6.280092e+03	4.991274e+03	2.233645e+03	6.003119e+02	
LASC015735	3.238766e+03	4.986566e+03	4.247589e+03	4.354820e+02	
rps14	5.705775e+03	4.833371e+03	2.714542e+03	9.322916e+02	
rps4	5.267638e+03	4.756701e+03	2.920397e+03	1.130535e+03	
SpiCE-LMa3	9.440123e+02	4.715860e+03	5.479245e+03	1.291057e+03	
pabpc1	5.107642e+03	4.684796e+03	2.756617e+03	1.883983e+03	
rplp1	6.241385e+03	4.670204e+03	2.219445e+03	7.922764e+02	
rps25	5.279566e+03	4.469241e+03	2.263057e+03	7.643132e+02	
rps18	4.993137e+03	4.423129e+03	2.460504e+03	9.295205e+02	
rps3	5.033971e+03	4.315981e+03	2.339229e+03	8.479464e+02	
rpl5_rpl5-a	5.700275e+03	4.284127e+03	2.210611e+03	6.602577e+02	
rpl24	4.500013e+03	4.194166e+03	2.617260e+03	1.524362e+03	
rpl27	4.566816e+03	4.085640e+03	2.036452e+03	4.967161e+02	
x975_2	2.645989e+03	4.033929e+03	2.341855e+03	1.917193e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proximal
rps23	4.333215e+03	4.000969e+03	2.437998e+03	7.803390e+02	
sp-1339#prd34368.1_3	3.934706e+01	3.995166e+03	4.209721e+04	3.022938e+04	
rps9	5.242629e+03	3.838395e+03	1.822672e+03	5.970441e+02	
rps15	4.049535e+03	3.814345e+03	2.339827e+03	1.021915e+03	
gapdh1	2.799750e+03	3.730576e+03	2.241797e+03	8.119706e+02	
rpl11	4.039912e+03	3.580023e+03	2.127937e+03	8.863829e+02	
LASC006378	3.141916e+03	3.571513e+03	3.090396e+03	8.834210e+02	
rpl15	3.939224e+03	3.548705e+03	2.180304e+03	8.151992e+02	
rpl18	4.191374e+03	3.530795e+03	1.953519e+03	6.723629e+02	
rps7	4.691306e+03	3.443894e+03	1.408120e+03	4.023182e+02	
rpl6_rpl-6	3.609236e+03	3.440912e+03	2.029982e+03	7.537894e+02	
rpl18a	3.600511e+03	3.437193e+03	2.073042e+03	9.001760e+02	
inx2_18	2.653961e+03	3.382001e+03	1.791177e+03	8.809944e+02	
rpl17	3.694896e+03	3.262826e+03	1.834984e+03	5.807225e+02	
rpl32	3.586279e+03	3.255933e+03	1.668209e+03	4.102618e+02	
LASC000281	3.239119e+03	3.144119e+03	2.056851e+03	1.118897e+03	
rpl40	3.550845e+03	3.108089e+03	1.706943e+03	5.417379e+02	
LASC014844	3.274807e+03	3.064764e+03	1.812864e+03	8.645707e+02	
rpl36	3.531896e+03	3.015615e+03	1.641157e+03	3.894920e+02	
ssr1	3.001715e+03	2.961336e+03	1.768426e+03	5.483586e+02	
rpl14	3.413967e+03	2.959963e+03	1.634055e+03	5.496894e+02	
rps12	4.002634e+03	2.951354e+03	1.347029e+03	5.052327e+02	
rps27a	3.348826e+03	2.923097e+03	1.456545e+03	4.537001e+02	
rpl23a	2.865478e+03	2.843059e+03	1.690241e+03	6.967059e+02	
eef1alpha1	3.042094e+03	2.835818e+03	2.082125e+03	1.300676e+03	
rps5	3.426291e+03	2.720631e+03	1.515564e+03	5.884930e+02	
rpl7a	3.002219e+03	2.720081e+03	1.563701e+03	4.761017e+02	
rps20	2.823585e+03	2.585307e+03	1.608064e+03	6.479603e+02	
rps2_2	2.744225e+03	2.543844e+03	1.562711e+03	4.384481e+02	
rpl10ab	2.390964e+03	2.456750e+03	1.419013e+03	5.349941e+02	
rpl4	3.010211e+03	2.401054e+03	1.229125e+03	3.018367e+02	
sec61g	2.942428e+03	2.384899e+03	1.086447e+03	4.199009e+02	
LASC000747	2.588727e+03	2.367609e+03	1.671650e+03	1.095225e+03	
rack1	2.705261e+03	2.359781e+03	1.191255e+03	3.780181e+02	
rpl37a_2	2.854301e+03	2.301187e+03	1.225884e+03	3.885148e+02	
rpl39	2.668301e+03	2.271383e+03	1.424822e+03	4.407092e+02	
rpl-7	2.798006e+03	2.252088e+03	1.351046e+03	5.043343e+02	
rpsa	2.451066e+03	2.220938e+03	1.496983e+03	7.279751e+02	
LASC003517	2.497764e+03	2.134884e+03	1.216913e+03	4.004177e+02	
rpl12	2.275863e+03	2.114875e+03	1.283786e+03	4.561903e+02	
fau_2	2.571358e+03	2.114468e+03	1.201876e+03	5.421842e+02	
idh2	2.241565e+03	2.043286e+03	1.117522e+03	2.503995e+02	
LASC019282	1.331869e+03	1.975534e+03	2.920165e+03	1.694832e+03	
rps19	2.033368e+03	1.963889e+03	1.185825e+03	3.893833e+02	
LASC010830	7.001938e+02	1.947254e+03	2.817684e+03	5.909000e+02	
eif4a1	2.052018e+03	1.925156e+03	1.612318e+03	9.043340e+02	
rps26	1.859787e+03	1.920549e+03	1.291126e+03	5.474589e+02	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
rpl13	2.313929e+03	1.916271e+03	1.082617e+03	3.595398e+02	
rpl13a	2.133122e+03	1.908968e+03	1.090757e+03	4.030042e+02	
rsf1_3	2.087682e+03	1.843381e+03	1.263377e+03	5.568574e+02	
rps17	1.963191e+03	1.825680e+03	1.165877e+03	3.954582e+02	
cnbp	2.227155e+03	1.768905e+03	6.857380e+02	1.999254e+02	
phgdh	1.424024e+03	1.765538e+03	1.277539e+03	6.124513e+02	
eef1d	2.023223e+03	1.743927e+03	1.098423e+03	5.544614e+02	
ktn1	2.088809e+03	1.692099e+03	8.886733e+02	1.070263e+02	
rps10b	1.651989e+03	1.629440e+03	9.343348e+02	3.225257e+02	
vdac2	1.840033e+03	1.617021e+03	1.003122e+03	3.771375e+02	
gcat	1.671232e+03	1.599536e+03	8.642849e+02	3.837795e+02	
rpl9_2	1.767976e+03	1.549019e+03	6.377237e+02	1.770673e+02	
psat1	1.661486e+03	1.529466e+03	1.099949e+03	6.858376e+02	
ssr2	1.309163e+03	1.463926e+03	8.783313e+02	2.867829e+02	
rpl26_2	1.572073e+03	1.440525e+03	8.214724e+02	2.326403e+02	
rpl35a	1.339839e+03	1.439396e+03	9.189608e+02	3.028023e+02	
rps24	1.743623e+03	1.439081e+03	7.612402e+02	3.112135e+02	
eef1b_2	1.581070e+03	1.438494e+03	8.694563e+02	5.658587e+02	
tdpoz4_15	1.138631e+03	1.418778e+03	2.907896e+02	4.710339e+00	
rpl9	1.474824e+03	1.333875e+03	8.233248e+02	2.201546e+02	
nupr1	1.418880e+03	1.323844e+03	6.128091e+02	6.865991e+01	
rpl28	1.372135e+03	1.292393e+03	6.860625e+02	2.535077e+02	
LASC011384	1.429568e+03	1.272965e+03	5.539032e+02	6.620781e+01	
ece2_2	1.009441e+03	1.238959e+03	6.750377e+02	1.132274e+02	
tsr	1.205258e+03	1.207308e+03	9.325760e+02	3.166658e+02	
slc25a3	1.148415e+03	1.205554e+03	8.930249e+02	4.559965e+02	
14-3-3zeta_ywhab	1.161784e+03	1.192857e+03	9.969935e+02	3.821224e+02	
cotl1	7.212832e+02	1.181740e+03	1.034819e+03	3.698011e+02	
znf346	8.488198e+02	1.150708e+03	4.433164e+02	9.810547e+01	
rpl22	1.475829e+03	1.145859e+03	5.302418e+02	1.599731e+02	
ncl	1.271038e+03	1.143310e+03	5.876654e+02	1.357487e+02	
atp5f1b	1.015138e+03	1.131951e+03	7.852353e+02	3.287952e+02	
tth-1	8.222284e+02	1.099035e+03	1.031890e+03	2.057618e+02	
rpl30	1.192866e+03	1.088259e+03	5.538807e+02	1.604600e+02	
ctsb	7.892519e+02	1.085404e+03	2.502580e+03	1.080898e+02	
surf4	1.097990e+03	1.082190e+03	5.529226e+02	1.515757e+02	
SpiCE-LMa6	2.884360e+02	1.045246e+03	7.289857e+02	1.381235e+01	
tgas006m08.1_2	8.640278e+02	1.025435e+03	8.683071e+02	5.499443e+02	
cp1	9.554652e+02	1.002129e+03	1.023376e+03	2.529452e+02	
LASC005273	1.058482e+01	9.966794e+02	1.037995e+04	4.002625e+03	
LASC015682	8.440166e+02	9.672189e+02	7.562104e+02	2.179048e+02	
b3galt5	7.344079e+02	9.429772e+02	1.009755e+03	7.981161e+02	
dap	1.027063e+03	9.428785e+02	4.816036e+02	1.470970e+02	
rps13	1.013400e+03	9.338692e+02	5.632335e+02	1.964171e+02	
slc25a4	4.895488e+02	9.327481e+02	1.726011e+03	7.930711e+02	
MiSpc	8.389111e+02	9.167020e+02	7.961199e+02	2.254457e+00	
LASC017036	1.154569e+03	9.119695e+02	5.023710e+02	1.057517e+02	
ubc	6.941793e+02	9.105985e+02	7.079861e+02	4.074325e+02	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
LASC019923	1.139647e+03	8.849211e+02	2.767666e+02	2.470095e+01	
yars	8.682145e+02	8.742728e+02	6.050401e+02	2.954640e+02	
eif3i_v1g239264_iscu	9.287081e+02	8.543819e+02	4.913088e+02	1.910303e+02	
ahcy-a	8.853719e+02	8.287623e+02	7.693912e+02	7.985565e+02	
blw_atp5fla	7.875506e+02	8.278781e+02	5.340686e+02	1.899518e+02	
rpl26	8.920361e+02	8.155895e+02	5.094201e+02	1.687445e+02	
eno1	9.031730e+02	7.958665e+02	5.495454e+02	1.853069e+02	
sec61b	6.542963e+02	7.715581e+02	5.854382e+02	2.463763e+02	
eif6	8.206798e+02	7.713433e+02	4.724915e+02	1.898155e+02	
pro5	6.825684e+02	7.577202e+02	5.842926e+02	1.779801e+02	
cyp1	7.773455e+02	7.445115e+02	5.553618e+02	2.293681e+02	
mdh2	6.464413e+02	7.378240e+02	5.516932e+02	2.236327e+02	
nars	8.010702e+02	7.375720e+02	5.310533e+02	2.405009e+02	
LASC000095	4.505734e+02	7.342267e+02	7.686301e+02	3.950411e+02	
khdrbs3	5.834267e+02	7.071307e+02	6.611084e+02	4.298305e+02	
eef5_tif51a_eif5a1	6.953763e+02	6.845812e+02	4.559286e+02	1.812604e+02	
LASC017035	8.123348e+02	6.769297e+02	3.543311e+02	8.236051e+01	
rpl35_2	8.052398e+02	6.491843e+02	3.246595e+02	8.438579e+01	
cth	5.069023e+02	6.467684e+02	4.977333e+02	2.356618e+02	
LASC007825	5.686616e+02	6.347043e+02	4.357928e+02	1.187350e+02	
atpalph4_4	4.905200e+02	6.093637e+02	4.057801e+02	2.709381e+02	
rpl35	8.016517e+02	5.993308e+02	3.129350e+02	1.540032e+02	
aldh1a2	5.762101e+02	5.843564e+02	6.494530e+02	6.191477e+02	
cher_2	1.777058e+02	5.812451e+02	6.370030e+02	1.850070e+02	
cd63	4.336476e+02	5.790821e+02	4.672547e+02	8.177259e+01	
oat	4.744089e+02	5.779865e+02	2.485892e+02	3.977353e+01	
uqcrfs1	5.341047e+02	5.687915e+02	4.029587e+02	2.046197e+02	
lama2_2	4.878510e+02	5.682540e+02	3.023442e+02	3.115506e+01	
ywrd_5	1.792646e+02	5.559756e+02	9.527992e+02	1.303269e+03	
rtn3-b	4.337573e+02	5.521147e+02	4.540171e+02	1.570880e+02	
cpib	3.940942e+02	5.506317e+02	7.208926e+02	2.653339e+02	
hspa9	6.151473e+02	5.345839e+02	3.270474e+02	1.104267e+02	
actn	4.176622e+02	5.324665e+02	4.484715e+02	1.704964e+02	
jagn	5.338198e+02	5.318154e+02	2.978081e+02	4.825499e+01	
rpl38	6.315685e+02	5.260636e+02	2.572198e+02	7.884473e+01	
maob_3	6.593681e+02	5.227153e+02	3.476382e+02	2.227875e+02	
rps16_2	5.372095e+02	5.161412e+02	3.085286e+02	1.451686e+02	
shmt2	4.674918e+02	5.027328e+02	2.085900e+02	3.272420e+01	
LASC021594	5.039247e+02	4.978891e+02	3.050889e+02	1.548188e+02	
ddx24	2.906285e+02	4.948327e+02	1.689519e+02	1.630511e+01	
slit1_2	4.659640e+02	4.931113e+02	6.026041e+02	3.990075e+02	
LASC019949	2.800246e+02	4.900082e+02	4.987474e+02	9.858061e+01	
LASC020361	4.432853e+02	4.900053e+02	3.623928e+02	1.296112e+02	
uba1	4.010347e+02	4.889265e+02	3.915018e+02	1.303439e+02	
reep5_4	3.868682e+02	4.881569e+02	3.614911e+02	9.257356e+01	
cox6a1	4.658269e+02	4.859792e+02	3.369973e+02	1.174701e+02	
tub2_2	3.462149e+02	4.840799e+02	4.073278e+02	1.105345e+02	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
nap1l1-b	4.592009e+02	4.771842e+02	3.746956e+02	2.024062e+02	
rpl27a	4.983130e+02	4.722436e+02	3.026477e+02	1.346497e+02	
slc3a1_3	5.214281e+02	4.714481e+02	2.764570e+02	1.242106e+02	
LASC016339	5.004968e+02	4.692902e+02	3.756594e+02	3.696597e+02	
hnrnpdl	4.754433e+02	4.590001e+02	3.311721e+02	1.351903e+02	
sqd	4.929743e+02	4.491334e+02	3.970586e+02	2.020574e+02	
hint1	4.078600e+02	4.467582e+02	3.324847e+02	1.135463e+02	
sar1a_zk180.4_sar1b	4.666107e+02	4.463615e+02	3.277532e+02	1.415062e+02	
arf1_2	4.600163e+02	4.440233e+02	3.602363e+02	1.784340e+02	
aldh1a2_2	4.543863e+02	4.356318e+02	3.809445e+02	8.148496e+00	
cg12163	4.285013e+02	4.220833e+02	3.569545e+02	9.337385e+01	
LASC007830	3.838242e+02	4.144642e+02	4.081849e+02	1.241551e+02	
naca	3.681738e+02	4.109615e+02	4.061818e+02	1.722763e+02	
tubb4b_3	2.842702e+02	4.082221e+02	5.632253e+02	1.789024e+02	
rab1a	3.658906e+02	4.080160e+02	3.675854e+02	1.741912e+02	
pdhb-1_pdhb	4.204228e+02	4.064049e+02	2.881849e+02	1.408120e+02	
tmem214	3.749099e+02	3.986995e+02	2.465037e+02	9.252115e+01	
sec23a	3.159866e+02	3.966222e+02	3.167432e+02	1.503782e+02	
rps21	4.810652e+02	3.885676e+02	1.772843e+02	4.806419e+01	
cct2	3.551722e+02	3.877623e+02	2.664812e+02	7.812312e+01	
arfgap2	4.183263e+02	3.870011e+02	1.959524e+02	5.351713e+01	
tubb4b_4	3.888266e+02	3.868957e+02	2.608104e+02	9.516426e+01	
LASC020761	4.349471e+02	3.867900e+02	2.592936e+02	1.007931e+02	
acadm	3.418360e+02	3.762416e+02	2.553295e+02	4.196436e+01	
acp7	1.177749e+02	3.699432e+02	9.369619e+02	4.246888e+02	
pdi	3.911554e+02	3.605869e+02	3.670348e+02	1.988474e+02	
EIF3-s7	3.949452e+02	3.493834e+02	2.143662e+02	8.715727e+01	
aco2	3.126650e+02	3.488910e+02	1.972541e+02	6.739692e+01	
psap	2.700437e+02	3.433623e+02	5.235035e+02	1.458062e+02	
fkbp12	2.831547e+02	3.414656e+02	2.983932e+02	1.124324e+02	
me1	1.507653e+02	3.365211e+02	3.423039e+02	2.019667e+02	
sdha	3.239891e+02	3.345447e+02	2.500826e+02	7.676042e+01	
lamp1_2	2.487179e+02	3.315119e+02	3.879061e+02	2.470036e+02	
atpsynf	3.327703e+02	3.265456e+02	2.024244e+02	9.829900e+01	
rsf1_2	2.654333e+02	3.159038e+02	3.213931e+02	1.228763e+02	
EIF4e1a EIF4e	3.622724e+02	3.147250e+02	2.331456e+02	1.375221e+02	
ywhae_2	2.099478e+02	3.146648e+02	3.548844e+02	1.101881e+02	
rps28	3.755135e+02	3.113035e+02	1.666177e+02	6.022058e+01	
serbp1	3.489336e+02	3.098707e+02	2.380547e+02	1.191235e+02	
blmh	3.119269e+02	3.094096e+02	1.925899e+02	7.535953e+01	
tktl1_tktl1	2.140249e+02	3.056538e+02	2.469977e+02	7.589603e+01	
ssr4	2.806837e+02	2.989417e+02	2.274241e+02	9.622065e+01	
LASC020665	1.606746e+02	2.964904e+02	1.484409e+02	2.680130e+01	
inx2_2	2.351609e+02	2.956445e+02	1.196902e+02	5.850295e+01	
got2	2.713138e+02	2.937176e+02	2.058738e+02	8.684417e+01	
jafrac1_2	2.284161e+02	2.908995e+02	2.868552e+02	1.465042e+02	
cg1354	3.413907e+02	2.885980e+02	1.511851e+02	2.846100e+01	
sdhb	2.678143e+02	2.883602e+02	2.109807e+02	7.802739e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
tbl2	2.600367e+02	2.851727e+02	2.294052e+02	6.381878e+01	
psmb7_2	2.343734e+02	2.840508e+02	2.185669e+02	5.799365e+01	
tpi	2.258686e+02	2.787804e+02	1.954628e+02	9.032453e+01	
his1.3	3.085513e+02	2.755890e+02	1.776564e+02	9.624604e+01	
cct7	2.754360e+02	2.742143e+02	1.910139e+02	7.181069e+01	
LASC021646	2.232949e+02	2.721283e+02	2.266607e+02	1.436723e+02	
sars	2.576588e+02	2.687609e+02	1.302117e+02	3.250157e+01	
LASC017315	1.577071e+02	2.648509e+02	2.346220e+02	3.443848e+01	
lec-1	2.210245e+02	2.641813e+02	1.879460e+02	1.041035e+01	
tmed2	2.713259e+02	2.623019e+02	1.976949e+02	9.322333e+01	
actr1b	2.180761e+02	2.617808e+02	1.994162e+02	6.237008e+01	
tpd52	2.134262e+02	2.611389e+02	2.222015e+02	8.201200e+01	
LASC005487	2.193178e+02	2.602361e+02	2.459089e+02	9.774299e+01	
pkm	2.350918e+02	2.600958e+02	1.632878e+02	3.673045e+01	
cox5b	2.133061e+02	2.577246e+02	2.063620e+02	7.273329e+01	
wos2	2.702216e+02	2.557866e+02	1.593677e+02	4.361417e+01	
sdcbp	1.367122e+02	2.529690e+02	4.585078e+02	3.655278e+01	
LASC002459	1.575321e+02	2.522940e+02	1.804402e+03	1.312907e+02	
mlc-c	1.989144e+02	2.507789e+02	2.806574e+02	1.115371e+02	
alyref2_2	2.459253e+02	2.406191e+02	9.052569e+01	1.956946e+01	
agap002499	2.348836e+02	2.366722e+02	1.390748e+02	4.827379e+01	
atpsyngamma	2.353828e+02	2.361431e+02	1.673095e+02	6.549387e+01	
sqstm1	1.864442e+02	2.348646e+02	1.927485e+02	6.172658e+01	
lgals4	9.067770e+01	2.342877e+02	5.253441e+02	8.355206e+01	
LASC021266	1.899634e+02	2.326952e+02	2.051916e+02	8.499233e+01	
ddx4_vasa	2.057381e+02	2.325473e+02	1.384829e+02	4.828150e+01	
LASC015439	1.775548e+02	2.302774e+02	2.502770e+02	6.892531e+01	
moe	1.884910e+02	2.285510e+02	2.136672e+02	7.630275e+01	
rho-1_3	1.550248e+02	2.278656e+02	2.010105e+02	2.352515e+01	
ybx1	2.383882e+02	2.262021e+02	1.657943e+02	7.409838e+01	
aldh1l2	2.191712e+02	2.261009e+02	9.238843e+01	2.304631e+01	
ael006169	1.320498e+02	2.254382e+02	3.804524e+02	1.296134e+02	
apl_8	3.453747e+02	2.250750e+02	4.197617e+02	2.660211e+02	
dld	2.252337e+02	2.214162e+02	1.221910e+02	4.995764e+01	
atpsyno	2.227189e+02	2.188587e+02	1.489423e+02	6.269613e+01	
cct3	2.051581e+02	2.176726e+02	1.494284e+02	4.544523e+01	
tmed6	2.432630e+02	2.134606e+02	1.004873e+02	2.641295e+01	
ppp2ca	1.993671e+02	2.124115e+02	1.931387e+02	1.045468e+02	
cct6	2.040175e+02	2.122738e+02	1.570614e+02	3.992166e+01	
nas-6	1.194740e+02	2.116466e+02	9.712121e+01	4.414467e+01	
nnmt_vlg200856	1.027978e+02	2.087188e+02	2.321938e+02	8.428360e+01	
cox5a	2.181458e+02	2.080486e+02	1.773621e+02	9.930467e+01	
eca	1.971728e+02	2.065277e+02	1.276108e+02	4.591378e+01	
psmb6	1.870921e+02	2.034198e+02	1.620039e+02	5.091968e+01	
LASC021418	1.576915e+02	2.023511e+02	3.190996e+02	1.622869e+02	
atp5f1d	2.179478e+02	2.014331e+02	1.197120e+02	4.652139e+01	
ppp2r1a	1.803173e+02	2.003364e+02	1.674347e+02	8.053355e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
aimp1	2.477578e+02	1.998239e+02	1.697078e+02	1.375162e+02	
LASC015247	1.208123e+02	1.987690e+02	3.210038e+03	1.882288e+03	
cndp2	1.243659e+02	1.986226e+02	2.558948e+02	7.736354e+01	
sorbs1_2	1.236150e+02	1.970412e+02	1.948828e+02	7.343749e+01	
pdia3	1.947074e+02	1.957402e+02	2.693346e+02	1.746619e+02	
kars1_2	1.744138e+02	1.938886e+02	1.553391e+02	5.939247e+01	
psme3	1.757385e+02	1.935151e+02	1.623703e+02	6.437860e+01	
psma4	1.898942e+02	1.923107e+02	1.271088e+02	4.345440e+01	
lap-2	1.676805e+02	1.908102e+02	1.429735e+02	4.742726e+01	
cct8	1.801201e+02	1.907828e+02	1.299543e+02	4.468331e+01	
hnrrp_2	1.747383e+02	1.900272e+02	1.711277e+02	5.335586e+01	
clta	1.651940e+02	1.888654e+02	1.461217e+02	5.867567e+01	
ube2l3_2	1.268216e+02	1.883582e+02	1.569782e+02	7.321934e+01	
gdi2_gdi1	1.537955e+02	1.880160e+02	1.348279e+02	4.943950e+01	
elf3j	2.121606e+02	1.849417e+02	1.032433e+02	3.040223e+01	
dlst	1.475523e+02	1.845627e+02	1.527249e+02	6.758332e+01	
eff_2	1.406535e+02	1.843336e+02	1.730040e+02	9.971943e+01	
aldh18a1	2.531466e+02	1.836862e+02	3.588153e+01	1.415092e+00	
hrb27c_2	1.522766e+02	1.835180e+02	2.052916e+02	8.807040e+01	
eff_9	1.585884e+02	1.813773e+02	1.462942e+02	6.427393e+01	
cpa_3	2.139978e+02	1.813579e+02	1.136580e+02	3.238752e+01	
stom	1.496481e+02	1.800400e+02	1.720773e+02	3.611352e+01	
crip1	1.121421e+02	1.799075e+02	4.883999e+02	5.160455e+02	
cat2_2	1.897120e+02	1.796085e+02	1.361236e+02	5.946195e+01	
hrb87f	1.938063e+02	1.782169e+02	1.153176e+02	3.819041e+01	
ndufa4	1.818287e+02	1.758889e+02	1.441839e+02	8.485642e+01	
tars1_tarsl2	2.132141e+02	1.754712e+02	1.233636e+02	7.903452e+01	
LASC022132	1.473741e+02	1.752538e+02	1.248536e+02	3.522916e+01	
sar1b	2.113294e+02	1.752294e+02	1.173104e+02	9.897770e+01	
sp-1339#prd34368.1_4	3.739237e+00	1.727780e+02	2.457643e+03	2.330909e+03	
gyg1_2	2.208442e+02	1.722740e+02	5.229854e+01	2.783090e+00	
atpsynb	1.531028e+02	1.712520e+02	1.275054e+02	4.524705e+01	
cpb_capzb	1.328737e+02	1.702595e+02	1.448032e+02	5.195346e+01	
uqcrb	1.802339e+02	1.693512e+02	1.394448e+02	5.902542e+01	
ndufb9	1.752272e+02	1.687197e+02	1.306034e+02	7.136109e+01	
cct5	1.478898e+02	1.655429e+02	1.421073e+02	5.483293e+01	
LASC003928	1.648231e+02	1.646801e+02	1.559610e+02	7.127568e+01	
ndufa5	1.509153e+02	1.634981e+02	1.411350e+02	4.717868e+01	
lamp1	1.444790e+02	1.626141e+02	1.299042e+02	6.381437e+01	
rho1_7	1.085290e+02	1.611562e+02	1.859371e+02	9.745645e+01	
enpp6_3	1.706120e+02	1.609641e+02	2.111448e+02	1.287146e+02	
llgl1	1.371714e+02	1.589143e+02	9.484759e+01	3.891253e+01	
cyc1	1.450329e+02	1.581542e+02	1.225060e+02	4.128172e+01	
uqcrc1	1.751568e+02	1.576393e+02	8.121852e+01	2.773465e+01	
rars	1.664962e+02	1.575407e+02	8.968954e+01	2.797655e+01	
elf3g-b	1.623132e+02	1.572146e+02	1.240467e+02	6.057821e+01	
set	1.545861e+02	1.548285e+02	1.219367e+02	5.406790e+01	
igf2bp1	9.926926e+01	1.532868e+02	1.340654e+02	5.297460e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
ehd1_2	1.117664e+02	1.522824e+02	1.068428e+02	3.025286e+01	
btf3l4	1.700244e+02	1.508782e+02	1.345898e+02	6.138680e+01	
metap1	1.461736e+02	1.503179e+02	9.350550e+01	2.995023e+01	
aldh7a1	1.470298e+02	1.490989e+02	1.222217e+02	5.578987e+01	
LASC018632	1.170296e+02	1.490125e+02	1.709004e+02	7.077570e+01	
nop58_2	1.769108e+02	1.487065e+02	7.185729e+01	2.572495e+01	
atox1	9.594630e+01	1.484490e+02	3.620904e+02	1.703261e+02	
pitpna	1.716358e+02	1.477333e+02	1.459553e+02	9.021379e+01	
sqh_2	1.494940e+02	1.477150e+02	1.759844e+02	8.521704e+01	
nsf1lc	1.240203e+02	1.456791e+02	9.229532e+01	2.683655e+01	
hspa4l_2	1.319117e+02	1.442715e+02	1.280083e+02	6.141951e+01	
acly	1.175405e+02	1.437509e+02	9.737545e+01	5.205520e+01	
fubp3_fubp1	1.133880e+02	1.433177e+02	1.293444e+02	5.149396e+01	
LASC005920	1.230577e+02	1.430310e+02	4.959052e+02	6.557318e+01	
ter94_vcp	1.101666e+02	1.428696e+02	1.484000e+02	6.387361e+01	
LASC016997	1.332940e+02	1.424388e+02	1.263653e+02	3.204567e+01	
lark	1.375017e+02	1.417707e+02	1.261704e+02	7.681644e+01	
LASC010482	1.465241e+02	1.415289e+02	1.071891e+02	3.115624e+01	
spg21	1.017649e+02	1.408624e+02	1.217248e+02	2.336850e+01	
LASC006288	7.476538e+01	1.397581e+02	1.203471e+02	2.300829e+01	
ap2m1	1.107715e+02	1.385479e+02	1.050839e+02	2.986022e+01	
slc7a6_slc7a8	1.292988e+02	1.363619e+02	5.439338e+01	2.947359e+01	
etfa	1.399906e+02	1.349763e+02	8.830702e+01	3.350658e+01	
arcn1	1.153843e+02	1.347869e+02	8.566852e+01	3.446644e+01	
fh	1.065980e+02	1.342602e+02	1.091644e+02	3.323898e+01	
uqcrc2	1.231987e+02	1.342524e+02	9.993174e+01	3.763488e+01	
ago3_ago2	7.662801e+01	1.335495e+02	9.121608e+01	2.103370e+01	
LASC012598	1.482771e+02	1.320697e+02	1.177392e+02	5.602609e+01	
ube2n_ubc-13	6.070631e+01	1.319743e+02	9.805138e+01	2.834745e+01	
cnp_2	9.891013e+01	1.299628e+02	6.191240e+01	2.234832e+01	
snx12	1.443188e+02	1.299299e+02	7.894200e+01	2.098467e+01	
ndufs4	1.441609e+02	1.299226e+02	7.811376e+01	2.796427e+01	
ran-4	1.046637e+02	1.287818e+02	9.356695e+01	3.203833e+01	
aimp2	1.004388e+02	1.286654e+02	8.578477e+01	2.484299e+01	
psma6	1.166416e+02	1.285036e+02	1.026455e+02	4.258928e+01	
nop60b	1.463575e+02	1.280859e+02	5.775841e+01	1.745643e+01	
anxa11	1.047984e+02	1.278274e+02	1.508138e+02	4.324253e+01	
acat1-b	1.274843e+02	1.275061e+02	6.890546e+01	2.141596e+01	
fib	1.658442e+02	1.265222e+02	7.527914e+01	2.254457e+01	
hadh	1.292184e+02	1.241174e+02	1.000673e+02	4.888321e+01	
prmt8	1.074875e+02	1.232350e+02	1.164125e+02	5.719898e+01	
zip	1.071451e+02	1.228602e+02	8.750638e+01	3.150209e+01	
atpsyncf6	1.162134e+02	1.221789e+02	7.238142e+01	2.498354e+01	
tuba1c	5.113909e+01	1.207570e+02	1.040423e+02	1.298181e+02	
sfpq	1.280007e+02	1.205366e+02	7.695873e+01	3.120092e+01	
erf1	1.424149e+02	1.173114e+02	8.225679e+01	4.029790e+01	
LASC017716	1.154116e+02	1.167461e+02	8.099414e+01	2.265053e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
phb	1.249876e+02	1.165909e+02	7.364000e+01	2.369586e+01	
sod1	9.306525e+01	1.165095e+02	9.461330e+01	3.906863e+01	
gs2_2	1.019605e+02	1.165032e+02	1.757867e+02	5.404834e+01	
plin2	9.640104e+01	1.148809e+02	2.753975e+02	1.283757e+02	
LASC009674	1.244825e+02	1.141307e+02	7.909732e+01	3.100416e+01	
uqcrq	1.263495e+02	1.134638e+02	6.483166e+01	1.681343e+01	
psma1	1.010566e+02	1.129923e+02	1.214189e+02	4.244031e+01	
l(2)efl_hspb1	5.562804e+01	1.127600e+02	1.814227e+02	1.750369e+01	
cltc	8.208446e+01	1.124246e+02	9.734010e+01	3.682779e+01	
gstd1_2	6.679775e+01	1.104521e+02	1.253918e+02	5.509489e+01	
ugp2_2	9.631013e+01	1.090531e+02	5.912473e+01	1.852552e+01	
lrrc47	1.031007e+02	1.087730e+02	7.648490e+01	2.391146e+01	
psmb5	1.036148e+02	1.084234e+02	9.443325e+01	3.498894e+01	
gbe1	9.548283e+01	1.083403e+02	9.299122e+01	3.277516e+01	
LASC017623	1.045105e+02	1.073041e+02	7.171765e+01	3.001400e+01	
rpn11	1.200407e+02	1.064932e+02	7.584429e+01	3.768058e+01	
cubn_3	7.272586e+01	1.063708e+02	6.329589e+01	2.457881e+01	
snx2	7.289933e+01	1.058317e+02	1.485063e+02	5.549082e+01	
efhd2	6.877881e+01	1.049192e+02	1.237748e+02	4.398867e+01	
qars	1.063465e+02	1.044294e+02	6.271900e+01	2.095358e+01	
ddx1	9.341208e+01	1.036616e+02	6.795102e+01	1.563402e+01	
LASC013621	8.086650e+01	1.031841e+02	1.091267e+02	1.460356e+01	
psmc6	8.508903e+01	1.027368e+02	7.903811e+01	2.354373e+01	
elf3-s8	1.155424e+02	1.026851e+02	7.656470e+01	3.014284e+01	
cfad	6.791216e+01	1.013106e+02	2.679622e+02	3.192747e+01	
atp5mg	9.266040e+01	1.008999e+02	6.596976e+01	2.426971e+01	
psmc5	1.023740e+02	1.008268e+02	7.022781e+01	2.741942e+01	
septin7	6.302202e+01	1.007377e+02	6.034877e+01	1.325639e+01	
rtca	9.042580e+01	1.005755e+02	6.848368e+01	1.954278e+01	
apip	1.022886e+02	1.001083e+02	7.469867e+01	4.281927e+01	
bai_tmed10	9.059270e+01	9.782703e+01	6.603232e+01	2.565807e+01	
ewsr1	1.062146e+02	9.701642e+01	9.941531e+01	4.598460e+01	
aass	6.849060e+01	9.686104e+01	7.658818e+01	4.154544e+01	
hibadh	1.098995e+02	9.641029e+01	5.889159e+01	2.167880e+01	
ptcd2	9.442420e+01	9.539906e+01	5.725533e+01	1.664743e+01	
zan_srfbp1	4.138588e+01	9.457160e+01	1.239629e+02	7.529002e+01	
ndufs1	8.939814e+01	9.450307e+01	5.497204e+01	2.029935e+01	
ube2v2	7.644467e+01	9.436383e+01	8.431754e+01	3.751521e+01	
LASC007277	1.131293e+02	9.422222e+01	5.609082e+01	2.305191e+01	
hnrnpk	1.003633e+02	9.362009e+01	6.640735e+01	3.218040e+01	
ywhae	8.663416e+01	9.351552e+01	7.991184e+01	3.889514e+01	
psmd8	7.911925e+01	9.334735e+01	8.984820e+01	4.291330e+01	
cd109	1.289019e+01	9.323696e+01	3.237037e+02	1.229451e+02	
vha26	8.215669e+01	9.288580e+01	8.551719e+01	3.145199e+01	
sirt5	8.143989e+01	9.256279e+01	6.124971e+01	2.738770e+01	
dnajc22	8.776560e+01	9.221642e+01	3.858384e+01	1.858188e+01	
atpsynd	7.814865e+01	9.150634e+01	7.787525e+01	3.917891e+01	
ppib_cyn-5	8.071204e+01	9.128913e+01	8.945018e+01	2.978957e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
ech1_2	8.488942e+01	9.073216e+01	5.690459e+01	3.973948e+01	
srp68	8.128537e+01	9.066824e+01	6.892770e+01	2.754457e+01	
LASC005698	6.331279e+01	9.062936e+01	7.578031e+01	2.441270e+01	
copa	8.389673e+01	9.035844e+01	8.198411e+01	4.781440e+01	
abat	6.930849e+01	9.007232e+01	8.832923e+01	5.673742e+01	
eif3a_2	9.453820e+01	8.912070e+01	5.066269e+01	1.821771e+01	
ndufs3	9.938402e+01	8.909766e+01	6.066391e+01	2.139627e+01	
ciz1	1.055464e+02	8.851883e+01	5.355642e+01	2.162193e+01	
rpn6	7.855348e+01	8.798186e+01	5.846747e+01	1.506468e+01	
sqh	5.511803e+01	8.777899e+01	1.091232e+02	2.440329e+01	
alpha-spec	6.311172e+01	8.720829e+01	6.558921e+01	1.849531e+01	
srsf12	8.235197e+01	8.684352e+01	7.476587e+01	2.213616e+01	
cbx3	8.875208e+01	8.675697e+01	5.818209e+01	2.382479e+01	
LASC011933	7.753796e+01	8.600627e+01	3.543027e+01	7.551167e+00	
vha55	6.212748e+01	8.553412e+01	7.647677e+01	3.103415e+01	
pebp1	6.669090e+01	8.525573e+01	8.632904e+01	3.814313e+01	
LASC011232	4.884040e+01	8.498752e+01	1.038646e+02	2.986354e+01	
mdh1	8.629178e+01	8.493086e+01	6.396042e+01	3.447188e+01	
gspt1	8.365658e+01	8.489730e+01	5.156848e+01	2.175650e+01	
ptbp1	5.810101e+01	8.416054e+01	9.621084e+01	9.883217e+01	
his2av	8.440260e+01	8.411422e+01	5.628638e+01	1.718303e+01	
tdh	8.826686e+01	8.390737e+01	6.825677e+01	1.555172e+01	
ipgm-1	6.071637e+01	8.383101e+01	6.671740e+01	2.755430e+01	
psma3	8.721430e+01	8.316409e+01	8.909233e+01	4.183688e+01	
rab2a	8.519815e+01	8.294054e+01	5.909981e+01	2.574271e+01	
cap2	5.706197e+01	8.251682e+01	1.301779e+02	4.715372e+01	
ldlr	5.804557e+01	8.248426e+01	5.862062e+01	3.386819e+01	
LASC016532	8.644582e+01	8.191672e+01	6.986105e+01	2.125341e+01	
psmd13	8.280502e+01	8.119586e+01	5.473992e+01	2.152439e+01	
gstm2	5.771469e+01	7.988369e+01	4.217416e+01	4.079762e+01	
psmd2	6.822474e+01	7.958177e+01	8.210673e+01	3.080395e+01	
ppif	6.071405e+01	7.908629e+01	1.029455e+02	1.551931e+01	
hsp83_hsp90aa1	7.171956e+01	7.835811e+01	6.289825e+01	1.400999e+01	
ndufa13	8.625203e+01	7.789552e+01	5.096613e+01	2.032997e+01	
slc9a3r1	6.891630e+01	7.744544e+01	9.687356e+01	7.233646e+01	
ael011789	7.879621e+01	7.726262e+01	6.017310e+01	2.667403e+01	
pkm_3	3.513977e+01	7.708408e+01	1.087029e+02	2.168607e+01	
copg2	5.823046e+01	7.643207e+01	5.557295e+01	3.169998e+01	
echs1	7.513797e+01	7.601499e+01	6.231691e+01	2.468592e+01	
LASC004161	7.566736e+01	7.588496e+01	5.862761e+01	2.171602e+01	
eif4b	6.829053e+01	7.538261e+01	4.648402e+01	2.084857e+01	
calr_crt	7.322388e+01	7.496886e+01	2.186318e+02	2.052425e+02	
hsrb1_3	7.470092e+01	7.480041e+01	1.282144e+01	1.044769e+00	
abce1	8.137014e+01	7.477455e+01	5.779245e+01	2.398338e+01	
LASC011347	9.529182e+01	7.453278e+01	9.696494e+01	5.938204e+01	
phb2	7.656571e+01	7.397891e+01	5.792648e+01	2.376905e+01	
glyp	5.483230e+01	7.392150e+01	3.782477e+01	6.385443e+00	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
gfpt2	5.393693e+01	7.319293e+01	8.106268e+01	2.116900e+01	
grn_6	1.932894e+01	7.279315e+01	5.964670e+02	6.065426e+01	
snd1	9.456420e+01	7.251202e+01	4.072289e+01	1.437295e+01	
folr3	5.198165e+01	7.234396e+01	2.439529e+01	5.977628e+00	
eprs	8.073413e+01	7.224590e+01	3.288394e+01	4.269090e+00	
prdx6	7.468277e+01	7.205775e+01	5.965702e+01	2.743465e+01	
eif3f	7.742899e+01	7.179471e+01	6.007022e+01	2.514557e+01	
snrpb	5.610393e+01	7.149826e+01	6.384668e+01	2.348668e+01	
psma5	7.037490e+01	7.134842e+01	6.414223e+01	2.216971e+01	
idh3a	7.515449e+01	7.112476e+01	4.199210e+01	1.187299e+01	
rad23b	5.869336e+01	7.105156e+01	5.334460e+01	1.835764e+01	
SpICE-LMa4	2.375463e+01	7.095375e+01	4.965271e+02	8.231669e+02	
psmc2	6.571048e+01	7.046054e+01	6.363098e+01	2.713948e+01	
LASC018834	4.423372e+01	6.966961e+01	5.898244e+01	1.106786e+01	
tbb-4_tubb4b	3.937167e+01	6.952657e+01	8.592298e+01	2.395230e+01	
aldh4a1	5.722946e+01	6.745350e+01	4.159913e+01	1.761169e+01	
stip1	5.681946e+01	6.732087e+01	5.363025e+01	1.573098e+01	
hsp60a	7.154050e+01	6.702567e+01	5.794820e+01	1.258654e+01	
lap3	7.334350e+01	6.645218e+01	6.106390e+01	1.762392e+01	
LASC015638	5.356170e+01	6.641860e+01	8.186537e+01	6.605260e+01	
psmd12	6.576671e+01	6.614811e+01	4.504458e+01	1.856992e+01	
abhd12_2	4.528002e+01	6.609558e+01	1.205903e+02	6.136898e+01	
cox6b1	6.577037e+01	6.583570e+01	5.097794e+01	2.106755e+01	
fas2_3	4.755633e+01	6.543802e+01	5.063426e+01	3.282707e+01	
LASC016132	5.340372e+01	6.536765e+01	4.439647e+01	1.588585e+01	
vdac2_2	7.464554e+01	6.422797e+01	4.514230e+01	1.603067e+01	
tth-1_2	8.739364e+01	6.410479e+01	9.051820e+01	5.622251e+01	
sod-2	5.594665e+01	6.410249e+01	5.058743e+01	2.401472e+01	
nurf-38_ipp1	7.519700e+01	6.312916e+01	4.279613e+01	2.592648e+01	
psmb3	5.938353e+01	6.263497e+01	5.208502e+01	1.795100e+01	
hsd17b10_2	7.437649e+01	6.233733e+01	4.228278e+01	2.051115e+01	
ndufv2	6.032751e+01	6.229097e+01	3.805934e+01	1.584048e+01	
lman1	4.668139e+01	6.196479e+01	6.293577e+01	3.766951e+01	
nop10	5.841275e+01	6.195133e+01	3.370732e+01	1.210905e+01	
hce_hcf	2.611374e+02	6.190657e+01	9.023178e+01	5.817548e+01	
aldh2_2	4.589182e+01	6.188997e+01	6.113566e+01	2.316680e+01	
csga	5.736359e+01	6.125217e+01	3.563938e+01	1.199614e+01	
arl8ba_arl8b	5.172575e+01	6.108076e+01	5.995118e+01	2.168371e+01	
LASC006772	4.089922e+01	6.102184e+01	5.701671e+01	3.801887e+01	
ppib_cypb	7.539340e+01	6.072080e+01	9.299532e+01	3.938483e+01	
gbeta1	4.597765e+01	6.066580e+01	5.233803e+01	2.824421e+01	
rab5b	5.558079e+01	6.065313e+01	5.227638e+01	2.694479e+01	
hadha	5.508113e+01	6.063261e+01	5.169238e+01	1.939663e+01	
nhp2	6.547057e+01	6.022769e+01	2.955704e+01	9.356746e+00	
ogdh	4.868809e+01	6.022255e+01	3.728676e+01	1.151532e+01	
rab7_rab7a_2	4.951889e+01	6.019980e+01	6.921317e+01	2.439222e+01	
beta-spec_2	3.707745e+01	5.994838e+01	4.996695e+01	1.839844e+01	
cpt1a_3	4.092493e+01	5.983268e+01	4.905395e+01	3.868263e+00	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
LASC013620	6.867557e+00	5.973722e+01	4.508155e+02	3.908906e+01	
ran_2	6.051251e+01	5.958899e+01	4.809012e+01	1.761870e+01	
tmco1	5.785415e+01	5.958786e+01	3.997472e+01	1.558391e+01	
LASC011026	5.773204e+01	5.941696e+01	4.823195e+01	2.012368e+01	
copb1	4.743590e+01	5.904487e+01	4.353677e+01	1.947772e+01	
kyat3	6.237056e+01	5.888951e+01	4.223796e+01	1.754958e+01	
tfg	4.801782e+01	5.872849e+01	6.180961e+01	3.125917e+01	
myo6_jar	3.027185e+01	5.843367e+01	3.140122e+01	7.962545e+00	
atp6v0a1	3.724972e+01	5.839219e+01	6.206730e+01	1.900210e+01	
pa2g4	6.165575e+01	5.834443e+01	3.996979e+01	1.837533e+01	
oplah	4.634534e+01	5.793159e+01	4.314766e+01	1.183588e+01	
dpp3_3	6.439822e+01	5.753319e+01	3.091482e+01	9.531837e+00	
nolc1	4.982688e+01	5.746499e+01	3.071224e+01	1.150665e+01	
anp32a_2	5.865375e+01	5.706500e+01	3.725154e+01	1.725966e+01	
sacm1l_sacm1lb	5.010287e+01	5.703651e+01	4.083427e+01	1.921710e+01	
sme	6.276026e+01	5.680049e+01	4.661024e+01	1.687636e+01	
cct4	5.417338e+01	5.678900e+01	5.226168e+01	2.184312e+01	
gpx4	5.638429e+01	5.644036e+01	5.184425e+01	2.640228e+01	
rala	3.872818e+01	5.640182e+01	6.877328e+01	2.251780e+01	
sec22b	5.825838e+01	5.639333e+01	4.244111e+01	2.013334e+01	
vapa	4.855326e+01	5.634233e+01	4.673524e+01	2.410704e+01	
tufm	5.353514e+01	5.623669e+01	4.307575e+01	1.966942e+01	
vat1l_2	4.431537e+01	5.622088e+01	3.869042e+01	1.842298e+01	
LASC013486	4.846077e+01	5.573778e+01	3.647123e+02	2.746604e+01	
rpt2_2	4.765077e+01	5.568100e+01	4.451635e+01	2.128589e+01	
gcdh	5.759013e+01	5.554328e+01	4.251725e+01	1.836434e+01	
chmp4b	4.749319e+01	5.551842e+01	4.293401e+01	1.703807e+01	
ndufa8	4.728749e+01	5.534124e+01	4.067758e+01	1.917525e+01	
LASC018874	4.864849e+01	5.506808e+01	4.502678e+01	2.195737e+01	
nop58	6.634553e+01	5.484493e+01	4.190524e+01	1.257612e+01	
serca_2	4.320967e+01	5.456573e+01	3.852995e+01	2.294934e+01	
psmd4	4.995928e+01	5.450986e+01	4.804947e+01	1.923888e+01	
smo-1	5.619196e+01	5.410273e+01	4.541218e+01	1.634652e+01	
edf1	5.217834e+01	5.399855e+01	4.295178e+01	2.094343e+01	
ndufs7	5.065442e+01	5.360101e+01	4.030374e+01	1.745056e+01	
sfpq_2	5.167675e+01	5.323942e+01	3.825543e+01	1.725390e+01	
serbp1_2	6.140815e+01	5.300823e+01	4.017149e+01	2.629200e+01	
sgtb	4.601519e+01	5.279309e+01	6.659661e+01	3.896962e+01	
ndufa7	4.770596e+01	5.253706e+01	4.268210e+01	1.956963e+01	
pgk	4.139920e+01	5.234978e+01	4.325333e+01	1.400946e+01	
snx6	3.842433e+01	5.209817e+01	5.061964e+01	1.497524e+01	
pdcd4	5.439263e+01	5.196755e+01	3.526923e+01	2.057870e+01	
nd-42	5.118541e+01	5.182413e+01	3.602909e+01	1.593640e+01	
chmp2a	4.317221e+01	5.177217e+01	5.352740e+01	1.620345e+01	
nuch2	3.912712e+01	5.162297e+01	9.222664e+01	4.944803e+01	
f42h10.3_lasp1	4.153856e+01	5.119875e+01	3.072713e+01	1.424082e+01	
dars	5.467924e+01	5.100848e+01	4.259061e+01	2.166764e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
gdh	5.492212e+01	5.092018e+01	3.526133e+01	2.143183e+01	
uso1	4.004902e+01	4.996905e+01	3.091968e+01	1.463051e+01	
ssb	5.833952e+01	4.996519e+01	2.341436e+01	8.762981e+00	
txnl1	4.235415e+01	4.995751e+01	4.815285e+01	1.674944e+01	
cct1	4.441782e+01	4.973651e+01	4.975602e+01	2.151848e+01	
fk506-bp2_fkbp1b	4.474364e+01	4.966365e+01	4.822113e+01	1.968523e+01	
psma2	4.590111e+01	4.909522e+01	4.383675e+01	1.205331e+01	
psma8_2	4.835945e+01	4.898605e+01	4.981532e+01	2.269853e+01	
dsp1	4.586687e+01	4.878897e+01	4.181851e+01	3.574085e+01	
rtcb	4.225887e+01	4.877824e+01	3.066427e+01	7.273486e+00	
impa1	6.657362e+01	4.875756e+01	2.692760e+01	1.081585e+01	
cd109_4	3.233081e+01	4.863896e+01	4.973256e+01	2.007409e+01	
zmpste24	3.558880e+01	4.811135e+01	3.880715e+01	1.007091e+01	
plekhf2	3.876339e+01	4.756501e+01	4.852625e+01	1.763825e+01	
hadhb	3.606152e+01	4.674371e+01	5.543662e+01	2.646024e+01	
etfb	4.728448e+01	4.653879e+01	3.770584e+01	1.685789e+01	
blvrb	4.586195e+01	4.637966e+01	4.710719e+01	2.573095e+01	
stx7	3.427806e+01	4.567480e+01	4.640815e+01	1.735725e+01	
prdx4_prdx1	5.026473e+01	4.540836e+01	3.388453e+01	1.237031e+01	
hsa5_2	3.767501e+01	4.538458e+01	1.347657e+02	8.652103e+01	
tecr_art-1	3.336173e+01	4.527809e+01	5.497461e+01	4.688180e+01	
eif4ebp2_eif4ebp1	3.583421e+01	4.511692e+01	3.688471e+01	2.004171e+01	
scsalpha1_os07g0577700	3.537005e+01	4.508735e+01	5.550133e+01	2.172550e+01	
ehd1_3	4.478822e+01	4.462509e+01	4.063636e+01	1.912733e+01	
psmb2	4.384369e+01	4.460300e+01	3.659089e+01	1.357856e+01	
atpalpha	3.971933e+01	4.458735e+01	5.041428e+01	4.630893e+01	
LASC016362	8.050800e+01	4.426125e+01	3.408435e+01	5.527464e+00	
hnrnpf_hnrnp2	4.480428e+01	4.422726e+01	3.663038e+01	2.380159e+01	
slc8a2	2.957003e+01	4.388509e+01	3.975127e+01	1.441417e+01	
ndufs6	4.082590e+01	4.350575e+01	3.823201e+01	1.690996e+01	
rbm4_rbm14	3.918220e+01	4.334960e+01	4.770984e+01	2.206269e+01	
LASC013967	3.841354e+01	4.318098e+01	2.503270e+02	6.670423e+01	
cs	4.919675e+01	4.308659e+01	3.119728e+01	1.577953e+01	
hibch	3.601622e+01	4.298430e+01	2.979828e+01	1.217613e+01	
LASC010684	1.588230e+01	4.291544e+01	6.211780e+01	1.465864e+01	
LASC009347	1.604242e+01	4.286305e+01	1.507582e+02	2.502160e+01	
cg3556_15	3.938035e+01	4.259192e+01	4.899107e+01	4.257804e+01	
uchl3_3	3.372649e+01	4.247213e+01	5.828436e+01	3.419913e+01	
park7	3.847765e+01	4.224926e+01	3.317357e+01	1.739806e+01	
elavl2	4.492146e+01	4.203004e+01	3.598782e+01	1.616730e+01	
cnm3	3.547173e+01	4.089217e+01	2.419417e+01	1.526902e+01	
LASC001129	3.299275e+01	4.087754e+01	2.365130e+01	1.184786e+01	
snu13	5.000885e+01	4.066397e+01	3.314528e+01	1.317404e+01	
gstm3	3.365764e+01	4.050708e+01	5.321926e+01	3.399145e+01	
nnt	3.491352e+01	4.033388e+01	2.032815e+01	7.584932e+00	
hsa5	4.203994e+01	4.026707e+01	3.992689e+01	1.837337e+01	
cope	4.422217e+01	4.011999e+01	3.310225e+01	1.348468e+01	
hnrnp_3	3.826592e+01	3.985951e+01	6.021063e+01	3.520104e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proximal
jafrac1	5.438229e+01	3.977882e+01	8.163891e+01	6.539529e+01	
lrp2_6	2.253935e+01	3.933907e+01	4.698338e+01	3.529299e+01	
snr-4	3.895682e+01	3.916578e+01	2.807272e+01	1.014823e+01	
esd_2	4.180114e+01	3.879872e+01	2.396903e+01	6.528271e+00	
prep	3.185138e+01	3.872937e+01	2.973353e+01	1.072527e+01	
acat1_2	4.011805e+01	3.857834e+01	3.676379e+01	1.923890e+01	
psmd1_rpn2	3.240286e+01	3.847812e+01	4.023171e+01	1.730169e+01	
ndufb4	3.346785e+01	3.833120e+01	3.515247e+01	1.624395e+01	
dcxr_2	3.774044e+01	3.799987e+01	4.366673e+01	1.295237e+01	
ppp1cc_ppp1cc-b	3.192852e+01	3.797087e+01	3.886377e+01	1.530708e+01	
kpnb1	3.182192e+01	3.784166e+01	2.668647e+01	8.962493e+00	
glyp_2	3.022686e+01	3.780781e+01	2.827515e+01	1.267898e+01	
ppp1cb	3.199767e+01	3.772856e+01	2.968729e+01	1.661394e+01	
cpt1a_2	2.500945e+01	3.766321e+01	5.047276e+01	1.475184e+01	
rpn1	2.849580e+01	3.761877e+01	5.778677e+01	3.699984e+01	
LASC002299	2.459649e+01	3.759424e+01	3.916109e+01	1.580766e+01	
hnrnpl	2.743165e+01	3.750986e+01	3.900789e+01	1.857381e+01	
mapre3	3.155023e+01	3.720415e+01	3.261323e+01	1.484909e+01	
cdc42_2	2.825073e+01	3.693582e+01	3.967733e+01	1.252328e+01	
ap2b1	2.427000e+01	3.678738e+01	3.379480e+01	1.093212e+01	
lrpap1	3.003218e+01	3.624698e+01	2.287822e+01	7.870042e+00	
nop56	4.465963e+01	3.623136e+01	1.867586e+01	6.973298e+00	
cand1	2.813020e+01	3.621637e+01	2.901548e+01	1.270289e+01	
tm_fabg	3.488533e+01	3.615779e+01	4.515798e+01	2.514454e+01	
mtap	2.072651e+01	3.608022e+01	4.285243e+01	6.065151e+00	
c05d11.1	2.964860e+01	3.605890e+01	3.140579e+01	1.286390e+01	
idh1	2.464809e+01	3.602077e+01	6.765352e+01	5.587103e+01	
acads	4.209147e+01	3.598246e+01	4.814411e+01	3.388852e+01	
suclg2	3.726957e+01	3.583283e+01	2.138252e+01	5.493142e+00	
itga9	3.007149e+01	3.575648e+01	2.789115e+01	7.426204e+00	
LASC005335	4.350776e+01	3.569367e+01	5.294944e+01	3.488582e+01	
pxn	2.473284e+01	3.534729e+01	4.338564e+01	2.317986e+01	
glrx3	3.551268e+01	3.519370e+01	2.197946e+01	8.174819e+00	
scfd1	3.348892e+01	3.499439e+01	2.371327e+01	9.853520e+00	
ap1m1	3.263187e+01	3.486944e+01	2.991261e+01	1.534345e+01	
rap1b	2.394520e+01	3.485220e+01	3.648070e+01	1.488529e+01	
pebp1_2	3.333586e+01	3.480952e+01	4.486287e+01	4.063422e+01	
capsl_2	3.097090e+01	3.471788e+01	5.060618e+01	1.092153e+01	
ablim1_ablim2	2.248603e+01	3.471648e+01	2.611547e+01	1.063240e+01	
aprt	2.968405e+01	3.453176e+01	2.854115e+01	1.205866e+01	
acadvl	3.174105e+01	3.451145e+01	2.766449e+01	9.213342e+00	
LASC001932	2.533004e+01	3.442216e+01	3.575804e+01	1.268820e+01	
igf2r	2.297534e+01	3.400647e+01	4.201370e+01	2.478652e+01	
spac4h3.07c	3.344369e+01	3.380164e+01	2.079791e+01	9.403682e+00	
anp32a	3.843241e+01	3.365801e+01	2.556382e+01	1.441416e+01	
hcd	1.350436e+02	3.341994e+01	4.448454e+01	3.045497e+01	
ehd1	2.738433e+01	3.336364e+01	2.272228e+01	1.177159e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
txn2	4.184542e+01	3.330447e+01	2.506127e+01	1.114805e+01	
hcg	1.519112e+02	3.310573e+01	4.961570e+01	3.287762e+01	
hca	1.445333e+02	3.281068e+01	5.274642e+01	3.247807e+01	
twf2	3.180715e+01	3.262304e+01	1.993250e+01	6.646324e+00	
vha14	2.615684e+01	3.218506e+01	3.697466e+01	1.500641e+01	
arf4	3.167071e+01	3.192456e+01	3.651061e+01	2.243113e+01	
alkbh3	2.580352e+01	3.191274e+01	2.269057e+01	7.240337e+00	
derl1	3.301273e+01	3.179253e+01	2.412707e+01	1.341644e+01	
pgd	2.534071e+01	3.099815e+01	8.307781e+01	7.186641e+01	
rab18	2.678686e+01	3.075850e+01	2.733737e+01	1.363738e+01	
hspe1	3.454576e+01	3.058355e+01	3.152441e+01	1.102808e+01	
rab7_rab7a	2.193289e+01	3.036389e+01	2.928356e+01	1.147394e+01	
zk1073.1_2	3.349108e+01	3.015462e+01	2.109850e+01	9.337559e+00	
slc25a11	2.966141e+01	3.008895e+01	2.210389e+01	1.148129e+01	
plxdc1	2.215509e+01	2.987985e+01	2.587491e+01	1.704023e+01	
srsf4_3	3.120838e+01	2.955157e+01	3.165517e+01	1.495057e+01	
ivd	2.771881e+01	2.943760e+01	2.343684e+01	6.350660e+00	
acdh-6_psmf1	2.417311e+01	2.940827e+01	2.948463e+01	1.591059e+01	
lgmn_2	2.065986e+01	2.936269e+01	5.428468e+01	2.298509e+01	
pdc6	2.083108e+01	2.933078e+01	3.634102e+01	1.083094e+01	
hsp90b1	2.505183e+01	2.932947e+01	9.205142e+01	6.506417e+01	
prdx3	2.137156e+01	2.882403e+01	2.389341e+01	1.204919e+01	
spa	1.567259e+01	2.861829e+01	9.927107e+03	2.215582e+04	
eif1a	2.749027e+01	2.856433e+01	2.453036e+01	1.344799e+01	
immt	2.932206e+01	2.835901e+01	2.149271e+01	8.327531e+00	
copb2	2.869017e+01	2.835799e+01	2.851645e+01	1.518394e+01	
dag1	2.351968e+01	2.828837e+01	1.267371e+01	5.792853e+00	
LASC009784	1.593736e+00	2.822944e+01	1.055895e+03	1.747359e+03	
eif3k	3.884074e+01	2.808525e+01	1.472230e+01	5.719697e+00	
gar1	3.192529e+01	2.807928e+01	1.996690e+01	6.790165e+00	
tln2	2.005436e+01	2.805841e+01	1.794516e+01	5.375099e+00	
lrrc59	3.022349e+01	2.803790e+01	1.841812e+01	4.815927e+00	
vha36-1	2.759574e+01	2.802778e+01	2.559224e+01	1.168610e+01	
acaa2	2.573686e+01	2.794760e+01	3.410584e+01	1.599571e+01	
cyt-b5	2.637135e+01	2.785106e+01	2.847554e+01	9.811713e+00	
clqbp	3.117850e+01	2.778511e+01	2.018273e+01	6.725072e+00	
piwil1	2.121806e+01	2.748965e+01	2.184846e+01	8.034032e+00	
ndufaf7	2.454606e+01	2.709160e+01	2.252546e+01	9.544454e+00	
eif2a	3.113171e+01	2.707566e+01	1.418652e+01	4.278179e+00	
b52	3.546756e+01	2.703211e+01	2.283689e+01	1.077677e+01	
hm13	2.407681e+01	2.695393e+01	1.670543e+01	6.981176e+00	
mthfs_2	2.511653e+01	2.674764e+01	1.464386e+01	3.720394e+00	
dhc64c	1.924972e+01	2.671667e+01	2.548044e+01	1.082060e+01	
fk506-bp1_fpr4	3.406146e+01	2.662512e+01	1.564293e+01	5.247026e+00	
ahcyl1	2.093076e+01	2.639916e+01	2.524893e+01	1.192594e+01	
cbr1	2.549862e+01	2.601794e+01	3.210306e+01	1.212718e+01	
igf2bp1_2	2.061797e+01	2.594905e+01	2.380781e+01	1.112466e+01	
dpm1	2.621799e+01	2.566995e+01	2.121958e+01	1.026341e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
snrpg	2.716009e+01	2.559826e+01	2.430332e+01	1.398064e+01	
arhgdb_ arhgdia	2.641521e+01	2.552351e+01	2.576240e+01	1.445142e+01	
arp3	2.403648e+01	2.542175e+01	2.341235e+01	9.544592e+00	
pcd_pcbd2	2.031703e+01	2.500501e+01	1.914677e+01	1.092935e+01	
vhaac39-1	1.636389e+01	2.493476e+01	3.495847e+01	1.610175e+01	
paip1	2.370178e+01	2.476280e+01	2.034169e+01	1.127436e+01	
rtraf	2.488584e+01	2.459971e+01	1.683294e+01	6.352639e+00	
dlat	2.148463e+01	2.455740e+01	2.109023e+01	7.631312e+00	
npc2a_2	4.283612e+01	2.453840e+01	2.665665e+02	4.822338e+02	
acad8	2.236383e+01	2.414572e+01	2.950506e+01	1.423691e+01	
hcb	9.273757e+01	2.389346e+01	3.253093e+01	2.101341e+01	
tmem68	1.661742e+01	2.388932e+01	4.871234e+01	9.757420e+01	
scepe1	1.524697e+01	2.371077e+01	2.353487e+01	9.983116e+00	
ist1	1.779513e+01	2.362848e+01	3.546051e+01	1.727298e+01	
rpt2	1.903765e+01	2.350840e+01	1.738819e+01	7.900515e+00	
nxnl2	1.710826e+01	2.345014e+01	2.168133e+01	7.799145e+00	
myg1	2.156361e+01	2.343665e+01	1.935729e+01	6.823743e+00	
hsp70b2_7	9.598573e+00	2.336965e+01	8.838892e+00	6.567539e-01	
syncrip_2	2.176544e+01	2.335023e+01	2.335882e+01	1.576529e+01	
xirp2	2.141675e+01	2.313588e+01	1.800413e+01	7.806293e+00	
fkbp7_fkbp3_fpr2	3.152804e+01	2.310617e+01	4.878976e+01	5.151777e+01	
psmc3	1.936561e+01	2.307890e+01	1.831887e+01	6.389740e+00	
g3bp2	1.731021e+01	2.295514e+01	1.649813e+01	8.946858e+00	
eif2s1	2.395330e+01	2.263764e+01	1.431023e+01	4.937966e+00	
gstz1_2	2.382683e+01	2.238580e+01	1.836802e+01	4.531537e+00	
psmb4	2.123782e+01	2.205744e+01	1.629771e+01	5.131187e+00	
hts	1.614355e+01	2.195045e+01	1.775546e+01	6.628000e+00	
dhtkd1	2.049912e+01	2.192934e+01	1.706737e+01	7.290957e+00	
rbm4	1.582991e+01	2.176473e+01	2.584735e+01	1.547924e+01	
tmx1	2.012372e+01	2.153455e+01	2.709595e+01	1.169532e+01	
ghra_2	1.852339e+01	2.153177e+01	1.283925e+01	4.101743e+00	
lypla2	1.886200e+01	2.149204e+01	1.960915e+01	9.578196e+00	
mtch2	1.348310e+01	2.132197e+01	1.833930e+01	4.617199e+00	
mys_2	1.327780e+01	2.123519e+01	2.272430e+01	7.471649e+00	
l(2)efl_5	1.952480e+01	2.098152e+01	3.102190e+01	4.101967e+00	
mecr	1.782893e+01	2.083685e+01	1.430952e+01	4.758940e+00	
canx	1.585523e+01	2.081514e+01	2.336303e+01	1.558338e+01	
ndufs2	1.766923e+01	2.047415e+01	1.677124e+01	6.923600e+00	
exc-4	1.964291e+01	2.035692e+01	1.440608e+01	8.078867e+00	
ctsa	1.399026e+01	2.033682e+01	4.691882e+01	1.084408e+01	
plxna4_2	1.439529e+01	2.020644e+01	1.857460e+01	1.232156e+01	
ddost	1.673370e+01	1.985662e+01	2.106953e+01	1.002952e+01	
hnrnp	1.810523e+01	1.973381e+01	1.759234e+01	8.572268e+00	
gpd1	1.368820e+01	1.932434e+01	2.571923e+01	1.482730e+01	
LASC019081	2.193947e+01	1.930001e+01	3.527971e+01	1.632230e+01	
arl1	1.844365e+01	1.922875e+01	1.787382e+01	1.202499e+01	
vhaa	1.519538e+01	1.921325e+01	2.792990e+01	1.221641e+01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
LASC013820	1.536878e+01	1.889300e+01	1.484402e+01	9.028514e+00	
hcc	7.659100e+01	1.885971e+01	2.505773e+01	1.720691e+01	
eif4a2	1.685809e+01	1.884084e+01	1.327370e+01	7.789470e+00	
crat	1.626557e+01	1.881058e+01	1.561874e+01	6.790666e+00	
lin7b	1.203958e+01	1.868257e+01	2.163006e+01	1.134944e+01	
hsp90ab1	1.768222e+01	1.865640e+01	1.134330e+01	4.260530e+00	
taldo1_taldo	1.894685e+01	1.862604e+01	1.575118e+01	9.604242e+00	
alpha-cat	1.343294e+01	1.856643e+01	2.492245e+01	9.772965e+00	
actg1	8.165669e+00	1.853913e+01	3.861314e+01	3.003290e+00	
snrpd3_2	2.028064e+01	1.822652e+01	1.849700e+01	8.189534e+00	
rpn2	1.098532e+01	1.819453e+01	2.715243e+01	1.778513e+01	
mrh1	9.787246e+00	1.813367e+01	1.586901e+01	5.345801e+00	
larpl	1.851545e+01	1.800668e+01	1.397920e+01	7.404584e+00	
gly3	1.394788e+01	1.795995e+01	1.590319e+01	6.590394e+00	
epb41l3_2	1.856868e+01	1.794913e+01	1.622950e+01	8.974977e+00	
cpm_2	2.287157e+01	1.794485e+01	1.915363e+01	1.837734e+00	
lman2	1.678683e+01	1.791241e+01	1.506612e+01	7.342953e+00	
myef2_2	1.526355e+01	1.782525e+01	1.482493e+01	8.233495e+00	
grn_7	1.359403e+01	1.773157e+01	1.937671e+01	1.156921e+01	
pdia6	1.508733e+01	1.758221e+01	4.121330e+01	2.319900e+01	
dctn2	1.441672e+01	1.756953e+01	1.750667e+01	7.938743e+00	
LASC019392	1.676326e+01	1.756661e+01	1.980769e+01	1.178820e+01	
LASC006421	1.659468e+01	1.715998e+01	7.435608e+00	1.569059e+00	
col4a1_2	4.157945e+01	1.705182e+01	2.103149e+01	1.437204e+01	
adk2	1.337666e+01	1.705061e+01	1.293195e+01	3.441000e+00	
gstml_3	2.056130e+01	1.696925e+01	2.799136e+00	1.214624e-01	
t05h10.6	1.424161e+01	1.694823e+01	1.354946e+01	6.446559e+00	
tmed7	1.827695e+01	1.685567e+01	1.577423e+01	9.546963e+00	
lcp1	1.075019e+01	1.669950e+01	1.998330e+01	9.728648e+00	
FlSpc	3.653870e+03	1.669434e+01	1.550348e+01	2.663770e+01	
dcd2b	1.876078e+01	1.667128e+01	3.262182e+02	3.850143e+02	
ubxn1_ubxn1-a	1.429643e+01	1.666874e+01	2.053999e+01	1.000656e+01	
pura	1.329482e+01	1.664651e+01	9.833090e+00	6.636944e+00	
ddx39b	1.568017e+01	1.640550e+01	1.430110e+01	8.636751e+00	
cisd1	1.771306e+01	1.632968e+01	1.190779e+01	5.269892e+00	
vps35	1.183141e+01	1.623908e+01	1.860856e+01	9.872430e+00	
hts_2	1.179788e+01	1.600037e+01	1.989198e+01	1.084402e+01	
gmpr_gmpr2	1.421713e+01	1.586701e+01	1.570699e+01	9.637144e+00	
reep6_reep5	1.892340e+01	1.580220e+01	1.961423e+01	4.003003e+00	
rpia	1.532840e+01	1.575232e+01	1.328758e+01	7.218258e+00	
hspl1_2	1.444623e+01	1.570060e+01	5.872568e+01	1.002270e+01	
erlin2-b	1.495659e+01	1.567858e+01	1.365475e+01	8.273330e+00	
h2afv	1.521536e+01	1.545340e+01	1.096854e+01	5.172081e+00	
metap2	1.268285e+01	1.542873e+01	1.241819e+01	5.461607e+00	
asna	1.416467e+01	1.542119e+01	1.990756e+01	1.895071e+01	
ddx3x	1.225312e+01	1.516492e+01	1.146477e+01	5.174555e+00	
LASC010290	1.390239e+01	1.484346e+01	8.842713e+00	3.175290e+00	
parn	1.507133e+01	1.431307e+01	1.511179e+01	7.397795e+00	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proximal
FlSpb	8.648069e+02	1.427164e+01	1.452682e+01	6.161220e+00	
mif	1.276682e+01	1.411039e+01	1.470968e+01	7.372769e+00	
eci1	1.453525e+01	1.389392e+01	2.625711e+01	1.252236e+01	
atl	1.188213e+01	1.380233e+01	1.064445e+01	3.742731e+00	
dpys	1.359389e+01	1.377705e+01	1.510477e+01	6.584633e+00	
nrg	9.487515e+00	1.375127e+01	1.675719e+01	9.701253e+00	
vhasfd	1.255121e+01	1.366304e+01	1.439936e+01	6.908120e+00	
apmap_3	9.356861e+00	1.352732e+01	2.257099e+01	1.092556e+01	
letm1	1.030375e+01	1.351493e+01	1.268183e+01	4.033587e+00	
ap2a2_2	1.113787e+01	1.347214e+01	9.733507e+00	3.899535e+00	
txn	1.178971e+01	1.322460e+01	1.301203e+01	3.889355e+00	
nipsnap_2	1.288021e+01	1.317564e+01	1.292034e+01	6.358307e+00	
awd	1.363837e+01	1.316497e+01	1.198417e+01	6.577936e+00	
etfdh	1.166401e+01	1.310882e+01	1.068708e+01	5.080400e+00	
usp5	9.672852e+00	1.305936e+01	1.235591e+01	4.693874e+00	
pgls	1.710347e+01	1.293490e+01	1.983296e+01	1.640840e+01	
LASC010973	1.152278e+01	1.277545e+01	6.726339e+00	2.565074e+00	
lam_2	1.042577e+01	1.276945e+01	1.173642e+01	4.479473e+00	
LASC010012	4.142206e+00	1.268568e+01	5.308173e+03	8.923669e+03	
mrpl12	1.398686e+01	1.258851e+01	9.316394e+00	3.242551e+00	
snb-1_vamp3_syb	9.691185e+00	1.234191e+01	1.122797e+01	4.735963e+00	
LASC005883	1.024662e+01	1.226133e+01	1.153951e+01	5.797754e+00	
manf_2	1.543189e+01	1.223381e+01	3.428602e+01	1.812982e+01	
eif4h	1.059306e+01	1.222197e+01	1.243384e+01	7.642291e+00	
sep2	9.555122e+00	1.221219e+01	1.481713e+01	4.373543e+00	
LASC005807	1.090040e+01	1.205426e+01	1.916540e+01	9.108138e+00	
mpp6	1.539797e+01	1.192427e+01	8.210651e+00	3.629616e+00	
bcap31	1.055888e+01	1.170806e+01	1.232139e+01	7.400169e+00	
nrg_2	9.501563e+00	1.157269e+01	1.063672e+01	8.333590e+00	
vinc	7.539276e+00	1.154763e+01	1.431660e+01	5.499201e+00	
unc-94	7.977830e+00	1.152845e+01	1.093955e+01	4.972205e+00	
smf	1.923310e+01	1.138781e+01	5.893708e+00	2.913181e+00	
bcat2_vlg246094	9.975791e+00	1.128157e+01	7.437265e+00	2.133681e+00	
sec11c	7.078830e+00	1.111870e+01	1.783741e+01	1.328575e+01	
tp02	1.303509e+01	1.109447e+01	9.540004e+00	5.305862e+00	
bola2	1.060704e+01	1.103431e+01	6.453986e+00	2.295609e+00	
myef2	8.526459e+00	1.102427e+01	1.162035e+01	6.857530e+00	
prtp	7.896034e+00	1.100612e+01	1.334862e+01	5.122978e+00	
ylpm1	9.964649e+00	1.096624e+01	1.521925e+01	8.474343e+00	
oxr1_2	1.451667e+01	1.094725e+01	9.417984e+00	4.400456e+00	
LASC022131	7.781072e+00	1.086593e+01	7.477409e+00	7.331692e-01	
srp14	1.392166e+01	1.077265e+01	5.929366e+00	2.360781e+00	
sfxn1	1.027508e+01	1.061238e+01	5.258449e+00	3.301891e+00	
LASC001730	3.948042e+00	1.059325e+01	4.963650e+00	3.057545e+01	
camki	7.079471e+00	1.041087e+01	7.070178e+00	4.038774e+00	
hexa	9.725197e+00	1.037508e+01	1.881726e+01	1.113389e+01	
ranbp1	1.026990e+01	1.030045e+01	9.915099e+00	5.384091e+00	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
syncrip	8.641029e+00	1.026209e+01	1.465356e+01	6.030564e+00	
hyou1	7.735117e+00	1.012046e+01	1.518707e+01	8.800694e+00	
LASC016483	1.247780e+01	1.006868e+01	3.824762e+00	1.714970e+00	
mtpn	1.075302e+01	1.004878e+01	1.153005e+01	6.861747e+00	
rplp1_2	1.091164e+01	9.956325e+00	7.788985e+00	1.943474e+00	
usp14	7.911903e+00	9.583048e+00	1.238456e+01	5.391153e+00	
xpnpep1	1.129742e+01	9.570600e+00	4.991866e+00	8.452610e-01	
LASC016310	4.867692e+00	9.385631e+00	1.997381e+01	2.440373e+00	
f26e4.3_2	1.155608e+01	9.259218e+00	7.745670e+00	3.077082e+00	
ppt1	8.614425e+00	9.160886e+00	1.430694e+01	9.468828e+00	
cse1l	7.144375e+00	9.139349e+00	7.458070e+00	2.314980e+00	
qdpr	1.004299e+01	9.105364e+00	5.501874e+00	2.038707e+00	
LASC004844	9.600537e+00	8.986198e+00	1.030720e+01	5.388562e+00	
sub1	8.914579e+00	8.963532e+00	1.376731e+01	8.878630e+00	
slc27a1_2	6.576591e+00	8.959367e+00	9.462917e+00	9.446564e+00	
LASC016530	6.869043e+00	8.805860e+00	8.492646e+00	5.248749e+00	
lana	1.076912e+01	8.687065e+00	8.728272e+00	4.566774e+00	
AgSp2b	2.416670e+01	8.427657e+00	7.235364e+00	5.628753e+00	
FlSpa	4.225487e+03	8.331317e+00	8.430056e+00	3.555453e+01	
idh3g	8.303031e+00	8.330169e+00	4.665898e+00	1.582514e+00	
lanb1	8.461396e+00	8.322587e+00	7.634905e+00	3.545362e+00	
adk_adk2	1.111454e+01	8.262927e+00	4.727273e+00	1.253303e+00	
nudt16	7.581271e+00	8.168648e+00	5.246643e+00	1.692398e+00	
pdc6_2	9.622940e+00	8.164585e+00	9.023194e+00	5.132209e+00	
hnrnpul1	6.927595e+00	8.145723e+00	7.686361e+00	3.210600e+00	
lamtor3	7.951067e+00	8.095609e+00	1.079934e+01	5.242961e+00	
atp6ap1	5.150629e+00	7.907051e+00	6.677195e+00	2.391172e+00	
slc25a3_2	6.966485e+00	7.885968e+00	6.464891e+00	3.962813e+00	
LASC003279	5.889157e+00	7.760315e+00	1.142397e+01	7.105887e+00	
hspg2	6.920718e+00	7.736276e+00	9.116715e+00	6.424149e+00	
spsc2	8.593003e+00	7.664120e+00	1.993844e+01	2.065169e+01	
pgm1_2	7.112453e+00	7.502367e+00	5.251313e+00	2.053211e+00	
pnliprp1	5.024910e+00	7.399604e+00	1.620751e+03	1.886213e+03	
emc1	6.935479e+00	7.277414e+00	1.032369e+01	7.238855e+00	
lam	1.995627e+00	7.203555e+00	4.916501e+00	1.598403e+00	
jana	7.511313e+00	7.199719e+00	5.802466e+00	3.158197e+00	
rac1_2	5.970879e+00	7.177289e+00	8.010635e+00	3.819765e+00	
nrx-iv	5.578704e+00	7.172127e+00	9.671941e+00	6.000013e+00	
khsrp	5.618270e+00	7.140167e+00	7.956276e+00	5.324244e+00	
tgas006m08.1_3	3.624568e+00	7.060506e+00	1.447096e+01	2.392895e+00	
tom1	4.219324e+00	7.055942e+00	1.055817e+01	9.030625e+00	
m6pr	4.249231e+00	7.024509e+00	7.147672e+00	3.253316e+00	
pccb	8.861935e+00	6.891268e+00	9.964151e+00	5.833163e+00	
LASC011699	2.697715e+00	6.751842e+00	1.529865e+01	1.666848e+00	
LASC017285	5.492412e+00	6.739408e+00	9.242252e+00	5.229904e+00	
golim4	5.977385e+00	6.689173e+00	1.151478e+01	1.016458e+01	
flo2_2	5.581439e+00	6.653492e+00	6.992364e+00	3.633150e+00	
sptbn5	5.345179e+00	6.646279e+00	4.227799e+00	1.698981e+00	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proximal
ergic1	6.318345e+00	6.473213e+00	6.126476e+00	4.242877e+00	
SpiCE-LMa2	3.156750e-01	6.334276e+00	2.153686e+02	1.085016e+03	
dek	7.263315e+00	6.309452e+00	5.812584e+00	2.850882e+00	
LASC020142	4.269932e+00	6.280675e+00	8.779409e+00	4.693874e+00	
cg3556_2	2.050786e+00	6.256283e+00	4.750672e+01	5.425887e+00	
LASC017000	4.014494e+00	6.106314e+00	4.240211e+00	1.976170e+00	
cana-14f	4.979417e+00	6.097427e+00	5.490394e+00	3.022681e+00	
LASC002717	5.109431e+00	6.053877e+00	6.697720e+00	4.004464e+00	
LASC015168	6.209359e+00	6.033909e+00	5.874969e+00	3.530299e+00	
f1r	3.760131e+00	6.024412e+00	1.049672e+01	6.432439e+00	
ube2l3	4.279918e+00	5.885560e+00	5.139586e+00	2.773452e+00	
tpd52l2	5.366337e+00	5.869341e+00	5.378351e+00	3.402682e+00	
pde12_4	4.643371e+00	5.843552e+00	4.010921e+00	1.701086e+00	
gpr158	4.301456e+00	5.832279e+00	7.049753e+00	4.151275e+00	
vat1l	3.878351e+00	5.642401e+00	4.641589e+00	7.243414e-01	
rab14	4.221556e+00	5.601662e+00	5.001201e+00	2.084135e+00	
cont	3.516598e+00	5.587869e+00	7.311451e+00	4.325604e+00	
meltf	4.665228e+00	5.470308e+00	7.451073e+00	4.633004e+00	
LASC003125	4.577344e+00	5.407179e+00	4.516094e+00	1.940635e+00	
cav1_3	1.014377e+01	5.385009e+00	5.690805e+00	3.544924e+00	
futsch	2.110880e+00	5.204687e+00	4.756971e+00	6.009499e-01	
ank3_8	2.603880e+00	5.125510e+00	6.753062e+00	2.621344e+00	
col4a1_let-2	1.002142e+01	5.100577e+00	8.054343e+00	4.421018e+00	
ccdc58	5.373391e+00	5.079876e+00	3.584553e+00	1.267185e+00	
LASC019275	9.091922e+00	5.024942e+00	9.663697e+00	5.973931e+00	
mvp	2.210799e+00	5.004975e+00	1.315672e+01	3.655167e+00	
atpsynd_2	4.089658e+00	4.931091e+00	5.385846e+00	0.000000e+00	
gyg1	3.567984e+00	4.848385e+00	5.974594e+00	2.428640e+00	
pabp2	4.756917e+00	4.827824e+00	4.231797e+00	2.596187e+00	
rab6a	4.051408e+00	4.825192e+00	4.693823e+00	2.102213e+00	
tgfb1_3	1.110970e+01	4.778324e+00	5.687117e+00	4.755874e+00	
LASC002397	3.381297e+00	4.715907e+00	4.018404e+00	2.059531e+00	
brafldraft	7.162348e+00	4.706128e+00	3.245592e+01	1.031311e+00	
LASC007511	5.357606e+00	4.670017e+00	3.455051e+00	2.285762e+00	
pka-c1	4.274990e+00	4.562762e+00	4.110179e+00	2.152017e+00	
sep1	3.098868e+00	4.521232e+00	6.530467e+00	3.443501e+00	
LASC020300	4.628780e+00	4.516004e+00	6.535006e+00	2.659538e+00	
LASC017532	2.466535e+00	4.459629e+00	1.340302e+03	3.099644e+03	
csrp2	5.193145e+00	4.451645e+00	3.670257e+00	3.204910e+00	
tom1l2	4.115138e+00	4.419770e+00	6.870686e+00	4.815702e+00	
SpiCE-LMa1	6.300507e+00	4.393849e+00	3.349885e+02	2.946811e+03	
LASC021415	2.022855e+00	4.232278e+00	6.464148e+00	5.000791e-01	
hpgd_14	2.375436e+00	4.210983e+00	3.148203e+00	3.272861e+00	
lsm10	3.355060e+00	4.110988e+00	5.240929e+00	2.604940e+00	
stt3a	3.156053e+00	4.084803e+00	7.374990e+00	7.193835e+00	
lmf2	3.472916e+00	4.064158e+00	3.767581e+00	2.511918e+00	
sh3beta	4.014526e+00	4.019328e+00	2.864224e+00	1.133902e+00	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxim
LASC005973	5.784125e+01	4.009059e+00	1.519581e+00	3.653641e+01	
dpp3_4	3.535416e+00	4.006331e+00	8.803925e+00	8.687393e+00	
pcca	4.587712e+00	3.978742e+00	3.601897e+00	2.291987e+00	
LASC002879	3.019906e+01	3.893781e+00	3.023025e+00	2.916424e+00	
ppm1e	2.577657e+00	3.780026e+00	4.857157e+00	3.280256e+00	
AcSpa	9.632925e+02	3.779316e+00	7.472366e+00	2.470037e+00	
AmSp-like1	4.472204e-01	3.767043e+00	5.782058e+01	8.341795e+01	
macrod1_macro2	2.766007e+00	3.718703e+00	3.362072e+00	2.098977e+00	
csga_5	2.650898e+00	3.693867e+00	7.358833e+00	5.390267e+00	
rheb	3.242307e+00	3.677049e+00	2.605109e+00	9.688332e-01	
txndc17	3.107773e+00	3.668888e+00	6.616226e+00	1.528339e+00	
LASC011075	1.305434e+00	3.654595e+00	2.800833e+00	1.209477e+00	
arl6ip5	3.462927e+00	3.541819e+00	2.358376e+00	6.003514e-01	
pld3	5.011139e+00	3.440686e+00	4.663561e+00	3.640742e+00	
capn5	3.170608e+00	3.339217e+00	3.102667e+00	1.232020e+00	
isoc2	3.131552e+00	3.243610e+00	4.109199e+00	2.598540e+00	
arf1	3.194345e+00	3.004070e+00	2.770517e+00	9.776984e-01	
lbr	2.259747e+00	2.919020e+00	2.493720e+00	1.373177e+00	
gstm1	5.745090e-01	2.822166e+00	1.830790e+01	1.993476e+00	
LASC005034	5.127779e+00	2.724970e+00	8.722820e+01	5.548076e+01	
lac_7	2.150091e+00	2.694068e+00	3.095605e+00	1.690835e+00	
itpr1	2.375447e+00	2.638620e+00	2.137442e+00	1.105000e+00	
jhamt_7	1.263698e+00	2.571721e+00	1.122557e+03	2.620255e+03	
lanb2_2	3.405268e+00	2.552855e+00	2.429367e+00	1.141035e+00	
mlec	2.098826e+00	2.475481e+00	2.368221e+00	1.413225e+00	
grhpr_7	9.169201e-01	2.391971e+00	3.325319e+01	2.944122e+00	
h2afy	1.469958e+00	2.338696e+00	1.858522e+00	8.802709e-01	
pka-c1_2	2.199624e+00	2.311521e+00	2.299942e+00	3.974000e-02	
bpnt1	1.744738e+00	2.296434e+00	4.245134e+00	2.629831e+00	
ncam1	4.798337e+00	2.244730e+00	7.484956e-01	3.300124e-01	
AmSp-like2	4.484882e-01	2.198826e+00	1.356592e+02	1.124443e+02	
galphaq	1.553399e+00	2.186822e+00	2.298771e+00	1.346446e+00	
trip11	2.357297e+00	2.109470e+00	1.199238e+00	5.432178e-01	
MaSp1a	3.852365e-01	2.104358e+00	9.529705e-01	8.221690e-02	
CaAn_20	2.675144e+00	2.095615e+00	2.577549e+00	2.302497e+00	
LASC004484	3.123417e+00	2.006522e+00	1.897089e+00	4.575980e+00	
gstm1_10	3.557983e-01	1.990937e+00	4.345055e+00	3.295226e-01	
adpgk	1.634082e+00	1.951794e+00	1.074462e+00	4.745172e-01	
pdia4	1.689033e+00	1.931802e+00	3.346863e+00	1.450538e+00	
MaSp1b	1.670411e-01	1.852871e+00	7.321799e-01	1.136868e-01	
AgSp1	7.390810e+00	1.764830e+00	1.662262e+00	6.318849e-01	
pgm2	1.041227e+00	1.634564e+00	1.185257e+00	6.301941e-01	
hspa8_hsp70	1.162799e+00	1.633533e+00	6.169979e-01	1.245943e-01	
LASC018595	8.985757e-01	1.623294e+00	6.552021e+02	1.754975e+03	
ndrg3_ndrg4-a	2.019552e+00	1.594025e+00	2.179168e+00	2.036141e+00	
LASC001726	4.000888e-01	1.488545e+00	2.643023e-01	1.273807e+00	
ftcd	8.439868e-01	1.480835e+00	9.037118e-01	1.129728e-01	
naga_3	6.608472e-01	1.468587e+00	6.711375e+00	4.387765e-01	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
LASC004993	8.659383e+00	1.458128e+00	9.102059e-01	5.146928e+00	
aldh5a1	1.138715e+00	1.436798e+00	2.065671e+00	1.729676e+00	
ntl	2.708979e+00	1.397513e+00	2.111999e+00	1.033975e+00	
LASC015175	1.238185e+00	1.385676e+00	1.693022e+01	8.215210e-02	
coll1a2	3.261556e+00	1.364068e+00	1.649742e+00	1.743150e+00	
SpiCE-LMa5	6.969383e+00	1.270252e+00	2.527746e-01	1.654265e-01	
lev-9_cr1_csmd3	7.423403e-01	1.217255e+00	1.078842e+00	5.324886e-01	
LASC009239	1.502414e+00	1.184116e+00	1.781350e+00	2.284103e+00	
mttp_2	4.960436e-01	1.129409e+00	2.053748e+02	5.170151e+02	
LASC016433	1.725965e+00	1.092815e+00	2.390000e+00	6.993604e-01	
LASC006408	1.483525e+00	1.082645e+00	6.832810e-01	3.046530e+00	
LASC000038	9.874670e-02	1.061442e+00	3.749659e-01	1.329428e+00	
MaSp2c	2.134172e-01	1.049104e+00	9.542661e-01	3.414290e-02	
MaSp1c	7.403110e-02	1.015544e+00	4.564221e-01	4.603820e-02	
ykv0_tm_fabg	9.543972e-01	1.001543e+00	3.547932e-01	4.070966e-01	
ywrd_2	7.081484e-01	9.654170e-01	4.208992e+00	3.407082e-01	
LASC004995	7.247190e+00	9.633645e-01	1.923890e+00	9.169045e-01	
erp44_3	1.363436e+00	9.598513e-01	9.123991e-01	7.737838e-01	
cmpk	1.002813e+00	9.107108e-01	1.234516e+00	7.814985e-01	
ptrhd1	1.158311e+00	8.770530e-01	1.516951e+00	1.018294e+00	
hdlbp_3	9.264422e-01	8.614856e-01	7.405118e-01	3.387594e-01	
LASC007099	2.510155e-01	8.500733e-01	2.003375e+02	4.715386e+02	
dst_2	1.601584e+00	7.982081e-01	8.259180e-01	1.439754e+00	
hsd17b6_hsd17b2	6.599200e-03	7.494253e-01	6.569571e+01	8.356221e+01	
lox12	6.328273e-01	7.281439e-01	3.041665e-01	1.672819e-01	
uchl3	5.094106e-01	7.184078e-01	4.615109e-01	1.797752e-01	
LASC012887	2.378027e+01	7.137610e-01	1.171444e-01	1.179553e+01	
pka-r2_2	4.838430e-01	6.958745e-01	6.447552e-01	2.299913e-01	
h2bc1	1.399240e-01	6.836910e-01	1.038909e+00	1.732029e-01	
nop58_3	0.000000e+00	6.506048e-01	0.000000e+00	0.000000e+00	
mas_ppaf2	7.799821e-01	5.906019e-01	7.896620e-01	2.979303e-01	
MaSp2f	2.026601e-01	5.643333e-01	9.937589e-01	8.344770e-02	
LASC010337	5.658667e-01	5.548056e-01	5.580884e-01	3.887139e-01	
mys	5.005272e-01	5.493866e-01	1.796859e-01	1.552715e-01	
LASC005488	4.628847e-01	5.489563e-01	9.112992e-01	1.098924e+00	
pdcd6_bche	4.032354e-01	5.457221e-01	6.448322e-01	5.442164e-01	
LASC009994	4.267100e-02	5.455811e-01	2.159586e+00	1.313359e-01	
fasn_8	1.565514e-01	5.221454e-01	2.285807e+01	6.523169e+01	
LASC019814	8.605323e-01	4.825610e-01	4.797588e+00	4.808752e-01	
zbtb48_3	4.860673e-01	4.770740e-01	3.618810e-01	3.038928e-01	
LASC003331	4.196980e-01	4.520933e-01	1.729154e+01	5.572972e-01	
LASC001732	0.000000e+00	4.450321e-01	1.606995e-01	3.943704e-01	
ywrd_3	1.357770e+00	4.230825e-01	6.721280e-01	4.384667e-01	
erp44_2	1.804024e-01	3.888684e-01	3.591891e-01	3.998110e-02	
LASC018776	3.360924e-01	3.273655e-01	5.542658e-01	1.407597e+00	
rpl10a	7.820230e-02	2.096301e-01	1.173799e-01	2.707400e-02	
MaSp3b	2.377380e-02	1.880581e-01	6.634400e-02	2.753070e-02	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proximal
hsp70b2_4	2.697590e-02	1.874930e-01	3.320238e-01	8.984460e-02	
LASC001731	6.284250e-02	1.608657e-01	1.419120e-01	7.632743e-01	
pnliprp3_pnlip	8.967960e-02	1.575634e-01	2.429920e+01	8.144285e+01	
nutf2	1.487588e-01	1.495934e-01	1.163591e-01	4.129110e-02	
LASC020883	8.152771e-01	1.475461e-01	7.580510e-02	7.918240e-02	
LASC003330	2.689160e-02	1.163060e-01	1.191591e+00	0.000000e+00	
beta-spec	1.897366e-01	1.086095e-01	9.889770e-02	5.637300e-02	
gstd1_gst1	0.000000e+00	8.936320e-02	3.527950e-02	0.000000e+00	
alphanub85e_tuba1c	2.832264e-01	7.684350e-02	5.962620e-02	8.339150e-02	
cd109_5	1.578284e-01	7.395480e-02	1.270410e-01	2.829590e-02	
gstml_5	7.983670e-02	6.521090e-02	3.089336e-01	7.305110e-02	
LASC015292	6.356880e-02	6.289280e-02	5.953790e-02	8.185710e-02	
epb41l3	7.026700e-03	5.482780e-02	9.942800e-02	1.559300e-02	
cmpk_2	0.000000e+00	4.869340e-02	3.826620e-02	1.070580e-02	
LASC015173	1.006079e-01	4.533990e-02	2.056669e+00	0.000000e+00	
ywrp_6	0.000000e+00	4.278870e-02	0.000000e+00	0.000000e+00	
vit-6_vg	0.000000e+00	4.244420e-02	1.148956e+00	1.158970e-02	
LASC000397	0.000000e+00	3.909000e-02	0.000000e+00	0.000000e+00	
fib_2	0.000000e+00	3.585600e-02	1.722157e-01	1.295320e-02	
cut-1_6	5.923330e-02	4.034000e-03	4.966230e-02	3.169660e-02	
LASC003332	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	
LASC005598	0.000000e+00	0.000000e+00	1.837870e-01	1.943400e-03	
LASC006407	0.000000e+00	0.000000e+00	0.000000e+00	2.126290e-02	
LASC007604	0.000000e+00	0.000000e+00	2.060800e-02	0.000000e+00	
LASC022503	0.000000e+00	0.000000e+00	1.835289e-01	0.000000e+00	
LASC022504	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	
TuSpb	0.000000e+00	0.000000e+00	4.665930e-02	0.000000e+00	
hpgd_12	0.000000e+00	0.000000e+00	1.036646e-01	0.000000e+00	
ywrp_7	0.000000e+00	0.000000e+00	6.941630e-02	0.000000e+00	
LASC008951	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	
LASC019165	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	
atpalpha_3	7.894600e-03	0.000000e+00	3.180500e-03	3.008200e-03	
psmb1	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	
psmb1_2	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	
AgSp2a	NA	NA	NA	NA	
LASC019470	NA	NA	NA	NA	
LASC021431	NA	NA	NA	NA	
LASC022267	NA	NA	NA	NA	
LASC022852	NA	NA	NA	NA	
LASC022854	NA	NA	NA	NA	
LASC022855	NA	NA	NA	NA	
LASC022856	NA	NA	NA	NA	
LASC022857	NA	NA	NA	NA	
LASC022858	NA	NA	NA	NA	
glyp_3	NA	NA	NA	NA	
h4-viii	NA	NA	NA	NA	
h4-viii_2	NA	NA	NA	NA	
hspa8_2	NA	NA	NA	NA	

(continued)

geneName	Tail_proximal	Sac_proximal	Sac_center	Sac_distal	log2FoldChange_Sac_proxin
pxt_3	NA	NA	NA	NA	
tgas006m08.1	NA	NA	NA	NA	
tubb4b	NA	NA	NA	NA	
ankrd50_2	NA	NA	NA	NA	
ap1b1	NA	NA	NA	NA	
ctsa_2	NA	NA	NA	NA	
eef1b	NA	NA	NA	NA	
esd	NA	NA	NA	NA	
flo2	NA	NA	NA	NA	
his1	NA	NA	NA	NA	
his1.3_2	NA	NA	NA	NA	
hrb27c_4	NA	NA	NA	NA	
hsp70b2_3	NA	NA	NA	NA	
hsp70b2_5	NA	NA	NA	NA	
hsp70b2_hsp70a2	NA	NA	NA	NA	
inx2_15	NA	NA	NA	NA	

Note:

Table 4: Differential expression results for gland parts.

Supplementary Table 5

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
resized.

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
MiSpa	ZoneA_MiSpabd	Minor	0	3.6283678	0.999	0.961
MiSpb	ZoneA_MiSpabd	Minor	0	3.0567054	1.000	0.969
MiSpd	ZoneA_MiSpabd	Minor	0	2.9685261	0.999	0.934
LASC007034	ZoneA_MiSpabd	Minor	0	2.7814425	0.992	0.869
x975_2	ZoneA_MiSpabd	Minor	0	2.7280464	0.952	0.699
lman2l	ZoneA_MiSpabd	Minor	0	2.4071500	0.946	0.735
LASC011995	ZoneA_MiSpabd	Minor	0	2.3186997	0.876	0.692
gdf-8	ZoneA_MiSpabd	Minor	0	1.9040064	0.813	0.638
ctsb_2	ZoneA_MiSpabd	Minor	0	1.5357608	0.834	0.679
SpiCE-LMa6	ZoneA_MiSpabd	Minor	0	1.3938746	0.889	0.687
tg_3	ZoneA_MiSpabd	Minor	0	1.2773108	0.924	0.792
ssr2	ZoneA_MiSpabd	Minor	0	1.2856407	0.805	0.695
LASC013613	ZoneA_MiSpabd	Minor	0	1.1365285	0.739	0.608
LASC005175	ZoneA_MiSpabd	Minor	0	1.2068707	0.822	0.701
LASC015194	ZoneA_MiSpabd	Minor	0	0.9862181	0.746	0.618
dscam2_dscam_4	ZoneA_MiSpabd	Minor	0	0.8896072	0.706	0.595
LASC009844	ZoneA_MiSpabd	Minor	0	1.1974227	0.708	0.614
adgrl3	ZoneA_MiSpabd	Minor	0	0.9991819	0.720	0.631
adamtsl5_2	ZoneA_MiSpabd	Minor	0	0.8183138	0.714	0.613
phm	ZoneA_MiSpabd	Minor	0	0.6288357	0.793	0.718
SpiCE-LMa3	ZoneA_MiSpabd	Minor	0	0.6556717	0.846	0.749
nell2	ZoneA_MiSpabd	Minor	0	0.5656807	0.665	0.632
kirrel1_kirrel2	ZoneA_MiSpabd	Minor	0	0.5200136	0.658	0.624
sp-1339#prd34368.1_3	ZoneB_MiSpc	Minor	0	2.5948830	1.000	0.944
MiSpc	ZoneB_MiSpc	Minor	0	2.0523338	0.891	0.620
LASC016881	ZoneB_MiSpc	Minor	0	1.9589224	0.893	0.626
zan	ZoneB_MiSpc	Minor	0	1.7381328	0.910	0.638
LASC000534	ZoneB_MiSpc	Minor	0	0.9985651	0.790	0.417
SpiCE-LMa6	ZoneB_MiSpc	Minor	0	1.5948218	0.958	0.765
SpiCE-LMa3	ZoneB_MiSpc	Minor	0	1.4484787	0.956	0.780
LASC008293	ZoneB_MiSpc	Minor	0	1.2971807	0.880	0.648
qpct	ZoneB_MiSpc	Minor	0	1.2964726	0.875	0.658
sp-1339#prd34368.1_4	ZoneB_MiSpc	Minor	0	1.5999900	0.939	0.658
tg_3	ZoneB_MiSpc	Minor	0	1.3844989	0.954	0.845
smoc1_2	ZoneB_MiSpc	Minor	0	0.6541140	0.934	0.713
phm	ZoneB_MiSpc	Minor	0	0.9922328	0.902	0.740
LASC009784	ZoneB_MiSpc	Minor	0	0.9943140	0.906	0.712
AmSp-like2	ZoneB_MiSpc	Minor	0	0.5898160	0.814	0.617
sod1_sod-1	ZoneB_MiSpc	Minor	0	0.5909028	0.770	0.611
LASC015194	ZoneB_MiSpc	Minor	0	0.7950572	0.860	0.662
adamtsl5_2	ZoneB_MiSpc	Minor	0	0.7744065	0.825	0.646
sele	ZoneB_MiSpc	Minor	0	0.8843063	0.799	0.642
LASC001164	ZoneB_MiSpc	Minor	0	0.9289889	0.733	0.608
LASC010012	ZoneC_SpICE-LMa1	Minor	0	5.4562232	0.986	0.700
spa	ZoneC_SpICE-LMa1	Minor	0	5.2562519	1.000	0.833

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC007099	ZoneC_SpICE-LMa1	Minor	0	4.0535632	0.959	0.617
smoc1_2	ZoneC_SpICE-LMa1	Minor	0	3.3790234	0.950	0.690
mttp_2	ZoneC_SpICE-LMa1	Minor	0	3.2814912	0.943	0.593
LASC009784	ZoneC_SpICE-LMa1	Minor	0	3.1732003	0.980	0.682
LASC017532	ZoneC_SpICE-LMa1	Minor	0	3.0782279	0.912	0.612
LASC020242	ZoneC_SpICE-LMa1	Minor	0	3.0616297	0.907	0.602
sp-1339#prd34368.1_4	ZoneC_SpICE-LMa1	Minor	0	2.9479293	0.968	0.628
sp-1339#prd34368.1_3	ZoneC_SpICE-LMa1	Minor	0	2.7566603	1.000	0.939
SpICE-LMa2	ZoneC_SpICE-LMa1	Minor	0	2.7181447	0.878	0.591
LASC017807	ZoneC_SpICE-LMa1	Minor	0	0.5303054	0.678	0.039
pxd_pxt_2	ZoneC_SpICE-LMa1	Minor	0	1.7296737	0.818	0.559
AmSp-like2	ZoneC_SpICE-LMa1	Minor	0	1.8945998	0.842	0.594
LASC012647	ZoneC_SpICE-LMa1	Minor	0	1.6116712	0.790	0.535
zan	ZoneC_SpICE-LMa1	Minor	0	1.2084913	0.859	0.622
npc2a_2	ZoneC_SpICE-LMa1	Minor	0	0.7528557	0.726	0.418
slc9a3	ZoneC_SpICE-LMa1	Minor	0	0.7240758	0.719	0.388
LASC013812	ZoneC_SpICE-LMa1	Minor	0	0.9013796	0.752	0.494
bmp1	ZoneC_SpICE-LMa1	Minor	0	0.7475668	0.715	0.476
SpICE-LMa4	ZoneC_SpICE-LMa1	Minor	0	1.0353129	0.763	0.569
SpICE-LMa1	ZoneC_SpICE-LMa1	Minor	0	0.6633167	0.677	0.390
AmSp-like1	ZoneC_SpICE-LMa1	Minor	0	0.5689905	0.710	0.492
LASC005800	ZoneC_SpICE-LMa1	Minor	0	0.9597897	0.745	0.549
adprm	ZoneC_SpICE-LMa1	Minor	0	0.7165561	0.735	0.574
con	ZoneC_SpICE-LMa1	Minor	0	0.6435884	0.695	0.445
cdh23_2	ZoneC_SpICE-LMa1	Minor	0	0.7593431	0.783	0.642
cd109_2	ZoneC_SpICE-LMa1	Minor	0	0.6319832	0.721	0.596
lysd	ZoneC_SpICE-LMa1	Minor	0	0.5967334	0.762	0.590
eml1	ZoneC_SpICE-LMa1	Minor	0	1.1827145	0.777	0.644
LASC005799	ZoneC_SpICE-LMa1	Minor	0	0.7804792	0.677	0.648
LASC001730	Duct_2	Minor	0	6.5347750	0.949	0.686
LASC013597	Duct_2	Minor	0	2.4663325	0.942	0.649
LASC005739	Duct_2	Minor	0	1.9443001	0.799	0.318
LASC001733	Duct_2	Minor	0	3.5855094	0.831	0.601
chp_7	Duct_2	Minor	0	1.6097599	0.688	0.267
svep1_11	Duct_2	Minor	0	1.8313576	0.782	0.376
tll2	Duct_2	Minor	0	2.3424255	0.828	0.594
tll1	Duct_2	Minor	0	2.2410352	0.876	0.593
cad87a	Duct_2	Minor	0	1.0466970	0.645	0.177
csmc3	Duct_2	Minor	0	1.1620517	0.583	0.166
zig-8_8	Duct_2	Minor	0	1.3996186	0.721	0.258
gluria_gria4	Duct_2	Minor	0	0.8475092	0.483	0.085
kcp_2	Duct_2	Minor	0	1.2378941	0.668	0.246
grm3_2	Duct_2	Minor	0	1.3712235	0.706	0.317
nphs1_4	Duct_2	Minor	0	0.5150363	0.568	0.153
lox12	Duct_2	Minor	0	1.5630484	0.718	0.394
LASC006408	Duct_2	Minor	0	3.4977391	0.768	0.530
LASC009203	Duct_2	Minor	0	1.2666394	0.675	0.331

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC000038	Duct_2	Minor	0	2.3391283	0.786	0.541
tsc22d1	Duct_2	Minor	0	1.9262211	0.875	0.712
LASC020961	Duct_2	Minor	0	3.2107009	0.744	0.546
nt1	Duct_2	Minor	0	1.4525072	0.902	0.670
LASC001080	Duct_2	Minor	0	1.1982936	0.631	0.260
ryk	Duct_2	Minor	0	1.6396103	0.828	0.672
LASC004993	Duct_2	Minor	0	1.7523707	0.639	0.281
LASC003059	Duct_2	Minor	0	1.4134085	0.736	0.387
hspg2	Duct_2	Minor	0	1.5717992	0.719	0.529
LASC000973	Duct_2	Minor	0	0.5024232	0.566	0.154
LASC001731	Duct_2	Minor	0	3.9766437	0.734	0.583
calu	Duct_2	Minor	0	0.9343377	0.652	0.257
agrn_6	Duct_2	Minor	0	0.9966195	0.671	0.329
igsf9b_hmcn1_nphs1	Duct_2	Minor	0	0.6505570	0.541	0.191
lrig3_4	Duct_2	Minor	0	1.6916732	0.777	0.561
LASC000692	Duct_2	Minor	0	1.3242995	0.723	0.513
LASC008645	Duct_2	Minor	0	1.6210907	0.778	0.577
cut-1_6	Duct_2	Minor	0	1.2752959	0.705	0.431
Spice_15	Duct_2	Minor	0	1.3951207	0.564	0.210
LASC013984	Duct_2	Minor	0	1.7028569	0.827	0.658
zig-8_3	Duct_2	Minor	0	1.8156629	0.744	0.520
ptp10d	Duct_2	Minor	0	0.8982880	0.636	0.362
cut-1	Duct_2	Minor	0	0.7801695	0.617	0.311
sas	Duct_2	Minor	0	1.1594923	0.619	0.286
LASC003054	Duct_2	Minor	0	0.7270396	0.555	0.256
col11a1	Duct_2	Minor	0	0.8903766	0.642	0.413
wnt16	Duct_2	Minor	0	0.8801153	0.642	0.400
LASC017285	Duct_2	Minor	0	1.4194599	0.788	0.686
gyc76c	Duct_2	Minor	0	1.2662537	0.706	0.585
gyc32e_2	Duct_2	Minor	0	0.8941126	0.628	0.497
gpc5	Duct_2	Minor	0	0.8529744	0.646	0.455
spn88ea	Duct_2	Minor	0	1.4564005	0.686	0.607
ddr2_3	Duct_2	Minor	0	0.7510446	0.620	0.417
LASC001726	Duct_2	Minor	0	1.8400153	0.596	0.399
rpl10	Duct_2	Minor	0	0.6772450	0.997	0.999
LASC005973	Duct_2	Minor	0	3.8721436	0.688	0.600
reck_2	Duct_2	Minor	0	0.5663960	0.597	0.369
let-653	Duct_2	Minor	0	0.8059852	0.544	0.248
LASC020962	Duct_2	Minor	0	0.9555754	0.750	0.577
calcr1a_2	Duct_2	Minor	0	1.2092252	0.859	0.775
kug	Duct_2	Minor	0	1.1739029	0.713	0.631
sema-2a	Duct_2	Minor	0	1.1558187	0.723	0.645
obst-e_2	Duct_2	Minor	0	0.6376058	0.697	0.514
LASC019092	Duct_2	Minor	0	1.2555169	0.678	0.517
LASC005883	Duct_2	Minor	0	1.0308330	0.730	0.670
LASC022034	Duct_2	Minor	0	0.7188223	0.653	0.445
LASC003252	Duct_2	Minor	0	1.2507272	0.725	0.629
dscam2_5	Duct_2	Minor	0	0.5959672	0.613	0.404

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
fgf8b	Duct_2	Minor	0	0.9978375	0.675	0.632
fur1	Duct_2	Minor	0	1.0974888	0.698	0.651
sema5a	Duct_2	Minor	0	0.7146583	0.611	0.541
LASC007251	Duct_2	Minor	0	0.5373414	0.613	0.529
LASC015905	Duct_2	Minor	0	0.7248157	0.694	0.595
dag1_2	Duct_2	Minor	0	0.7002999	0.612	0.504
LASC005504	Duct_2	Minor	0	0.6139299	0.671	0.466
alpl_8	Duct_2	Minor	0	0.7788623	0.756	0.692
LASC016138	Duct_2	Minor	0	1.0624308	0.718	0.643
gpc5_2	Duct_2	Minor	0	0.8145251	0.694	0.651
n_2	Duct_2	Minor	0	0.5733842	0.608	0.567
fth1-a	Duct_2	Minor	0	0.5443345	0.612	0.565
tmprss15_3	Duct_2	Minor	0	0.7570887	0.671	0.649
adam8_adam12	Duct_2	Minor	0	0.5180029	0.615	0.595
cht10_3	Duct_2	Minor	0	0.5689933	0.651	0.640
naga_3	Duct_2	Minor	0	0.6341810	0.646	0.615
atp6ap1	Duct_2	Minor	0	0.5156191	0.740	0.707
LASC008293	Duct_2	Minor	0	1.2625620	0.703	0.658
robo2_2	Duct_2	Minor	0	0.6597925	0.636	0.639
nrk	Duct_2	Minor	0	0.5554599	0.619	0.618
LASC015386	Duct_1	Minor	0	5.2551807	0.917	0.595
LASC019275	Duct_1	Minor	0	5.0812736	0.977	0.626
LASC016362	Duct_1	Minor	0	4.8243203	0.993	0.664
LASC005034	Duct_1	Minor	0	4.3420029	0.967	0.585
LASC012976	Duct_1	Minor	0	3.7840618	1.000	0.874
gluclalpha_2	Duct_1	Minor	0	3.5773157	0.927	0.454
rpl10	Duct_1	Minor	0	3.1348031	1.000	0.999
LASC004790	Duct_1	Minor	0	3.0845007	0.941	0.612
LASC009204	Duct_1	Minor	0	2.9800612	0.905	0.585
nt1	Duct_1	Minor	0	2.9702142	0.984	0.655
arsi	Duct_1	Minor	0	2.9440817	0.924	0.527
LASC020962	Duct_1	Minor	0	2.4521508	0.888	0.551
nid2	Duct_1	Minor	0	2.4331544	0.857	0.468
LASC005504	Duct_1	Minor	0	2.1533005	0.877	0.428
LASC013597	Duct_1	Minor	0	1.8875241	0.980	0.642
LASC009350	Duct_1	Minor	0	1.5459641	0.781	0.151
tmem165_2	Duct_1	Minor	0	0.8811664	0.698	0.151
LASC013984	Duct_1	Minor	0	1.8847392	0.909	0.643
obst-e_2	Duct_1	Minor	0	1.9163828	0.833	0.489
LASC018571	Duct_1	Minor	0	0.9294594	0.754	0.312
LASC014807	Duct_1	Minor	0	0.6553379	0.675	0.217
tll1	Duct_1	Minor	0	1.5284948	0.893	0.590
col15a1	Duct_1	Minor	0	1.1643414	0.752	0.376
alpl_8	Duct_1	Minor	0	1.5116726	0.869	0.671
LASC013681	Duct_1	Minor	0	0.7238204	0.668	0.230
atp6ap1	Duct_1	Minor	0	1.4834646	0.866	0.683
LASC019092	Duct_1	Minor	0	1.5621320	0.813	0.492

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC015292	Duct_1	Minor	0	0.6120437	0.705	0.288
LASC005739	Duct_1	Minor	0	0.9341160	0.803	0.318
LASC011983	Duct_1	Minor	0	0.8153992	0.684	0.261
LASC001525	Duct_1	Minor	0	0.6596294	0.681	0.330
lrig3_4	Duct_1	Minor	0	1.1997343	0.816	0.554
LASC022034	Duct_1	Minor	0	1.0218034	0.745	0.429
calu	Duct_1	Minor	0	0.5258092	0.694	0.250
zig-8_8	Duct_1	Minor	0	0.7052641	0.709	0.261
npc2	Duct_1	Minor	0	0.9397608	0.688	0.393
LASC020957	Duct_1	Minor	0	1.0519980	0.649	0.256
LASC003059	Duct_1	Minor	0	0.8869828	0.757	0.384
p4ha2	Duct_1	Minor	0	0.5385231	0.666	0.261
LASC008645	Duct_1	Minor	0	1.1453764	0.802	0.573
LASC017285	Duct_1	Minor	0	1.0651889	0.850	0.675
dscam2_5	Duct_1	Minor	0	0.8116732	0.706	0.387
atk_2	Duct_1	Minor	0	0.5147532	0.656	0.304
LASC015905	Duct_1	Minor	0	1.2500886	0.772	0.581
LASC000038	Duct_1	Minor	0	0.9442064	0.791	0.540
efnb2a	Duct_1	Minor	0	1.4106899	0.784	0.622
zig-8_3	Duct_1	Minor	0	0.9078994	0.749	0.519
syg-2_3	Duct_1	Minor	0	0.6441143	0.648	0.395
LASC006841	Duct_1	Minor	0	0.6580988	0.630	0.302
glra4_glra3	Duct_1	Minor	0	0.6752465	0.620	0.351
naga_3	Duct_1	Minor	0	0.9921156	0.730	0.600
tsc22d1	Duct_1	Minor	0	1.2801197	0.855	0.716
cht10_3	Duct_1	Minor	0	0.7387279	0.736	0.625
awd	Duct_1	Minor	0	0.5746844	0.694	0.615
gyc76c	Duct_1	Minor	0	0.6637519	0.720	0.582
LASC017513	Duct_1	Minor	0	2.4154223	0.486	0.199
ryk	Duct_1	Minor	0	0.5564410	0.795	0.678
sema-2a	Duct_1	Minor	0	0.6029816	0.726	0.645
calcr1a_2	Duct_1	Minor	0	0.8780346	0.841	0.779

Note:

Table 5: Supplementary data for Figure 5.

Supplementary Table 6

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
resized.

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
MaSp2b	ZoneA_MaSp2	Major	0.0000000	10.3128508	0.952	0.223
MaSp2e	ZoneA_MaSp2	Major	0.0000000	5.9960285	0.948	0.288
MaSp2a	ZoneA_MaSp2	Major	0.0000000	5.7455471	0.744	0.201
MaSp2f	ZoneA_MaSp2	Major	0.0000000	5.5667879	0.973	0.314
LASC005175	ZoneA_MaSp2	Major	0.0000000	3.8382147	0.723	0.342
MaSp2c	ZoneA_MaSp2	Major	0.0000000	2.5105196	0.963	0.342
LASC009844	ZoneA_MaSp2	Major	0.0000000	2.0636077	0.648	0.373
MaSp4	ZoneA_MaSp2	Major	0.0000000	6.1751120	0.463	0.112
MaSp1a	ZoneA_MaSp2	Major	0.0000000	0.9340020	0.454	0.209
lman2l	ZoneA_MaSp2	Major	0.0000000	1.4197412	0.658	0.388
LASC021139	ZoneA_MaSp2	Major	0.0000000	1.8562457	0.478	0.231
gdf-8	ZoneA_MaSp2	Major	0.0000000	5.5075576	0.563	0.329
LASC016431	ZoneA_MaSp2	Major	0.0000000	3.9747906	0.280	0.108
nup210	ZoneA_MaSp2	Major	0.0000000	0.7846790	0.236	0.095
tmed6	ZoneA_MaSp2	Major	0.0000000	0.5934953	0.470	0.287
kiaa1324l	ZoneA_MaSp2	Major	0.0000000	1.6920958	0.313	0.166
LASC007183	ZoneA_MaSp2	Major	0.0000000	1.0234210	0.266	0.084
x975_2	ZoneA_MaSp2	Major	0.0000000	3.0081265	0.479	0.325
vnn3	ZoneA_MaSp2	Major	0.0000000	2.5059556	0.221	0.093
pxt_6	ZoneA_MaSp2	Major	0.0000000	1.4621234	0.236	0.075
LASC012923	ZoneA_MaSp2	Major	0.0000000	1.5322308	0.252	0.150
reln	ZoneA_MaSp2	Major	0.0000000	0.5437338	0.206	0.098
LASC011347	ZoneA_MaSp2	Major	0.0000000	0.4076077	0.059	0.148
LASC001164	ZoneA_MaSp2	Major	0.0000000	1.4911427	0.174	0.160
fst	ZoneA_MaSp2	Major	0.0000000	3.6351804	0.236	0.074
dag1	ZoneA_MaSp2	Major	0.0000000	1.5961311	0.322	0.228
madd-4	ZoneA_MaSp2	Major	0.0000000	1.1108528	0.154	0.059
LASC013808	ZoneA_MaSp2	Major	0.0000000	1.6973140	0.210	0.147
acpp	ZoneA_MaSp2	Major	0.0000000	1.2040544	0.195	0.149
lcch3_2	ZoneA_MaSp2	Major	0.0000000	0.9781392	0.200	0.134
hig	ZoneA_MaSp2	Major	0.0000000	1.5555285	0.117	0.038
LASC016428	ZoneA_MaSp2	Major	0.0000000	0.3902467	0.181	0.116
kcp	ZoneA_MaSp2	Major	0.0000000	0.5466526	0.149	0.070
LASC012458	ZoneA_MaSp2	Major	0.0000000	1.4037781	0.122	0.065
LASC015891	ZoneA_MaSp2	Major	0.0000000	1.2073171	0.083	0.171
adgrl3_2	ZoneA_MaSp2	Major	0.0000000	1.5768757	0.176	0.108
avr2	ZoneA_MaSp2	Major	0.0000000	2.0776624	0.103	0.216
svep1_9	ZoneA_MaSp2	Major	0.0000000	0.5671419	0.104	0.134
pofut2	ZoneA_MaSp2	Major	0.0000000	1.0065325	0.192	0.118
let-4_3	ZoneA_MaSp2	Major	0.0000000	1.1218775	0.125	0.075
LASC003142	ZoneA_MaSp2	Major	0.0000000	2.4634029	0.224	0.162
cd74	ZoneA_MaSp2	Major	0.0000000	0.6220272	0.151	0.121
kirrel1_kirrel2	ZoneA_MaSp2	Major	0.0000005	0.6407272	0.197	0.137
adgrl3	ZoneA_MaSp2	Major	0.0000130	0.6809760	0.212	0.149
tgoln2	ZoneA_MaSp2	Major	0.0000172	0.6621994	0.188	0.200

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC015767	ZoneA_MaSp2	Major	0.0000481	0.9588684	0.125	0.078
LASC016433	ZoneA_MaSp2	Major	0.0001167	1.4709821	0.147	0.105
LASC008662	ZoneA_MaSp2	Major	0.0004665	1.2781752	0.033	0.126
fat4_2	ZoneA_MaSp2	Major	0.0081721	1.2209847	0.109	0.093
nell2	ZoneA_MaSp2	Major	1.0000000	1.4299389	0.226	0.182
LASC022037	ZoneA_MaSp2	Major	1.0000000	1.1579080	0.132	0.076
LASC002635	ZoneA_MaSp2	Major	1.0000000	0.8303458	0.170	0.139
cut-1_3	ZoneA_MaSp2	Major	1.0000000	0.6293054	0.114	0.111
nt1_spz3	ZoneA_MaSp2	Major	1.0000000	0.6208570	0.149	0.079
gpatch1	ZoneA_MaSp2	Major	1.0000000	0.4185702	0.128	0.187
LASC015194	ZoneA_SpiCE-LMa3	Major	0.0000000	5.7813702	0.658	0.330
SpiCE-LMa6	ZoneA_SpiCE-LMa3	Major	0.0000000	3.8765610	0.659	0.342
LASC011995	ZoneA_SpiCE-LMa3	Major	0.0000000	3.2632309	0.680	0.363
LASC005801	ZoneA_SpiCE-LMa3	Major	0.0000000	1.6162145	0.741	0.442
SpiCE-LMa5	ZoneA_SpiCE-LMa3	Major	0.0000000	5.8244049	0.614	0.322
tg_3	ZoneA_SpiCE-LMa3	Major	0.0000000	1.9251554	0.601	0.388
SpiCE-LMa3	ZoneA_SpiCE-LMa3	Major	0.0000000	2.6364046	0.525	0.310
LASC013613	ZoneA_SpiCE-LMa3	Major	0.0000000	8.8830339	0.362	0.160
abhd14a	ZoneA_SpiCE-LMa3	Major	0.0000000	5.1351338	0.569	0.395
LASC007034	ZoneA_SpiCE-LMa3	Major	0.0000000	9.1340196	0.620	0.429
ctsb_2	ZoneA_SpiCE-LMa3	Major	0.0000000	6.0973170	0.314	0.175
LASC000784	ZoneA_SpiCE-LMa3	Major	0.0000000	3.6985044	0.411	0.245
LASC011991	ZoneA_SpiCE-LMa3	Major	0.0000000	3.4229198	0.327	0.147
LASC009844	ZoneA_SpiCE-LMa3	Major	0.0000000	1.4106679	0.532	0.384
LASC015735	ZoneA_SpiCE-LMa3	Major	0.0000000	6.6831943	0.519	0.350
slc39a13	ZoneA_SpiCE-LMa3	Major	0.0000000	1.3016819	0.295	0.141
LASC005157	ZoneA_SpiCE-LMa3	Major	0.0000000	0.3835407	0.271	0.130
LASC009135	ZoneA_SpiCE-LMa3	Major	0.0000000	2.3467605	0.300	0.150
lman2l	ZoneA_SpiCE-LMa3	Major	0.0000000	0.8063647	0.518	0.410
creld2	ZoneA_SpiCE-LMa3	Major	0.0000000	0.6930033	0.190	0.102
lrrn2	ZoneA_SpiCE-LMa3	Major	0.0000000	0.7362375	0.204	0.080
LASC013964	ZoneA_SpiCE-LMa3	Major	0.0000000	0.5117984	0.148	0.099
flt4	ZoneA_SpiCE-LMa3	Major	0.0000000	3.3687170	0.244	0.162
LASC008889	ZoneA_SpiCE-LMa3	Major	0.0000000	1.9150062	0.210	0.142
flz	ZoneA_SpiCE-LMa3	Major	0.0000000	0.4314045	0.219	0.153
LASC021139	ZoneA_SpiCE-LMa3	Major	0.0000000	0.4444962	0.180	0.314
ctsb	ZoneA_SpiCE-LMa3	Major	0.0000000	4.2867535	0.158	0.093
LASC012179	ZoneA_SpiCE-LMa3	Major	0.0000000	1.2605925	0.222	0.160
LASC005800	ZoneA_SpiCE-LMa3	Major	0.0000000	6.1894049	0.230	0.172
fgfr1l_5	ZoneA_SpiCE-LMa3	Major	0.0000000	0.7400612	0.223	0.151
LASC018509	ZoneA_SpiCE-LMa3	Major	0.0000000	1.0538997	0.266	0.183
pgrp	ZoneA_SpiCE-LMa3	Major	0.0000000	2.2572924	0.234	0.187
LASC019920	ZoneA_SpiCE-LMa3	Major	0.0000000	0.8375471	0.169	0.088
p4hb	ZoneA_SpiCE-LMa3	Major	0.0000000	0.6685034	0.096	0.157
LASC005799	ZoneA_SpiCE-LMa3	Major	0.0000000	0.7091084	0.175	0.144
LASC017572	ZoneA_SpiCE-LMa3	Major	0.0000000	3.7356575	0.188	0.145
slit3_5	ZoneA_SpiCE-LMa3	Major	0.0000000	2.9816440	0.229	0.211

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC006600	ZoneA_SpiCE-LMa3	Major	0.0000000	3.0383817	0.238	0.199
btd	ZoneA_SpiCE-LMa3	Major	0.0000000	0.5459727	0.124	0.067
LASC013495	ZoneA_SpiCE-LMa3	Major	0.0000000	4.0588420	0.222	0.182
AgSp2b	ZoneA_SpiCE-LMa3	Major	0.0000000	1.3662624	0.323	0.233
sli_7	ZoneA_SpiCE-LMa3	Major	0.0000000	0.2912830	0.126	0.068
gpc6	ZoneA_SpiCE-LMa3	Major	0.0000000	0.4321686	0.154	0.148
lanb2	ZoneA_SpiCE-LMa3	Major	0.0000000	1.0200876	0.145	0.125
LASC020966	ZoneA_SpiCE-LMa3	Major	0.0000000	3.1632072	0.162	0.103
zan	ZoneA_SpiCE-LMa3	Major	0.0000001	0.2703316	0.047	0.150
lrig3_5	ZoneA_SpiCE-LMa3	Major	0.0000010	0.3402567	0.145	0.104
ptpn7	ZoneA_SpiCE-LMa3	Major	0.0000013	3.3611754	0.230	0.146
a2m_a2ml1	ZoneA_SpiCE-LMa3	Major	0.0000024	5.0034142	0.186	0.215
efnb2a	ZoneA_SpiCE-LMa3	Major	0.0000028	1.0391026	0.086	0.134
LASC015537	ZoneA_SpiCE-LMa3	Major	0.0000113	0.6640556	0.176	0.149
AgSp1	ZoneA_SpiCE-LMa3	Major	0.0000502	2.2929432	0.328	0.257
MaSp2c	ZoneA_SpiCE-LMa3	Major	0.0000731	0.7977481	0.448	0.464
nrf-6_32	ZoneA_SpiCE-LMa3	Major	0.0001199	1.7450036	0.177	0.098
nt1_spz3	ZoneA_SpiCE-LMa3	Major	0.0003294	1.2195285	0.122	0.081
chadl	ZoneA_SpiCE-LMa3	Major	0.0010218	0.5837008	0.123	0.141
adamtsl5_2	ZoneA_SpiCE-LMa3	Major	0.0011782	0.9105935	0.137	0.119
adamts9_adamts20	ZoneA_SpiCE-LMa3	Major	0.0013173	0.3060018	0.130	0.083
LASC008294	ZoneA_SpiCE-LMa3	Major	0.0018974	0.7484299	0.075	0.114
LASC009364	ZoneA_SpiCE-LMa3	Major	0.0045064	0.5627302	0.125	0.120
LASC007183	ZoneA_SpiCE-LMa3	Major	0.0099372	1.2065708	0.118	0.119
LASC011347	ZoneA_SpiCE-LMa3	Major	0.0182300	0.3903915	0.103	0.142
kug	ZoneA_SpiCE-LMa3	Major	0.0350738	1.1669125	0.112	0.088
lrp6	ZoneA_SpiCE-LMa3	Major	0.0674943	2.5089921	0.117	0.113
oaf	ZoneA_SpiCE-LMa3	Major	0.5292079	8.1117692	0.204	0.176
LASC011993	ZoneA_SpiCE-LMa3	Major	0.5635137	0.5416420	0.128	0.145
LASC020958	ZoneA_SpiCE-LMa3	Major	0.9327918	0.3987612	0.153	0.107
LASC022638	ZoneA_SpiCE-LMa3	Major	1.0000000	1.9192223	0.170	0.118
fur1	ZoneA_SpiCE-LMa3	Major	1.0000000	1.3152608	0.106	0.127
MaSp1c	ZoneA_MaSp1	Major	0.0000000	6.3822681	0.847	0.192
MaSp1b	ZoneA_MaSp1	Major	0.0000000	6.3412488	0.909	0.269
MaSp1a	ZoneA_MaSp1	Major	0.0000000	4.0266011	0.921	0.205
dscam2_dscam_4	ZoneA_MaSp1	Major	0.0000000	7.8172174	0.420	0.108
adgrl3	ZoneA_MaSp1	Major	0.0000000	4.1040875	0.446	0.139
adamtsl5_2	ZoneA_MaSp1	Major	0.0000000	4.0235422	0.389	0.104
hmcn1_robo1	ZoneA_MaSp1	Major	0.0000000	5.6541267	0.475	0.188
trabd2b_3	ZoneA_MaSp1	Major	0.0000000	2.8980267	0.370	0.091
slc39a13	ZoneA_MaSp1	Major	0.0000000	3.2697643	0.479	0.161
bmp2_2	ZoneA_MaSp1	Major	0.0000000	4.8335358	0.457	0.148
dag1	ZoneA_MaSp1	Major	0.0000000	1.7528158	0.489	0.227
LASC017572	ZoneA_MaSp1	Major	0.0000000	1.0985992	0.400	0.139
lrp2_6	ZoneA_MaSp1	Major	0.0000000	8.0742272	0.528	0.305
mgl-1	ZoneA_MaSp1	Major	0.0000000	6.0440449	0.440	0.164
csmd2	ZoneA_MaSp1	Major	0.0000000	1.4684072	0.284	0.077
tutl	ZoneA_MaSp1	Major	0.0000000	3.1332207	0.382	0.148

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
plxna4_2	ZoneA_MaSp1	Major	0.0000000	0.5002040	0.366	0.165
sema-1a_2	ZoneA_MaSp1	Major	0.0000000	1.3614088	0.400	0.178
CaAn_8	ZoneA_MaSp1	Major	0.0000000	3.2280390	0.353	0.092
LASC018202	ZoneA_MaSp1	Major	0.0000000	2.4882895	0.466	0.252
LASC002684	ZoneA_MaSp1	Major	0.0000000	4.4955572	0.424	0.177
stan	ZoneA_MaSp1	Major	0.0000000	2.1885107	0.262	0.078
tutl_2	ZoneA_MaSp1	Major	0.0000000	4.1864063	0.295	0.100
egfr_2	ZoneA_MaSp1	Major	0.0000000	3.8720879	0.401	0.168
LASC013964	ZoneA_MaSp1	Major	0.0000000	1.4001311	0.274	0.100
LASC022305	ZoneA_MaSp1	Major	0.0000000	0.2936275	0.224	0.096
con	ZoneA_MaSp1	Major	0.0000000	5.6766484	0.351	0.138
nemp1	ZoneA_MaSp1	Major	0.0000000	1.6140047	0.291	0.126
LASC017480	ZoneA_MaSp1	Major	0.0000000	3.0473804	0.350	0.138
moxd1_2	ZoneA_MaSp1	Major	0.0000000	2.4797692	0.280	0.115
cadn_2	ZoneA_MaSp1	Major	0.0000000	1.2519927	0.354	0.195
LASC020733	ZoneA_MaSp1	Major	0.0000000	2.1186623	0.244	0.126
SpiCE-LMa3	ZoneA_MaSp1	Major	0.0000000	2.3025353	0.531	0.357
lcch3_2	ZoneA_MaSp1	Major	0.0000000	1.1111371	0.315	0.134
mew	ZoneA_MaSp1	Major	0.0000000	4.0528032	0.275	0.092
sele	ZoneA_MaSp1	Major	0.0000000	4.6621797	0.261	0.109
cut-1_3	ZoneA_MaSp1	Major	0.0000000	1.9633760	0.281	0.099
cd109_3	ZoneA_MaSp1	Major	0.0000000	5.4933677	0.317	0.147
cd109_6	ZoneA_MaSp1	Major	0.0000000	2.4977725	0.293	0.116
LASC019920	ZoneA_MaSp1	Major	0.0000000	1.5816626	0.210	0.102
gyc76c	ZoneA_MaSp1	Major	0.0000000	0.6831521	0.309	0.094
LASC009135	ZoneA_MaSp1	Major	0.0000000	1.7617453	0.331	0.180
rxfp1	ZoneA_MaSp1	Major	0.0000000	0.7831577	0.161	0.042
cad96ca	ZoneA_MaSp1	Major	0.0000000	3.0203962	0.259	0.125
nup210	ZoneA_MaSp1	Major	0.0000000	2.1118841	0.194	0.116
if	ZoneA_MaSp1	Major	0.0000000	1.0719737	0.228	0.079
LASC000308	ZoneA_MaSp1	Major	0.0000000	3.0785259	0.287	0.145
unc5c_unc5b	ZoneA_MaSp1	Major	0.0000000	0.5931071	0.241	0.095
ptp10d_2	ZoneA_MaSp1	Major	0.0000000	3.4362353	0.295	0.132
kirrel3	ZoneA_MaSp1	Major	0.0000000	2.6070518	0.152	0.073
tmed6	ZoneA_MaSp1	Major	0.0000000	0.2770355	0.425	0.314
ror2	ZoneA_MaSp1	Major	0.0000000	3.5686270	0.281	0.090
dld_dl	ZoneA_MaSp1	Major	0.0000000	2.7532129	0.131	0.089
fat4_2	ZoneA_MaSp1	Major	0.0000000	0.8419014	0.199	0.089
mmp16_3	ZoneA_MaSp1	Major	0.0000000	0.6176769	0.168	0.101
slc31a1_3	ZoneA_MaSp1	Major	0.0000000	4.8987970	0.252	0.143
kirrel1_kirrel2	ZoneA_MaSp1	Major	0.0000000	4.3293003	0.339	0.134
fst	ZoneA_MaSp1	Major	0.0000000	0.6297101	0.174	0.099
adam10_4	ZoneA_MaSp1	Major	0.0000000	3.4873550	0.217	0.097
adam8_adam12	ZoneA_MaSp1	Major	0.0000000	2.0881680	0.186	0.072
LASC005077	ZoneA_MaSp1	Major	0.0000000	1.2707074	0.277	0.181
flt4	ZoneA_MaSp1	Major	0.0000000	0.8682620	0.272	0.178
rxfp2	ZoneA_MaSp1	Major	0.0000000	0.8181128	0.141	0.079

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
gyc76c_2	ZoneA_MaSp1	Major	0.0000000	0.8405924	0.268	0.117
LASC009004	ZoneA_MaSp1	Major	0.0000000	0.5866490	0.168	0.084
postn	ZoneA_MaSp1	Major	0.0000000	0.6807268	0.174	0.067
ptk7	ZoneA_MaSp1	Major	0.0000000	3.0768186	0.209	0.135
ptprm_ptprt	ZoneA_MaSp1	Major	0.0000000	2.9975657	0.333	0.195
nell2	ZoneA_MaSp1	Major	0.0000000	3.4525927	0.333	0.180
prtq	ZoneA_MaSp1	Major	0.0000000	8.1514749	0.260	0.126
n_3	ZoneA_MaSp1	Major	0.0000000	1.0341877	0.195	0.101
dome_2	ZoneA_MaSp1	Major	0.0000000	1.3353787	0.294	0.145
LASC002635	ZoneA_MaSp1	Major	0.0000000	1.3609093	0.256	0.136
ft	ZoneA_MaSp1	Major	0.0000000	0.7953131	0.276	0.155
ddr2	ZoneA_MaSp1	Major	0.0000000	0.8660591	0.221	0.104
LASC003142	ZoneA_MaSp1	Major	0.0000000	1.8105702	0.274	0.166
gpc6_2	ZoneA_MaSp1	Major	0.0000000	4.5486711	0.251	0.127
slc9a3_2	ZoneA_MaSp1	Major	0.0000004	1.4260136	0.241	0.180
adamts9_adamts20	ZoneA_MaSp1	Major	0.0000021	1.7369116	0.240	0.085
nlgn4x	ZoneA_MaSp1	Major	0.0000022	0.9779346	0.213	0.094
LASC022638	ZoneA_MaSp1	Major	0.0000054	0.6969528	0.238	0.124
nrf-6_32	ZoneA_MaSp1	Major	0.0000095	0.7324497	0.139	0.118
p3h1_p3h2	ZoneA_MaSp1	Major	0.0000186	0.9434030	0.124	0.070
LASC011249	ZoneA_MaSp1	Major	0.0000777	1.9472398	0.166	0.139
nrk	ZoneA_MaSp1	Major	0.0001778	7.1419829	0.183	0.105
pxt_6	ZoneA_MaSp1	Major	0.0002475	0.7921283	0.146	0.103
creld2	ZoneA_MaSp1	Major	0.0007096	0.8488549	0.180	0.122
lar_2	ZoneA_MaSp1	Major	0.0010869	1.2403713	0.192	0.139
fgfr1	ZoneA_MaSp1	Major	0.0456492	1.5289854	0.201	0.138
lana	ZoneA_MaSp1	Major	0.1707831	0.4626722	0.150	0.099
adgrb3_ttn	ZoneA_MaSp1	Major	0.2889286	4.0111074	0.233	0.145
LASC001164	ZoneA_MaSp1	Major	0.2950436	0.6667201	0.212	0.159
LASC012923	ZoneA_MaSp1	Major	0.3363887	3.0165282	0.256	0.162
lar	ZoneA_MaSp1	Major	0.5715456	2.1142225	0.248	0.200
LASC018303	ZoneA_MaSp1	Major	1.0000000	3.6264260	0.162	0.087
calcr1a_2	ZoneA_MaSp1	Major	1.0000000	2.4930125	0.251	0.145
arsb_7	ZoneA_MaSp1	Major	1.0000000	1.7229918	0.145	0.114
unc-71_adam11	ZoneA_MaSp1	Major	1.0000000	1.7128943	0.150	0.119
LASC021766	ZoneA_MaSp1	Major	1.0000000	1.3284966	0.143	0.146
sup-17	ZoneA_MaSp1	Major	1.0000000	1.0894417	0.226	0.128
LASC009427	ZoneABC	Major	0.0000000	1.6747274	0.678	0.162
spa	ZoneABC	Major	0.0000000	0.9098610	0.774	0.149
cd74	ZoneABC	Major	0.0000000	1.4541895	0.226	0.112
LASC011517	ZoneABC	Major	0.0000000	0.3903665	0.246	0.111
adgrg6_2	ZoneABC	Major	0.0000000	2.3703716	0.293	0.073
ppaf2_3	ZoneABC	Major	0.0000000	0.9372264	0.284	0.118
LASC009204	ZoneABC	Major	0.0000000	1.5668676	0.138	0.087
LASC013615	ZoneABC	Major	0.0000000	3.5447496	0.228	0.140
ddr2_3	ZoneABC	Major	0.0000000	1.5521453	0.173	0.064
LASC005077	ZoneABC	Major	0.0000000	1.4479624	0.352	0.164
htra4	ZoneABC	Major	0.0000000	4.5700545	0.227	0.100

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
serpinb1_2	ZoneABC	Major	0.0000000	2.0145051	0.179	0.074
zig-4	ZoneABC	Major	0.0000000	1.1181274	0.207	0.138
bmpr1b	ZoneABC	Major	0.0000000	1.3672570	0.209	0.130
LASC012179	ZoneABC	Major	0.0000000	1.1739294	0.224	0.170
corin_sorl1	ZoneABC	Major	0.0000000	1.8809249	0.145	0.093
kcp	ZoneABC	Major	0.0000000	0.9050962	0.112	0.081
fgfr1	ZoneABC	Major	0.0000000	1.1954590	0.192	0.135
lana	ZoneABC	Major	0.0000000	1.1942462	0.152	0.096
sema-1a_2	ZoneABC	Major	0.0000000	3.3695009	0.135	0.202
lrp1_cblif_2	ZoneABC	Major	0.0000000	1.5408770	0.172	0.134
slc31a1_3	ZoneABC	Major	0.0000000	0.8536114	0.150	0.151
LASC010178	ZoneABC	Major	0.0000000	0.4792133	0.106	0.098
LASC021016	ZoneABC	Major	0.0000000	0.8982575	0.176	0.146
unc5c_unc5b	ZoneABC	Major	0.0000000	0.6752261	0.196	0.092
LASC002333	ZoneABC	Major	0.0000000	0.6754772	0.105	0.076
LASC001338	ZoneABC	Major	0.0000000	0.9651993	0.161	0.114
LASC012206	ZoneABC	Major	0.0000000	1.5372836	0.175	0.111
cblif	ZoneABC	Major	0.0000000	0.3333544	0.171	0.129
LASC005799	ZoneABC	Major	0.0000000	2.0462452	0.219	0.143
LASC017517	ZoneABC	Major	0.0000004	1.3300528	0.231	0.247
slit1_2	ZoneABC	Major	0.0000022	1.5785912	0.166	0.127
LASC018509	ZoneABC	Major	0.0000044	1.0518218	0.262	0.197
LASC005497	ZoneABC	Major	0.0000458	1.1245335	0.160	0.129
LASC007798	ZoneABC	Major	0.0000544	0.2552787	0.202	0.152
LASC016138	ZoneABC	Major	0.0002985	0.7158904	0.113	0.121
LASC009364	ZoneABC	Major	0.0003435	0.7976067	0.160	0.115
LASC022305	ZoneABC	Major	0.0005528	1.8729388	0.171	0.095
LASC016866	ZoneABC	Major	0.0010699	1.2329453	0.221	0.151
reln	ZoneABC	Major	0.0017392	1.4098476	0.101	0.121
adprm	ZoneABC	Major	0.0042855	1.2438588	0.088	0.134
flz	ZoneABC	Major	0.0108277	1.6523510	0.152	0.174
LASC015537	ZoneABC	Major	0.0545577	2.1092821	0.094	0.166
cp1	ZoneABC	Major	0.2926502	6.1502477	0.260	0.310
cubn_4	ZoneABC	Major	0.3615061	0.7439664	0.041	0.161
mmut	ZoneABC	Major	1.0000000	4.5741392	0.050	0.143
tm7sf3	ZoneABC	Major	1.0000000	1.8140284	0.116	0.135
LASC020958	ZoneABC	Major	1.0000000	1.7311207	0.148	0.115
nlgn4x	ZoneABC	Major	1.0000000	1.0688887	0.042	0.112
acpp	ZoneABC	Major	1.0000000	0.9297187	0.137	0.160
sfrp5	ZoneABC	Major	1.0000000	0.7743546	0.131	0.114
LASC022638	ZoneABC	Major	1.0000000	0.3299840	0.110	0.136
AmSp-like2	ZoneB_MaSp3	Major	0.0000000	9.4580991	0.620	0.086
tnxb_7	ZoneB_MaSp3	Major	0.0000000	9.2230561	0.554	0.014
MaSp3a	ZoneB_MaSp3	Major	0.0000000	8.7822520	0.727	0.032
qpct	ZoneB_MaSp3	Major	0.0000000	7.8782285	0.587	0.071
sp-1339#prd34368.1_4	ZoneB_MaSp3	Major	0.0000000	6.5653855	0.745	0.159
MaSp3b	ZoneB_MaSp3	Major	0.0000000	5.9700879	0.911	0.086

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
SpiCE-LMa2	ZoneB_MaSp3	Major	0.0000000	3.4551909	0.684	0.183
LASC008293	ZoneB_MaSp3	Major	0.0000000	3.3836995	0.653	0.131
LASC009784	ZoneB_MaSp3	Major	0.0000000	6.7402479	0.577	0.135
AmSp-like1	ZoneB_MaSp3	Major	0.0000000	11.1599644	0.469	0.089
LASC015905	ZoneB_MaSp3	Major	0.0000000	9.7964727	0.473	0.058
zan	ZoneB_MaSp3	Major	0.0000000	3.6639490	0.492	0.087
LASC008294	ZoneB_MaSp3	Major	0.0000000	3.8109102	0.396	0.075
LASC008296	ZoneB_MaSp3	Major	0.0000000	7.3043937	0.330	0.054
LASC016881	ZoneB_MaSp3	Major	0.0000000	1.0833557	0.449	0.151
tg_3	ZoneB_MaSp3	Major	0.0000000	2.6561790	0.725	0.420
cpm_2	ZoneB_MaSp3	Major	0.0000000	10.4733902	0.363	0.102
LASC000534	ZoneB_MaSp3	Major	0.0000000	7.5592863	0.199	0.014
adgrl3_2	ZoneB_MaSp3	Major	0.0000000	3.2629106	0.267	0.107
dnajc1	ZoneB_MaSp3	Major	0.0000000	5.0910248	0.213	0.074
LASC008291	ZoneB_MaSp3	Major	0.0000000	4.8786502	0.330	0.142
fgfr11_5	ZoneB_MaSp3	Major	0.0000000	3.3002514	0.375	0.151
LASC016514	ZoneB_MaSp3	Major	0.0000000	2.3328583	0.161	0.046
stan	ZoneB_MaSp3	Major	0.0000000	2.0164510	0.205	0.080
SpiCE-LMa4	ZoneB_MaSp3	Major	0.0000000	4.3807851	0.300	0.100
LASC005177	ZoneB_MaSp3	Major	0.0000000	0.9404728	0.402	0.235
LASC005077	ZoneB_MaSp3	Major	0.0000000	2.0637848	0.332	0.174
cfad	ZoneB_MaSp3	Major	0.0000000	13.6696725	0.517	0.322
cd109	ZoneB_MaSp3	Major	0.0000000	1.2369464	0.316	0.160
slc9a3_2	ZoneB_MaSp3	Major	0.0000000	2.3285828	0.342	0.169
sp-1339#prd34368.1_3	ZoneB_MaSp3	Major	0.0000000	4.4630012	0.127	0.046
sele	ZoneB_MaSp3	Major	0.0000000	0.7917744	0.246	0.108
LASC008662	ZoneB_MaSp3	Major	0.0000000	0.4505469	0.154	0.104
nrf-6_32	ZoneB_MaSp3	Major	0.0000000	0.7045048	0.183	0.113
zig-4	ZoneB_MaSp3	Major	0.0000000	2.8075369	0.234	0.139
mug1	ZoneB_MaSp3	Major	0.0000000	2.4022469	0.173	0.063
LASC021016	ZoneB_MaSp3	Major	0.0000000	3.9999512	0.222	0.143
moxd1_2	ZoneB_MaSp3	Major	0.0000000	1.6414835	0.195	0.120
LASC008642	ZoneB_MaSp3	Major	0.0000000	2.1495492	0.170	0.115
cpa2_3	ZoneB_MaSp3	Major	0.0000000	2.3510096	0.122	0.094
dld_dl	ZoneB_MaSp3	Major	0.0000000	2.5536458	0.180	0.084
SpiCE-LMa6	ZoneB_MaSp3	Major	0.0000000	0.8582548	0.508	0.421
adamts9_adamts20	ZoneB_MaSp3	Major	0.0000000	1.1346640	0.118	0.093
egfr_2	ZoneB_MaSp3	Major	0.0000000	0.7278876	0.192	0.184
csmd2	ZoneB_MaSp3	Major	0.0000000	1.3488511	0.138	0.088
LASC022305	ZoneB_MaSp3	Major	0.0000000	1.5515922	0.116	0.104
rep	ZoneB_MaSp3	Major	0.0000000	1.1701052	0.106	0.090
p4hb	ZoneB_MaSp3	Major	0.0000000	0.5624569	0.176	0.137
LASC013964	ZoneB_MaSp3	Major	0.0000000	1.7141654	0.155	0.108
cd74	ZoneB_MaSp3	Major	0.0000000	1.6046775	0.168	0.123
stim_2	ZoneB_MaSp3	Major	0.0000000	6.9429335	0.249	0.150
LASC013808	ZoneB_MaSp3	Major	0.0000000	0.6559283	0.213	0.154
bmp1	ZoneB_MaSp3	Major	0.0000000	0.7001918	0.130	0.095
LASC001164	ZoneB_MaSp3	Major	0.0000000	1.3402391	0.213	0.158

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
tgoln2	ZoneB_MaSp3	Major	0.0000000	2.3414674	0.209	0.196
tm7sf3	ZoneB_MaSp3	Major	0.0000000	1.5285236	0.126	0.133
LASC011995	ZoneB_MaSp3	Major	0.0000000	0.7564624	0.306	0.464
cd109_6	ZoneB_MaSp3	Major	0.0000000	1.2845340	0.176	0.124
lanb2	ZoneB_MaSp3	Major	0.0000000	2.3106556	0.183	0.125
cpe	ZoneB_MaSp3	Major	0.0000000	2.5983695	0.178	0.151
ddr2	ZoneB_MaSp3	Major	0.0000000	1.8222061	0.097	0.114
LASC007798	ZoneB_MaSp3	Major	0.0000000	0.9035743	0.157	0.159
burs	ZoneB_MaSp3	Major	0.0000000	3.7326412	0.150	0.127
LASC015891	ZoneB_MaSp3	Major	0.0000000	2.5275763	0.144	0.155
pxt_6	ZoneB_MaSp3	Major	0.0000000	2.2396931	0.051	0.111
AgSp2b	ZoneB_MaSp3	Major	0.0000000	0.2729660	0.333	0.251
LASC020958	ZoneB_MaSp3	Major	0.0000000	1.7242260	0.134	0.118
nup210	ZoneB_MaSp3	Major	0.0000000	0.5437625	0.089	0.125
LASC012206	ZoneB_MaSp3	Major	0.0000000	1.5223565	0.128	0.118
LASC018303	ZoneB_MaSp3	Major	0.0000000	2.2382916	0.119	0.090
LASC009364	ZoneB_MaSp3	Major	0.0000000	0.9566417	0.118	0.121
LASC007183	ZoneB_MaSp3	Major	0.0000000	1.1052089	0.071	0.123
fur2	ZoneB_MaSp3	Major	0.0000000	1.6619302	0.113	0.128
LASC017480	ZoneB_MaSp3	Major	0.0000000	1.1803736	0.121	0.156
adgrb3_ttn	ZoneB_MaSp3	Major	0.0000004	0.7910246	0.172	0.149
tmed6	ZoneB_MaSp3	Major	0.0000004	0.3258103	0.331	0.321
ppib_cypb	ZoneB_MaSp3	Major	0.0000011	1.4355994	0.160	0.156
LASC016866	ZoneB_MaSp3	Major	0.0000012	1.0544207	0.147	0.162
nemp1	ZoneB_MaSp3	Major	0.0000717	0.7963343	0.123	0.139
LASC005497	ZoneB_MaSp3	Major	0.0000867	0.7695113	0.113	0.135
LASC013602	ZoneB_MaSp3	Major	0.0001169	1.5544053	0.165	0.164
LASC018509	ZoneB_MaSp3	Major	0.0016758	0.4340656	0.189	0.207
lcch3_2	ZoneB_MaSp3	Major	0.0017009	0.9455646	0.071	0.154
LASC021766	ZoneB_MaSp3	Major	0.0194102	0.3297997	0.072	0.152
nlgn4x	ZoneB_MaSp3	Major	1.0000000	0.7954053	0.050	0.108
LASC007099	ZoneC_SpiCE-LMa1	Major	0.0000000	14.7697915	0.947	0.069
npc2a_2	ZoneC_SpiCE-LMa1	Major	0.0000000	14.1868629	0.873	0.083
spz3_2	ZoneC_SpiCE-LMa1	Major	0.0000000	13.4027659	0.794	0.053
pxd_pxt_2	ZoneC_SpiCE-LMa1	Major	0.0000000	13.3342014	0.901	0.067
LASC017807	ZoneC_SpiCE-LMa1	Major	0.0000000	12.5537784	0.847	0.074
LASC004595	ZoneC_SpiCE-LMa1	Major	0.0000000	11.3283618	0.998	0.100
lysd	ZoneC_SpiCE-LMa1	Major	0.0000000	11.3175536	0.816	0.167
LASC020242	ZoneC_SpiCE-LMa1	Major	0.0000000	9.4817937	0.967	0.113
LASC010016	ZoneC_SpiCE-LMa1	Major	0.0000000	9.3713282	0.999	0.106
LASC017532	ZoneC_SpiCE-LMa1	Major	0.0000000	8.6401777	0.993	0.109
LASC007995	ZoneC_SpiCE-LMa1	Major	0.0000000	6.6332052	0.942	0.112
SpiCE-LMa1	ZoneC_SpiCE-LMa1	Major	0.0000000	6.1881025	1.000	0.128
spa	ZoneC_SpiCE-LMa1	Major	0.0000000	5.1471964	1.000	0.154
LASC016881	ZoneC_SpiCE-LMa1	Major	0.0000000	5.0615792	0.823	0.113
LASC009427	ZoneC_SpiCE-LMa1	Major	0.0000000	4.3091886	0.974	0.156
SpiCE-LMa2	ZoneC_SpiCE-LMa1	Major	0.0000000	3.1550229	0.973	0.154

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC005034	ZoneC_SpiCE-LMa1	Major	0.0000000	0.5026288	0.773	0.107
mttp_2	ZoneC_SpiCE-LMa1	Major	0.0000000	10.4550644	0.716	0.040
LASC013967	ZoneC_SpiCE-LMa1	Major	0.0000000	7.1810655	0.615	0.063
SpiCE-LMa4	ZoneC_SpiCE-LMa1	Major	0.0000000	1.8693211	0.523	0.078
LASC003333	ZoneC_SpiCE-LMa1	Major	0.0000000	4.2528313	0.648	0.306
slit3_5	ZoneC_SpiCE-LMa1	Major	0.0000000	0.3633843	0.108	0.227
LASC005801	ZoneC_SpiCE-LMa1	Major	0.0000000	1.3297073	0.756	0.501
kcp_2	ZoneC_SpiCE-LMa1	Major	0.0000000	7.4494532	0.472	0.028
sema5a	ZoneC_SpiCE-LMa1	Major	0.0000000	2.5343090	0.060	0.126
zan	ZoneC_SpiCE-LMa1	Major	0.0000000	0.6872607	0.498	0.085
LASC000308	ZoneC_SpiCE-LMa1	Major	0.0000000	3.7483194	0.086	0.161
LASC012647	ZoneC_SpiCE-LMa1	Major	0.0000000	11.8005797	0.487	0.027
corin_sorl1	ZoneC_SpiCE-LMa1	Major	0.0000000	0.5505518	0.049	0.105
ppib_cypb	ZoneC_SpiCE-LMa1	Major	0.0000000	0.3498937	0.119	0.160
LASC013968	ZoneC_SpiCE-LMa1	Major	0.0000000	2.8730985	0.111	0.059
LASC020733	ZoneC_SpiCE-LMa1	Major	0.0000000	1.3620265	0.058	0.142
kug	ZoneC_SpiCE-LMa1	Major	0.0000000	1.5668531	0.035	0.100
gpatch1	ZoneC_SpiCE-LMa1	Major	0.0000000	0.6140494	0.108	0.182
pgrp	ZoneC_SpiCE-LMa1	Major	0.0000000	1.7654826	0.184	0.201
unc-71_adam11	ZoneC_SpiCE-LMa1	Major	0.0000000	1.5201465	0.083	0.125
unc5c_unc5b	ZoneC_SpiCE-LMa1	Major	0.0000000	0.5579278	0.036	0.112
kiaa1324l	ZoneC_SpiCE-LMa1	Major	0.0000000	2.9148533	0.129	0.200
ncam1	ZoneC_SpiCE-LMa1	Major	0.0000000	2.0536131	0.067	0.165
arsb_7	ZoneC_SpiCE-LMa1	Major	0.0000000	0.7981511	0.124	0.115
svep1_9	ZoneC_SpiCE-LMa1	Major	0.0000000	0.8791508	0.058	0.135
hspa5_2	ZoneC_SpiCE-LMa1	Major	0.0000000	7.8246545	0.448	0.120
LASC013808	ZoneC_SpiCE-LMa1	Major	0.0000000	0.2560329	0.082	0.167
gpc6_2	ZoneC_SpiCE-LMa1	Major	0.0000000	2.0374329	0.065	0.143
avr2	ZoneC_SpiCE-LMa1	Major	0.0000000	0.6818487	0.154	0.199
cdh23_2	ZoneC_SpiCE-LMa1	Major	0.0000000	6.0437611	0.451	0.107
adam10_4	ZoneC_SpiCE-LMa1	Major	0.0000000	1.0095333	0.113	0.104
kirrel3	ZoneC_SpiCE-LMa1	Major	0.0000000	0.7423491	0.104	0.076
gusb	ZoneC_SpiCE-LMa1	Major	0.0000000	1.5747831	0.155	0.067
timp	ZoneC_SpiCE-LMa1	Major	0.0000000	5.4550056	0.344	0.100
serpinb1_2	ZoneC_SpiCE-LMa1	Major	0.0000000	1.5558798	0.131	0.084
ephb2	ZoneC_SpiCE-LMa1	Major	0.0000000	7.1707238	0.384	0.086
LASC010178	ZoneC_SpiCE-LMa1	Major	0.0000000	1.8116205	0.121	0.097
cd109	ZoneC_SpiCE-LMa1	Major	0.0000000	4.7156132	0.493	0.142
mmp16	ZoneC_SpiCE-LMa1	Major	0.0000000	3.5016156	0.112	0.035
LASC016428	ZoneC_SpiCE-LMa1	Major	0.0000000	1.9738977	0.173	0.123
adgrg6_2	ZoneC_SpiCE-LMa1	Major	0.0000000	2.6091120	0.178	0.094
calr_crt	ZoneC_SpiCE-LMa1	Major	0.0000000	4.3068909	0.531	0.219
LASC005799	ZoneC_SpiCE-LMa1	Major	0.0000000	1.2381145	0.174	0.151
AgSp2b	ZoneC_SpiCE-LMa1	Major	0.0000000	0.5710114	0.148	0.269
LASC010533	ZoneC_SpiCE-LMa1	Major	0.0000000	2.6604828	0.201	0.048
fkbp7_fkbp3_fpr2	ZoneC_SpiCE-LMa1	Major	0.0000000	2.1344164	0.400	0.112
lrp1_cblif_2	ZoneC_SpiCE-LMa1	Major	0.0000000	1.0523434	0.190	0.134
cblif	ZoneC_SpiCE-LMa1	Major	0.0000000	2.5481653	0.214	0.126

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
n_3	ZoneC_SpiCE-LMa1	Major	0.0000000	1.4353803	0.135	0.105
mmp16_3	ZoneC_SpiCE-LMa1	Major	0.0000000	0.9030511	0.117	0.104
LASC006810	ZoneC_SpiCE-LMa1	Major	0.0000000	7.9258851	0.370	0.128
p3h1_p3h2	ZoneC_SpiCE-LMa1	Major	0.0000000	0.5639457	0.141	0.067
slc9a3	ZoneC_SpiCE-LMa1	Major	0.0000000	6.4327461	0.195	0.019
smpd1	ZoneC_SpiCE-LMa1	Major	0.0000000	5.1154229	0.368	0.053
csmd2	ZoneC_SpiCE-LMa1	Major	0.0000000	1.0509505	0.117	0.090
oxt_xytl1	ZoneC_SpiCE-LMa1	Major	0.0000000	1.9481682	0.128	0.042
LASC013615	ZoneC_SpiCE-LMa1	Major	0.0000000	1.6897026	0.172	0.150
duox2_duox	ZoneC_SpiCE-LMa1	Major	0.0000000	5.8296746	0.361	0.158
ft	ZoneC_SpiCE-LMa1	Major	0.0000000	1.8014288	0.188	0.162
adprm	ZoneC_SpiCE-LMa1	Major	0.0000000	1.5238132	0.181	0.123
LASC015891	ZoneC_SpiCE-LMa1	Major	0.0000000	0.3921894	0.165	0.153
LASC016514	ZoneC_SpiCE-LMa1	Major	0.0000000	3.9696313	0.183	0.043
LASC017210	ZoneC_SpiCE-LMa1	Major	0.0000000	3.1173400	0.161	0.048
igf2r	ZoneC_SpiCE-LMa1	Major	0.0000001	6.0144401	0.409	0.154
dome_2	ZoneC_SpiCE-LMa1	Major	0.0000001	3.8721546	0.307	0.141
lrig3_5	ZoneC_SpiCE-LMa1	Major	0.0000001	0.8768924	0.130	0.114
LASC007994	ZoneC_SpiCE-LMa1	Major	0.0000004	6.8303029	0.347	0.026
LASC018202	ZoneC_SpiCE-LMa1	Major	0.0000044	4.4877352	0.344	0.260
ddr2_3	ZoneC_SpiCE-LMa1	Major	0.0000051	1.3733496	0.187	0.068
LASC022638	ZoneC_SpiCE-LMa1	Major	0.0000069	0.3379182	0.143	0.131
sfrp5	ZoneC_SpiCE-LMa1	Major	0.0000076	0.3394130	0.186	0.109
tutl_2	ZoneC_SpiCE-LMa1	Major	0.0000236	0.2623277	0.154	0.110
cpe	ZoneC_SpiCE-LMa1	Major	0.0000477	2.4772288	0.218	0.147
LASC013004	ZoneC_SpiCE-LMa1	Major	0.0000719	2.5046490	0.172	0.157
mys_2	ZoneC_SpiCE-LMa1	Major	0.0001145	7.4631246	0.169	0.190
hmcn1_3	ZoneC_SpiCE-LMa1	Major	0.0002697	0.8305822	0.235	0.164
CaAn_8	ZoneC_SpiCE-LMa1	Major	0.0003308	1.6729283	0.107	0.111
hsp90b1	ZoneC_SpiCE-LMa1	Major	0.0004291	3.2307846	0.180	0.066
smoc1_2	ZoneC_SpiCE-LMa1	Major	0.0009774	1.5941640	0.240	0.070
LASC012458	ZoneC_SpiCE-LMa1	Major	0.0019380	1.4147057	0.201	0.063
dnajc1	ZoneC_SpiCE-LMa1	Major	0.0023837	1.3702445	0.336	0.062
adgrl3_adgrl2_adgrl1	ZoneC_SpiCE-LMa1	Major	0.0512460	0.6133085	0.191	0.047
calcr1a_2	ZoneC_SpiCE-LMa1	Major	0.4590431	2.9092363	0.283	0.140
LASC007660	ZoneC_SpiCE-LMa1	Major	1.0000000	4.2352087	0.255	0.064
gyc76c_2	ZoneC_SpiCE-LMa1	Major	1.0000000	1.2206776	0.265	0.115
hmcn1_robo1	ZoneC_SpiCE-LMa1	Major	1.0000000	1.1337487	0.243	0.205
LASC001733	Duct_2	Major	0.0000000	13.4668336	0.958	0.072
LASC001731	Duct_2	Major	0.0000000	13.1712532	0.848	0.074
Spice_15	Duct_2	Major	0.0000000	13.0374357	0.827	0.073
LASC001732	Duct_2	Major	0.0000000	8.1866373	0.767	0.085
LASC020961	Duct_2	Major	0.0000000	7.5883850	0.884	0.115
LASC019092	Duct_2	Major	0.0000000	7.1111114	0.824	0.104
LASC020962	Duct_2	Major	0.0000000	6.7669023	0.784	0.085
obst-e_2	Duct_2	Major	0.0000000	6.0815606	0.889	0.105
LASC019275	Duct_2	Major	0.0000000	5.4304419	0.793	0.118

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC016362	Duct_2	Major	0.0000000	5.3326719	0.991	0.137
LASC001730	Duct_2	Major	0.0000000	4.7114704	0.982	0.109
LASC005739	Duct_2	Major	0.0000000	12.7367625	0.688	0.040
LASC004790	Duct_2	Major	0.0000000	3.4173257	0.701	0.097
ifi30_2	Duct_2	Major	0.0000000	11.8201732	0.704	0.104
LASC006408	Duct_2	Major	0.0000000	18.5785897	0.679	0.026
rpl10	Duct_2	Major	0.0000000	6.8716893	0.789	0.238
LASC013597	Duct_2	Major	0.0000000	6.8139697	0.687	0.156
LASC005973	Duct_2	Major	0.0000000	11.4543666	0.614	0.060
LASC015386	Duct_2	Major	0.0000000	3.2044942	0.572	0.091
LASC012976	Duct_2	Major	0.0000000	2.2389684	0.721	0.345
mcf2_4	Duct_2	Major	0.0000000	6.1314993	0.534	0.141
calu	Duct_2	Major	0.0000000	4.8942985	0.486	0.072
lar	Duct_2	Major	0.0000000	1.9721987	0.046	0.217
LASC003909	Duct_2	Major	0.0000000	1.7168359	0.196	0.442
ptk7	Duct_2	Major	0.0000000	0.7205197	0.015	0.150
tll1	Duct_2	Major	0.0000000	3.4902682	0.421	0.046
sog	Duct_2	Major	0.0000000	2.8981726	0.049	0.123
csmd3_2	Duct_2	Major	0.0000000	10.8940993	0.423	0.016
lrrn2	Duct_2	Major	0.0000000	1.7021846	0.058	0.119
sfrp5	Duct_2	Major	0.0000000	1.1927736	0.048	0.122
LASC019920	Duct_2	Major	0.0000000	0.9831785	0.034	0.116
lcch3_2	Duct_2	Major	0.0000000	1.2744415	0.032	0.156
LASC021766	Duct_2	Major	0.0000000	2.2074746	0.066	0.152
LASC016428	Duct_2	Major	0.0000000	1.2398707	0.030	0.136
fgfr1	Duct_2	Major	0.0000000	1.9711923	0.053	0.150
calub	Duct_2	Major	0.0000000	1.9635449	0.097	0.161
LASC002684	Duct_2	Major	0.0000000	2.4716614	0.125	0.201
tll2	Duct_2	Major	0.0000000	4.8714862	0.386	0.031
LASC016138	Duct_2	Major	0.0000000	2.3322530	0.081	0.123
LASC013984	Duct_2	Major	0.0000000	4.2338870	0.321	0.036
lrp1_2	Duct_2	Major	0.0000000	4.3910422	0.074	0.156
cpa2_3	Duct_2	Major	0.0000000	1.5417554	0.050	0.100
LASC001726	Duct_2	Major	0.0000000	14.3378054	0.411	0.015
csmd3	Duct_2	Major	0.0000000	9.5100206	0.322	0.022
nemp1	Duct_2	Major	0.0000000	1.1291377	0.053	0.144
svep1_9	Duct_2	Major	0.0000000	1.1045727	0.066	0.133
LASC001452	Duct_2	Major	0.0000000	6.3379365	0.065	0.203
LASC010178	Duct_2	Major	0.0000000	1.6330608	0.065	0.102
unc-71_adam11	Duct_2	Major	0.0000000	1.7028809	0.011	0.130
ace_3	Duct_2	Major	0.0000000	11.5600657	0.362	0.029
slc9a3_2	Duct_2	Major	0.0000000	1.2077751	0.148	0.187
unc5c_unc5b	Duct_2	Major	0.0000000	0.9697978	0.041	0.111
sup-17	Duct_2	Major	0.0000000	3.5474166	0.085	0.139
LASC004703	Duct_2	Major	0.0000000	2.0178959	0.109	0.064
ost-1_2	Duct_2	Major	0.0000000	0.6371466	0.069	0.123
corin_sorl1	Duct_2	Major	0.0000000	0.8897785	0.103	0.099
slit1_2	Duct_2	Major	0.0000000	0.8335177	0.053	0.138

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC001080	Duct_2	Major	0.0000000	6.0188015	0.123	0.059
LASC000692	Duct_2	Major	0.0000000	6.3377590	0.290	0.041
lrig3_5	Duct_2	Major	0.0000000	0.8842061	0.065	0.120
hmcn1_3	Duct_2	Major	0.0000000	2.8195714	0.145	0.172
slc39a4_slc39a12	Duct_2	Major	0.0000000	3.6719817	0.107	0.083
cblif	Duct_2	Major	0.0000000	0.7234084	0.086	0.138
LASC016433	Duct_2	Major	0.0000000	0.6408136	0.080	0.116
cut-1_3	Duct_2	Major	0.0000000	2.4944769	0.097	0.113
mew	Duct_2	Major	0.0000000	1.2350276	0.018	0.113
alpl_8	Duct_2	Major	0.0000000	0.3912655	0.410	0.157
LASC007251	Duct_2	Major	0.0000000	5.0960877	0.288	0.053
ptp10d	Duct_2	Major	0.0000000	6.6547840	0.134	0.050
spn88ea	Duct_2	Major	0.0000000	6.4003059	0.293	0.072
htra4	Duct_2	Major	0.0000000	0.8900238	0.101	0.118
adprm	Duct_2	Major	0.0000000	0.9945249	0.099	0.130
calr_crt	Duct_2	Major	0.0000000	0.4402821	0.254	0.246
dld_dl	Duct_2	Major	0.0000000	1.4459095	0.104	0.091
cad87a	Duct_2	Major	0.0000000	4.3543075	0.287	0.022
ror2	Duct_2	Major	0.0000000	3.4103517	0.109	0.103
csmd1_4	Duct_2	Major	0.0000000	1.9317569	0.147	0.091
rxfp2	Duct_2	Major	0.0000000	1.7189273	0.107	0.081
LASC005034	Duct_2	Major	0.0000000	1.0254239	0.177	0.166
hebp1_4	Duct_2	Major	0.0000000	2.9689684	0.128	0.146
col11a1	Duct_2	Major	0.0000000	3.2699587	0.109	0.049
LASC004993	Duct_2	Major	0.0000000	11.7518561	0.328	0.042
LASC008293	Duct_2	Major	0.0000000	2.9125604	0.386	0.160
LASC012886	Duct_2	Major	0.0000000	9.6133588	0.338	0.040
gpc5	Duct_2	Major	0.0000002	5.0361186	0.150	0.074
arsb	Duct_2	Major	0.0000003	4.5628151	0.145	0.050
n_3	Duct_2	Major	0.0000010	1.5628865	0.088	0.110
LASC019081	Duct_2	Major	0.0000010	2.1476878	0.138	0.103
cadm3	Duct_2	Major	0.0000011	3.9700507	0.287	0.062
ntn1	Duct_2	Major	0.0000011	4.7606579	0.109	0.013
LASC001585	Duct_2	Major	0.0000013	2.1774391	0.139	0.110
svep1_cfb_c2_plau_csmd1	Duct_2	Major	0.0000025	1.6609080	0.146	0.090
calcr1a_2	Duct_2	Major	0.0000094	1.3472833	0.146	0.153
LASC009203	Duct_2	Major	0.0000380	8.5019182	0.268	0.007
LASC009362	Duct_2	Major	0.0000643	1.4449889	0.167	0.242
sema5a	Duct_2	Major	0.0001296	1.2046506	0.138	0.119
adgrl3_adgrl2_adgrl1	Duct_2	Major	0.0001392	4.1702479	0.151	0.053
lar_2	Duct_2	Major	0.0009339	3.9324393	0.112	0.145
fur1	Duct_2	Major	0.0020598	1.6362624	0.109	0.122
trabd2b_3	Duct_2	Major	0.0024465	1.8234448	0.111	0.111
LASC015767	Duct_2	Major	0.0068264	0.5049309	0.104	0.085
chadl	Duct_2	Major	0.0194177	2.4831273	0.167	0.133
LASC005883	Duct_2	Major	0.0196996	5.4883678	0.390	0.226
LASC004413	Duct_2	Major	0.0653867	5.5345515	0.292	0.051

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
igsf9b_hmcn1_nphs1	Duct_2	Major	0.0659988	3.7608793	0.255	0.036
gpc5_2	Duct_2	Major	0.0805386	5.1263145	0.238	0.151
reck_2	Duct_2	Major	0.0908641	3.2915862	0.138	0.051
LASC016829	Duct_2	Major	0.0909136	6.4058779	0.292	0.081
fstl5_2	Duct_2	Major	0.7110686	5.4998205	0.241	0.016
LASC007798	Duct_2	Major	0.8749776	1.5461874	0.109	0.163
agrn_6	Duct_2	Major	1.0000000	5.9985391	0.260	0.026
tsc22d1	Duct_2	Major	1.0000000	4.6033467	0.269	0.125
dome	Duct_2	Major	1.0000000	2.8742999	0.135	0.149
LASC010652	Duct_2	Major	1.0000000	1.9120637	0.262	0.063
moxd1_2	Duct_2	Major	1.0000000	1.5475722	0.157	0.124
LASC022640	Duct_2	Major	1.0000000	1.1886569	0.133	0.145
gluc1alpha_2	Duct_2	Major	1.0000000	0.7264756	0.157	0.037
LASC009364	Duct_2	Major	1.0000000	0.5235786	0.065	0.126
fkbp7_fkbp3_fpr2	Duct_2	Major	1.0000000	0.3281313	0.225	0.131
LASC001730	Duct_1	Major	0.0000000	2.2278247	0.665	0.131
LASC016362	Duct_1	Major	0.0000000	1.6940848	0.932	0.135
LASC019275	Duct_1	Major	0.0000000	1.6089137	0.643	0.126
LASC004790	Duct_1	Major	0.0000000	3.6723306	0.624	0.099
LASC020962	Duct_1	Major	0.0000000	0.3249077	0.578	0.097
obst-e_2	Duct_1	Major	0.0000000	0.8190991	0.587	0.126
LASC012976	Duct_1	Major	0.0000000	4.1441934	0.670	0.347
calu	Duct_1	Major	0.0000000	2.1904350	0.505	0.067
zig-4	Duct_1	Major	0.0000000	0.4032139	0.029	0.158
ncam1	Duct_1	Major	0.0000000	2.0341962	0.047	0.166
csmd1_4	Duct_1	Major	0.0000000	0.8328357	0.043	0.100
vnn3	Duct_1	Major	0.0000000	0.4644023	0.045	0.123
LASC015386	Duct_1	Major	0.0000000	3.3157011	0.422	0.101
tll1	Duct_1	Major	0.0000000	2.4549316	0.286	0.056
lrrn2	Duct_1	Major	0.0000000	1.5623791	0.032	0.122
calub	Duct_1	Major	0.0000000	2.8652976	0.067	0.164
LASC011249	Duct_1	Major	0.0000000	0.2870120	0.042	0.150
unc-71_adam11	Duct_1	Major	0.0000000	1.6160141	0.030	0.129
cd109_6	Duct_1	Major	0.0000000	1.1038227	0.042	0.136
slc9a3_2	Duct_1	Major	0.0000000	1.2725696	0.144	0.188
cad96ca	Duct_1	Major	0.0000000	0.4478569	0.097	0.138
LASC003909	Duct_1	Major	0.0000000	0.8679261	0.298	0.435
gluc1alpha_2	Duct_1	Major	0.0000000	5.2758938	0.249	0.028
LASC005034	Duct_1	Major	0.0000000	4.8005064	0.354	0.150
ppaf2_3	Duct_1	Major	0.0000000	0.8973659	0.158	0.138
LASC013004	Duct_1	Major	0.0000000	0.5287866	0.075	0.166
lrp1_cblif_2	Duct_1	Major	0.0000000	1.0438701	0.134	0.139
unc5c_unc5b	Duct_1	Major	0.0000000	0.9168795	0.043	0.111
LASC005177	Duct_1	Major	0.0000000	5.1623613	0.171	0.256
gyc76c	Duct_1	Major	0.0000000	0.3469681	0.043	0.115
cg3556_23	Duct_1	Major	0.0000000	1.0796959	0.104	0.052
LASC017517	Duct_1	Major	0.0000000	0.3802457	0.204	0.249
LASC017480	Duct_1	Major	0.0000000	1.3075819	0.122	0.156

(continued)

geneName	cluster	gland	p_val_adj	avg_log2FC	pct.1	pct.2
LASC013984	Duct_1	Major	0.0000000	2.2030566	0.245	0.040
alpl_8	Duct_1	Major	0.0000000	6.2572987	0.320	0.163
p4hb	Duct_1	Major	0.0000000	0.7547931	0.103	0.144
reln	Duct_1	Major	0.0000000	0.4506504	0.172	0.114
LASC016428	Duct_1	Major	0.0000000	0.8350180	0.090	0.131
LASC008889	Duct_1	Major	0.0000000	0.2743655	0.101	0.166
pisd	Duct_1	Major	0.0000000	1.1002327	0.103	0.129
LASC001585	Duct_1	Major	0.0000000	1.0492433	0.151	0.109
sup-17	Duct_1	Major	0.0000000	1.8073899	0.089	0.139
serpinb1_2	Duct_1	Major	0.0000000	1.2807954	0.161	0.081
LASC008662	Duct_1	Major	0.0000000	0.8969285	0.152	0.104
LASC016433	Duct_1	Major	0.0000000	0.6155981	0.104	0.114
LASC011517	Duct_1	Major	0.0000000	0.7430060	0.162	0.126
dag1_2	Duct_1	Major	0.0000000	1.0606285	0.119	0.064
LASC001338	Duct_1	Major	0.0000001	2.0463238	0.152	0.117
fur2	Duct_1	Major	0.0000005	1.3399013	0.108	0.129
ddr2	Duct_1	Major	0.0000007	0.4069982	0.069	0.116
btd	Duct_1	Major	0.0000010	1.0330542	0.116	0.079
tgoln2	Duct_1	Major	0.0000163	0.7315724	0.191	0.198
fur1	Duct_1	Major	0.0000177	1.8340359	0.076	0.125
svep1_9	Duct_1	Major	0.0000260	1.7255803	0.191	0.123
col4a1	Duct_1	Major	0.0000670	1.7332871	0.122	0.066
gyc32e_2	Duct_1	Major	0.0006377	1.2034549	0.129	0.158
sema5a	Duct_1	Major	0.0007203	0.6533192	0.178	0.115
LASC011347	Duct_1	Major	0.0015646	1.0274841	0.320	0.114
tsc22d1	Duct_1	Major	0.0020747	1.4328416	0.166	0.133
mmp16_3	Duct_1	Major	0.0080998	1.0461438	0.160	0.101
hebp1_4	Duct_1	Major	0.0226233	0.8926490	0.273	0.133
LASC016866	Duct_1	Major	0.0804030	0.4629327	0.121	0.164
LASC019081	Duct_1	Major	0.0917943	0.3312641	0.138	0.102
cad87a	Duct_1	Major	0.1050057	2.4055290	0.125	0.034
unc5c	Duct_1	Major	1.0000000	3.2307617	0.175	0.051
LASC015767	Duct_1	Major	1.0000000	1.0961965	0.135	0.082
fth1-a	Duct_1	Major	1.0000000	0.9307719	0.235	0.128

Note:

Table 6: Supplementary data for Figure 6.